

Recent advances, challenges and future perspectives of polycyclic aromatic hydrocarbons as materials for metal-ion batteries – A comprehensive review

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ABSTRACT

Polycyclic aromatic hydrocarbons (PAHs) have become a promising choice for the development of metal-ion batteries in the future, thanks to their exceptional redox properties, versatile molecular structures, and affordability. This comprehensive review explores recent advances in the application of PAHs as anode and cathode materials for Li-ion, Na-ion, Al-ion and dual-ion batteries, highlighting their exceptional electrochemical properties such as high specific capacity, long-term cycle life, and environmental sustainability. The review also discusses innovative pre-metallization strategies using PAH-based reagents to enhance battery performance by reducing irreversible ion loss and improving initial Coulombic efficiency. Furthermore, the integration of machine learning and high-throughput screening to optimize PAH-based materials is discussed, providing insights into future design principles. Despite significant progress, challenges such as low redox activity and solubility issues persist, necessitating further research to unlock the full potential of PAHs in energy storage technologies. This review provides a roadmap for advancing PAH-based battery systems towards practical applications.

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1. Introduction

Polycyclic aromatic hydrocarbons (PAHs) is a broad class of organic compounds with unique electronic, optical, and structural properties, making them attractive for a wide range of applications, from organic electronics to materials science [1–4]. Due to their ability to self-assemble and their role as molecular precursors for carbon nanostructures, a vast amount of research has been dedicated to the synthesis and functionalization of PAHs [5–16]. Interest in these compounds remains strong, as evidenced by recent reviews covering both the fundamental aspects of their chemistry [17] and new synthetic methods, including modern electrochemical approaches [18,19].

PAHs possess unique physical and chemical properties that make them suitable for energy storage. Their extended π -conjugated systems facilitate reversible electron transfer and ion intercalation. The energy levels of the lowest unoccupied molecular orbital (LUMO) and highest occupied molecular orbital (HOMO) govern their redox behavior, where low LUMO energies favor reduction (anode applications) and low HOMO energies enable oxidation (cathode applications). The fundamental properties including electron affinity and binding energy with metal ions are exploited across all battery components. Building on these attributes, the past decade has witnessed a steady increase in interest in applying PAHs to electrochemical energy storage technologies, particularly in metal-ion batteries [20–24]. Initial research focused on using PAHs as components for the pre-metallization of anodes to compensate for irreversible ion loss and improve initial Coulombic efficiency [25]. Simultaneously, research has been conducted on PAHs as potential candidates for organic cathodes in innovative battery designs, such as dual-ion batteries, where they exhibit impressive performance capabilities. [26].

This growing research effort is underpinned by significant market trends. It is anticipated that global demand for lithium-ion batteries will experience substantial growth over the course of the next decade. It is anticipated that demand will increase from approximately 700 GWh in 2022 to around 4.7 TWh in 2030 [27]. The majority of this demand is expected to be for batteries used in transportation, particularly in electric vehicles, which are projected to account for approximately 4300 GWh in 2030. The elevated demand for lithium-ion batteries has precipitated a substantial escalation in the cost of metallic lithium. Therefore, it is imperative to create batteries that utilize less expensive metal cations as mobile ions, as well as to develop novel battery types with enhanced gravimetric or volumetric energy density [28,29].

Despite the impressive progress in the field, existing reviews tend to focus on isolated aspects of this broad area. For example, the review by Su and Chang [25] thoroughly examines the use of PAHs in pre-lithiation and pre-sodiation processes but does not address their role as standalone electrode materials. The work by Hu and co-authors [26] is dedicated to organic cathodes for dual-ion batteries, where PAHs are just one of many considered classes of compounds. The fundamental review by Stępień and co-authors [3] serves as an exhaustive guide to the chemistry of heterocyclic PAHs, however, its primary focus is on synthetic methodology rather than electrochemical applications. The review by Wang and co-authors [18] describes modern electrochemical approaches to PAH synthesis, which is an important but instrumental aspect preceding the creation of functional materials.

Thus, to date, there is no single, comprehensive review that unites the latest advances in the use of PAHs as active materials for all key components of metal-ion batteries—anodes, cathodes, and pre-metallization agents. The present work aims to fill this gap. In contrast to previous reviews, we offer a holistic view of PAHs as a versatile platform for creating high-performance and environmentally sustainable energy storage systems. We not only systematize the most recent information on the role of PAHs in lithium-, sodium-, aluminum-ion, and dual-ion batteries but also thoroughly examine innovative strategies to enhance their performance, including pre-metallization. Furthermore, for the first time within a single review, we discuss the integration of machine learning methods and high-throughput screening for the accelerated design and optimization of new PAH-based materials. By critically analyzing current challenges, such as low redox activity and solubility issues, we propose a roadmap for future research aimed at fully unlocking the potential of PAHs in next-generation energy storage technologies (see Fig. 1).

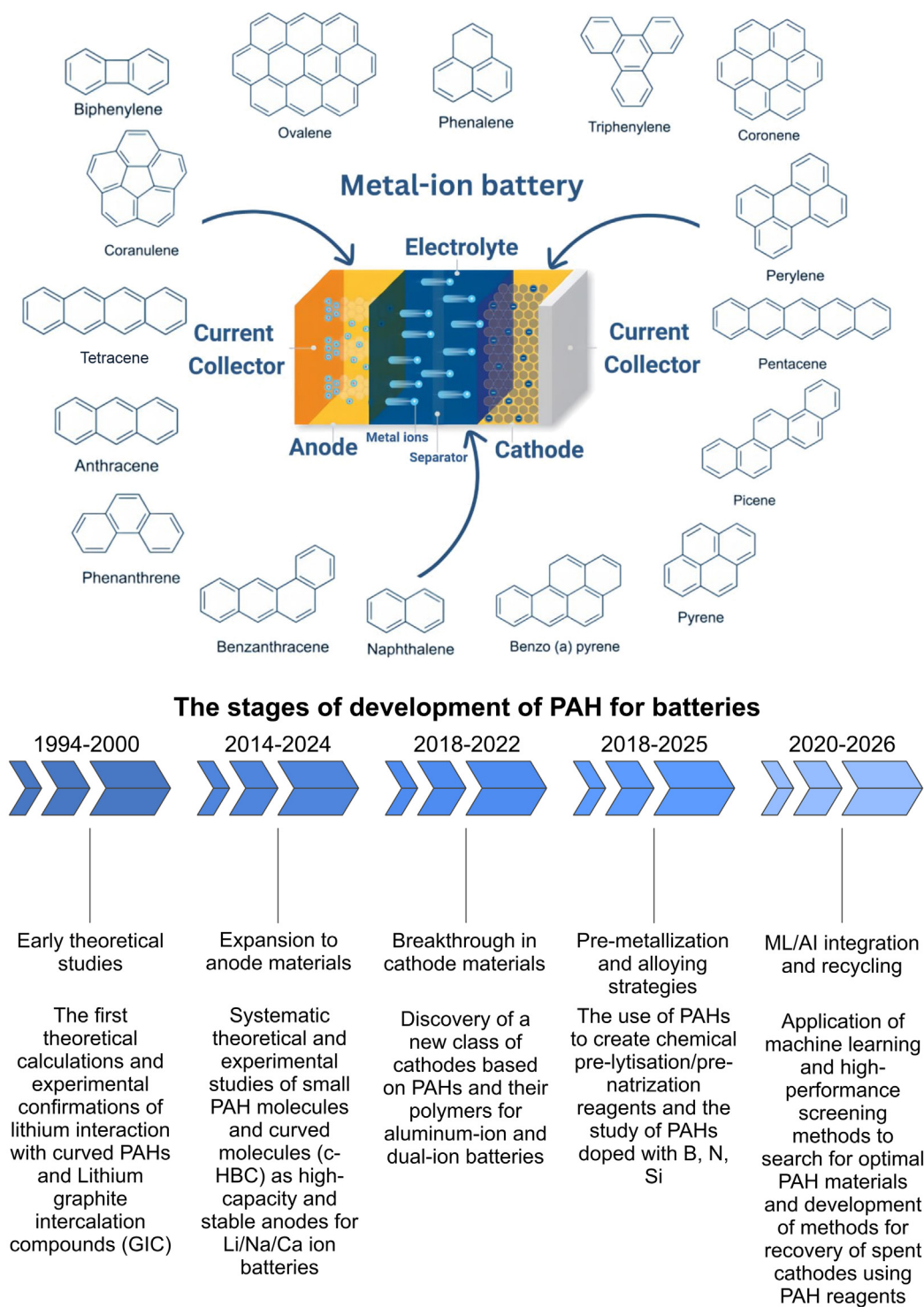


Fig. 1. The general scheme of applications of polyaromatic hydrocarbons (PAHs) in electrochemical metal-ion storage devices, together with a timeline of the main stages in the development of PAHs for batteries.

2. PAH as anode

2.1. Introduction to PAH-based anodes

To meet the growing demand for high-capacity and high-energy-density batteries, significant research efforts have focused on developing advanced anode materials [30–39]. Carbon-based materials have emerged as promising candidates due to their ability to reversibly intercalate lithium ions at low electrochemical potentials [35,40]. While synthetic graphite remains the dominant anode material in lithium-ion batteries—owing to its good specific capacity and cycling stability—its production involves an energy-intensive graphitization process at temperatures exceeding 2500 °C, leading to high manufacturing costs.

An alternative approach involves exploring organic materials as sustainable, high-performance anodes for metal-ion batteries [41]. Compared to inorganic counterparts, organic anode materials offer several advantages: they are derived from abundant natural sources or biomass, reducing reliance on critical raw materials. Their tunable molecular structures enable enhanced electrochemical performance, while their eco-friendly nature and lower toxicity make them safer alternatives. With their structural flexibility and cost-effectiveness, organic anode materials represent a compelling option for next-generation energy storage systems [13–15,21].

2.2. Fundamental mechanisms and theoretical insights

In particular PAH molecules, such as benzene, naphthalene, anthracene and tetracene, hexa-peri-hexabenzocoronene et al. have been extensively studied as potential candidates for high-performance battery materials [6,25,42–55]. The process of Li and Na atoms being absorbed by PAH has shown the possibility of using these materials in Li/Na-ion batteries [56–61]. Ishikawa et al. explored the process of lithium (Li) atoms attaching to pyrene, anthracene, and phenanthrene. It was revealed that the most favorable position for lithium doping is the ring-over site. Additionally, Li can also occupy edge sites on PAHs. The extent of electron transfer from alkali metals to PAHs depends on the size and structure of the PAH. Furthermore, the incorporation of Li atoms induces a loss of planarity in the PAH molecules [58]. The geometric and electronic configuration of carbon materials with lithium impurities have been explored using ovalene as a prototypical PAH model, employing semi-empirical molecular orbital calculations [62]. Pattarapongdilok et al. examined the adsorption of Li⁺ and Li atoms on coronene, circumcoronene, and circum-circumcoronene–graphene quantum dots of varying sizes in positive, negative, and neutral charge states. Their findings revealed that smaller graphene quantum dots exhibit a higher Li-loading capacity compared to larger ones [63]. Friedlein et al. [64] conducted a comparative analysis of the charge storage capacity of PAHs using semi-quantitative techniques. Their findings indicate that small and medium-sized PAHs may possess the highest energy storage capacity among all pure carbon systems (see Fig. 2 a). Ramya and Suresh [43] conducted an investigation into the electron distribution in PAHs, using the topology of the molecular electrostatic potential (MESP) for analysis (Fig. 2 b). This approach was used to classify the different types of rings found in PAHs, specifically the sextet rings (Rs), naphthalene-type rings (Rn), rings with a single double bond (Rd), rings with a butadiene-like distribution (Rb), and empty rings (Re), which are without any formal double bonds. In the case of polyacenes, such as pentacene, heptacene, or higher polyacenes with an odd number of hexagonal rings, the central ring appeared as Rs type, while those having an even number of hexagonal rings showed an Rn-type ring at the center. An evaluation of each type of ring in a PAH was conducted through a detailed study of its interactive behavior with Li⁺ cations and Li atoms. With the exception of chrysene, the Rs or Rn ring of a PAH is the most absorbent structure for both Li⁺ and Li. Ramya and Suresh [43] showed that PAHs made of small molecules, viz. naphthalene, pyrene, phenanthrene, chrysene, triphenylene, benzene, and circularly extended “carbon nanoflakes” such as coronene, circumbiphenyl, C₄₂H₁₆, and C₅₀H₁₈, are good anode materials for applications in LIBs [43]. Wang et al. [5] have systemically explored PAH/PAH^{n−} as anode redox-active couples for nonaqueous redox flow batteries. Nap/Nap^{n−} (Nap is naphthalene) identified as a promising option due to its low redox potential (−2.559 V vs. SHE), high solubility, acceptable diffusion coefficient, fast kinetics, and excellent stability (Fig. 2 c). Panigrahi and Sastry [56] investigated lithium adsorption on naphthalene, anthracene, and tetracene, employing multiple approaches to analyze lithium’s reduction capability (Fig. 2 d,e). Their study revealed that the binding energy of PAH–Li complexes is strongly influenced by both the Li atom’s position and the PAH’s size. The analysis of sequential binding energy revealed that the second lithium atom is more readily adsorbed, as indicated by its stronger binding force. Furthermore, a relationship between the degree of charge transfer, binding, and deformation energy has been established.

Chepkasov and Krashennnikov [49] conducted theoretical calculations to investigate the energy aspects of alkali metal (AM) atom insertion into bulk tetracene crystals. Unlike the adsorption of Li and Na atoms on individual tetracene molecules, the insertion of these and other AM atoms (K, Rb, Cs) into bulk tetracene crystals is energetically favorable compared to the formation of an infinite AM crystal, across a broad range of AM concentrations. Moreover, the insertion energy is significantly lower than that for graphite intercalation, a material currently used in AM battery anodes. In the case of Li, there is no swelling of the intercalated crystals. For Na, the crystal volume increases by less than 10%, making crystalline tetracene particularly appealing for energy storage due to its higher capacity compared to graphite. By studying the interaction of AM atoms with tetracene molecules, it was found that the intercalation energy (E_i) decreases with decreasing electronegativity, which can be attributed to stronger bonding resulting from greater charge transfer. However, Li deviates from this trend (Fig. 2 f). Similar behavior has been reported for AM atoms intercalated into bilayer graphene and graphite [36,65,66]. This anomaly arises from the strong covalent bonding between Li and carbon atoms [67], as well as the ionic contribution to the bonding. Maltsev et al. [48] investigated anthracene, tetracene, and pentacene as potential anode materials for Ca-ion batteries. Their research showed that the absorption of Ca atoms onto individual polyacene molecules is energetically advantageous (Fig. 2 g–i), as is their incorporation into bulk crystals across a broad spectrum of Ca concentrations. Notably, volume expansion during intercalation remained below 20% for all three materials, with pentacene exhibiting the smallest expansion (< 8%). These results highlight the promising potential of anthracene, tetracene, and pentacene as high-performance anode materials for Ca-ion batteries.

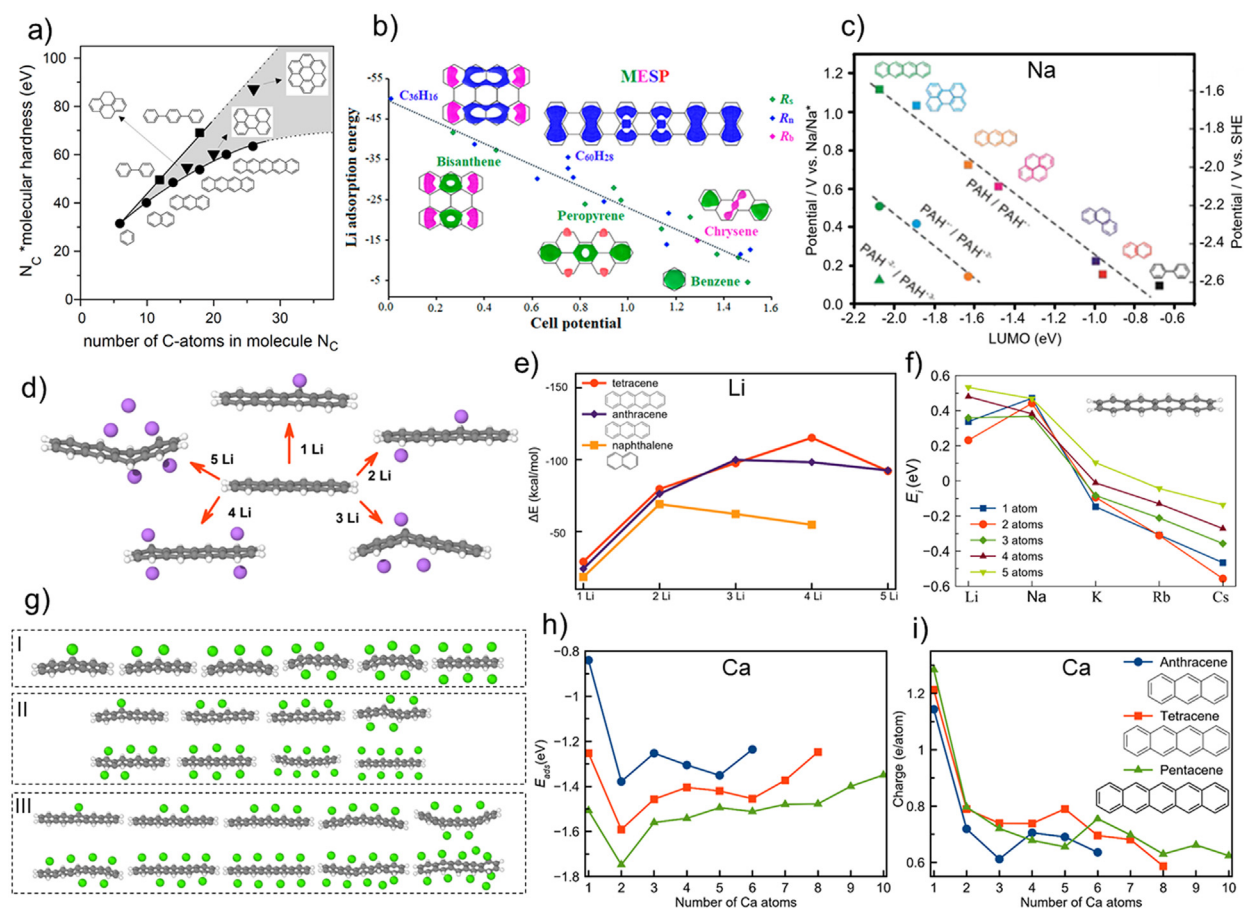


Fig. 2. (a) The product of the molecular electronic hardness η and the number of carbon atoms in the molecule N_C of acenes (full circles), non-linear polyaromatic hydrocarbon molecules (open triangle) and oligophenyls (full squares) as a function of N_C are shown (Ref. [64]). (b) Dependence of Li adsorption energy on cell potential of PAHs. (Ref. [43]) (c) The relationship between the redox potentials and the LUMO energies of PAH molecules (Ref. [5]). (d, e) Presentation of the adsorption of lithium atoms on the PAH and binding energy (Ref. [56]). (f) Energy required to take an AM atom from the bulk crystal and adsorb it on a free-standing tetracene molecule (Ref. [49]). (g) Optimized geometries of polyacene molecules with adsorbed Ca atoms: (I) anthracene, (II) tetracene, and (III) pentacene molecules. (h) Energy required to take a Ca atom from the bulk crystal and adsorb it on free-standing polyacene molecules, and (i) charge transfer from Ca atoms to polyacene molecules as a function of the number of adsorbed atoms (Ref. [48]).

2.3. Evaluation of PAHs as anode materials

Arya et al. [20] explored the possibility of using small-molecule polycyclic aromatic hydrocarbons (SMPAHs) as anode materials for lithium-ion batteries. Through a thorough experimental process that included electrode production, material analysis, and electrochemical testing, they thoroughly investigated the electrochemical performance of SMPAHs such as naphthalene (Naph), biphenyl (BP), 9,9-dimethylfluorene (DiMF), phenanthrene (Phen), p-terphenyl (p-Ter), and pyrene (Py). Since SMPAHs are known to dissolve in ether-based solvents (e.g., tetrahydrofuran and dimethoxyethane), making them incompatible with battery electrolytes, the study employed a commercial carbonate electrolyte for the SMPAH anode. To evaluate material stability, a dissolution test was performed, confirming no detectable dissolution after 24 h in the carbonate electrolyte (Fig. 3 a, b). Galvanostatic charge/discharge curves were obtained from Li||SMPAH batteries to determine specific capacities (Fig. 3 c–h). The measured initial lithiation/delithiation (reversible) capacities were:

Naph: 435/213 mAhg⁻¹

BP: 478/220 mAhg⁻¹

DiMF: 522/243 mAhg⁻¹

Phen: 458/201 mAhg⁻¹

p-Ter: 186/61 mAhg⁻¹

Py: 453/58 mAhg⁻¹ The corresponding initial Coulombic efficiencies (ICEs) were 49.0%, 46.0%, 46.6%, 43.8%, 32.8%, and 12.8%, respectively. While conventional graphite anodes (LiC₆) store one lithium ion per six-carbon ring, several SMPAHs demonstrated

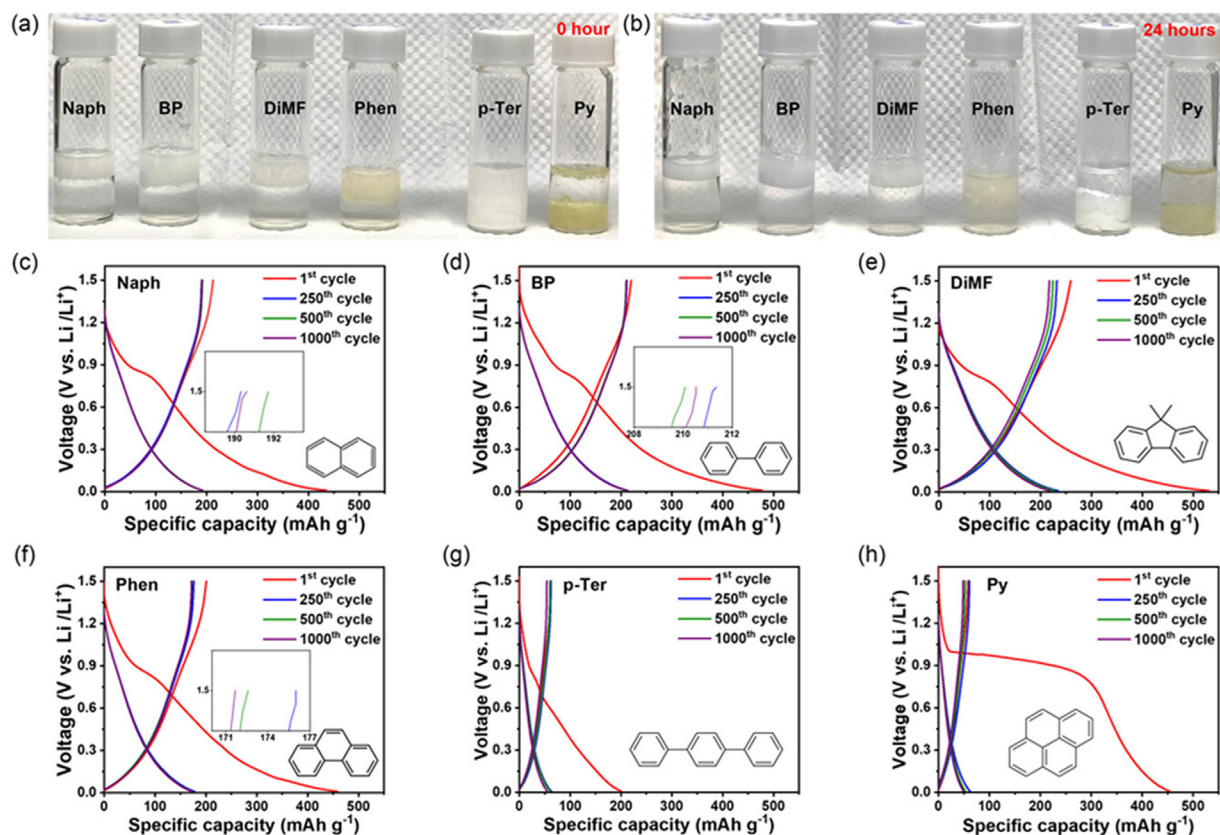


Fig. 3. Dissolution testing for various SMPAHs in the carbonate electrolyte (1 M LiPF₆ in an EC/EMC ratio of 1:2, volume ratio; 0.4 mg powder in 4 mL electrolyte) (a) before resting and (b) after 24 h of resting. Galvanostatic charge/discharge curves of lithium cells with (c) naphthalene, (d) biphenyl, (e) 9,9-dimethylfluorene, (f) phenanthrene, (g) *p*-terphenyl, and (h) pyrene anode materials at C/3 (Ref. [20]).

superior storage of Li, namely Naph (1.02), BP (1.26), and Phen (1.33). This enhanced lithium storage capability highlights the promise of SMPAHs as high-performance anode materials for Li-ion batteries.

Additionally, it was observed that the Py electrode exhibited a significant weight reduction compared to the other SMPAH anodes, indicating a high level of dissolution in the electrolyte during the cycling process. In the case of *p*-Ter, there were also observed changes in the UV-vis spectra, which appeared to be linked to substantial variations in the electrolyte components, suggesting unstable electrochemical reactions at the interface of the *p*-Ter anodes. Conversely, Naph, BP, DiMF, and Phen have shown great potential as anode materials in LIBs due to their high reversible capacity and their insolubility in the carbonate electrolyte [20].

The study of lithiation and delithiation has shown that lithium insertion into SMPAHs primarily takes place through the interaction with the π -electron systems of aromatic rings, without the release of hydrogen atoms. In order to study this process in detail, Arya et al. [20] carried out a DFT study on the interaction between individual molecules and PAH crystals and Li. All the rings of carbon in PAHs are flat, except for those in benzopyrene and *p*-terphenyl, which have non-zero torsional angles. The electrostatic potential is at its lowest in the center of the carbon ring for all polycyclic aromatic hydrocarbon molecules, suggesting a higher likelihood of Li⁺ ion attachment in that location. Calculations of adsorption have shown that most Li atoms or ions adsorb preferentially at the centers of carbon rings, with varying degrees of distortion in PAH molecules. Multiple Li atom adsorption on PAH molecules is possible. However, the strong electrostatic repulsion between ions restricts the maximum adsorption to two lithium ions per polyaromatic hydrocarbon molecule.

The crystal structures of both Naph and Phen feature dimers in their unit cells. These dimers are aligned with minimal π - π stacking but exhibit significant CH- π interactions, along with a tilting angle between adjacent molecules (Fig. 4). In contrast, bulk Py contains two dimers per unit cell, displaying strong π - π interactions within each dimer. Additionally, the dimer pairs are oriented almost perpendicular to each other (Fig. 4). The calculation of adsorption of different concentrations of Li on three bulk PAH systems showed that for bulk Naph and Phen, the energy of adsorption was comparable for the same quantity of adsorbed lithium, regardless of the system or the precise location of adsorption. Conversely, Li tends to adsorb more strongly onto Py molecules in the bulk compared to Naph or Phen molecules, in almost all instances. The adsorption of Li results in varying degrees of volume reduction in Py. It was determined that, during the electrochemical cycle, the Py dimer may initially undergo a complete rotation during the process of lithiation. Subsequently, during the process of delithiation, two molecules may expand vertically (from 3.4 Å to 3.8 Å),

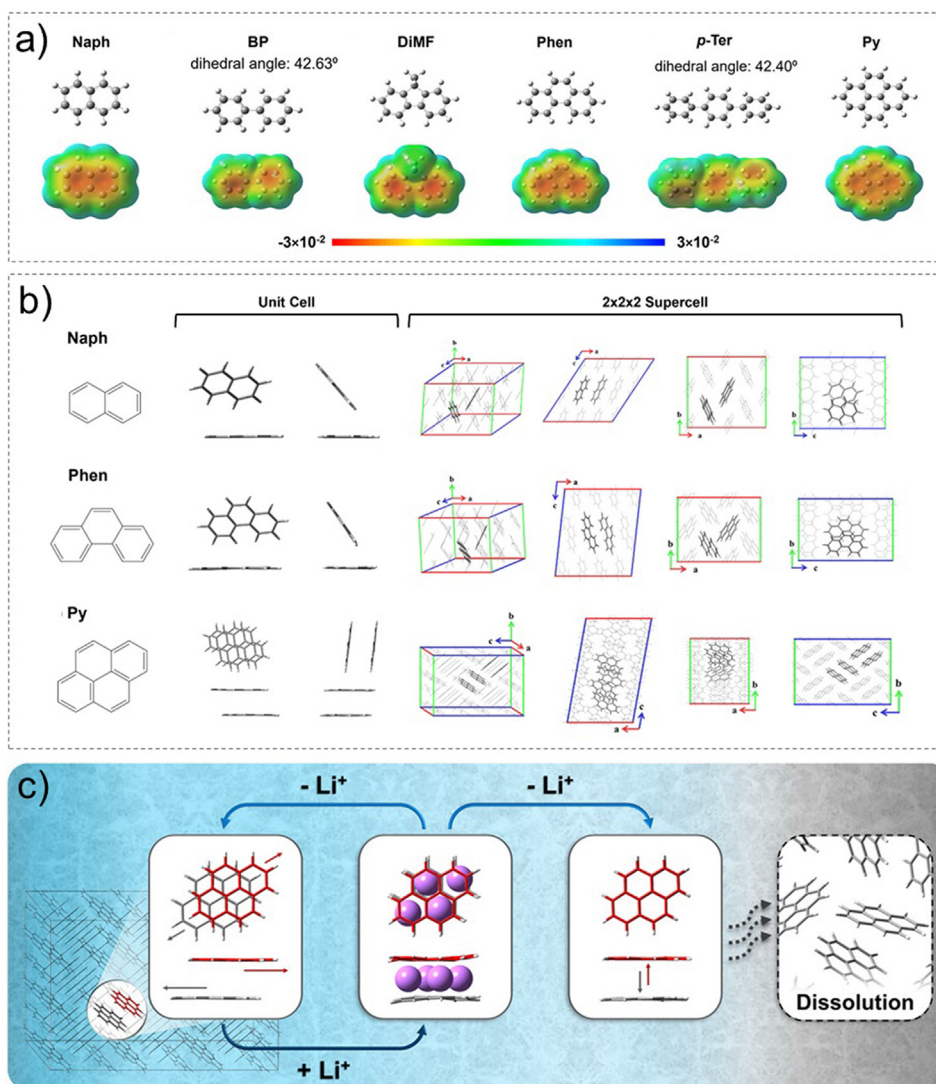


Fig. 4. (a) Optimized geometries and electrostatic potentials (ESPs) (isovalue: $0.0004 \text{ } |e|/\text{Bohr}^3$) of naphthalene, biphenyl, (c) 9,9-dimethylfluorene, phenanthrene, p-terphenyl, and pyrene single molecules. (b) The single molecule, the optimized molecular geometries in the unit cell, and the 2x2x2 supercells of the crystal naphthalene, phenanthrene, and pyrene systems. (c) Proposed geometry variations of pyrene crystals during the electrochemical process. The irreversible structural collapse due to the vertical expansion of the pyrene dimer results in the dissolution of pyrene into the electrolyte. (Ref. [20]).

as illustrated in Fig. 4c. These changes in structure may lead to irreversible dissolution of Py into the electrolyte, causing strong dissolution in battery cells, as shown in Fig. 3.

Cycling performance analysis revealed that SMPAHs demonstrate exceptional long-term cyclability, exhibiting negligible capacity fading. Capacity retention of 100.1%, 95.3%, 92.3%, 93.4%, and 93.8% was maintained for Naph, BP, DiMF, Phen, and p-Ter, respectively, with coulombic efficiencies close to 100% even after 1500 cycles. By contrast, the Py electrode showed significantly lower stability, retaining only 68.4% of its capacity under the same conditions. This is attributed to progressive dissolution of the active material during prolonged cycling. Outstanding cycle stability of SMPAH can be linked to a well-balanced interaction between Li^+ ions and PAH rings [6], enabling reversible binding and efficient release of active ions, optimizing cycling performance. The Naph, BP, DiMF, and Phen electrodes exhibit outstanding rate performance, maintaining their capacities at 121, 150, 174, and 95 mAh per gram, respectively, even at a high rate of 5C. When we switch back to the C/15 rate, these SMPAH electrodes retain their initial capacities, showcasing their remarkable reversibility.

Thus, in the work of Arya et al. [20], a combination of experimental and theoretical approaches were employed to demonstrate that Naph, BP, and DiMF are promising candidates for anode Li-ion battery materials. The findings of this investigation demonstrate that these materials possess a significant capacity, remarkable cycle durability, and exceptional performance at varying rates.

Investigation of the electrochemical behavior of SMPAH revealed their superior long-term cycleability, with capacity retention exceeding 92% after 1500 cycles and CE approaching 100%. This outstanding performance is a result of the stable connections between Li^+ ions and the aromatic hydrocarbon rings on the surface, which enable reversible attachment and detachment processes, thereby enhancing the longevity of the electrode.

To assess the potential of PAHs as anode materials for metal-ion batteries, Chepkasov et al. [68] conducted a systematic first-principles study to investigate the intercalation of Li, Na, K, Rb, Mg, and Ca into PAH crystals, such as naphthalene (NP), anthracene (AN), tetracene (TN), and pentacene (PN). The calculation of the intercalation energy, E_i , for metal atoms on the PAH molecules showed that E_i diminishes as the number of benzene rings in the PAH molecule rises, with the lowest value observed for a pentacene molecule (Fig. 5 a). The reduction in the electronegativity from magnesium to rubidium, as measured by the thermochemical electronegativity scale [69], results in a decrease in E_i . However, this is not the case for lithium. The behavior of alkali metal atoms intercalated in a bilayer of graphene and graphite has been previously discussed in the literature [36,65,66]. This phenomenon is attributed to the strong covalent bond between lithium and carbon atoms [67] and the ionic component of the bond. Similar to individual PAH molecules, the intercalation energy in PAH crystals diminishes as the number of benzene rings increases, as illustrated in Fig. 5b. In contrast to isolated molecules, the most significant deviation in intercalation energy is not observed for lithium, but for calcium in crystals. This is because in the context of limited interatomic spacing, the strong Coulomb interaction resulting from the presence of a divalent charge becomes more pronounced.

To investigate the chemical bonding of adsorbed or intercalated metal atoms in PAH crystals or isolated PAH molecules Chepkasov et al. [68] performed crystal orbital Hamilton population (COHP) calculations [70,71]. COHP analysis reveals that Mg and Ca exhibit a higher number of bonding states compared to alkali metals. The bond strength was evaluated using the integrated COHP (ICOHP), where more negative values indicate stronger bonds. For alkali metals, the ICOHP values remained relatively consistent, ranging from -0.24 to -0.17 eV/bond in the pentacene crystal and from -0.18 to -0.11 eV/bond in the isolated pentacene molecule (see Fig. 5 c). In contrast, Mg and Ca form significantly stronger bonds, with ICOHP values of -1.57 eV/bond (Mg) and -1.28 eV/bond (Ca) in the pentacene crystal, and -0.74 eV/bond (Mg) and -0.63 eV/bond (Ca) in isolated pentacene molecules. These findings suggest that stronger interactions between Mg and PAH systems could hinder the reversibility of insertion/extraction processes, especially at high charge/discharge rates - a phenomenon observed in other Mg battery host materials [72,73].

Furthermore, the intercalation of the maximum possible number of metal atoms into the PAH crystals was calculated to determine the volume change, intercalation energy, and capacity of the studied PAH crystals. The maximum number of intercalated metal atoms into the PAH was calculated based on the ratio of one metal atom per benzene ring in the PAH unit cell (4, 6, 8, and 10 metal atoms for NP, AN, TN, PN unit cells). At this concentration, metal clustering is avoided, preventing excessive structural expansion. Intercalation energy calculations showed that Mg incorporation into PAH crystals was energetically unfavorable, rendering these structures unstable (see Fig. 5 e). In contrast, other metals exhibited negative intercalation energies, suggesting feasible intercalation. However, phonon calculations confirmed dynamic stability (no imaginary phonon modes) only for Ca-, Li-, Na-, and K-intercalated tetracene and pentacene. The smallest volume change occurs when Ca, Li and Na atoms are intercalated into the pentacene crystal (5.2%, 0.8%, 3.4% for Ca, Li, Na respectively), which is an important result indicating the possibility of using pentacene crystals with Ca, Li or Na in batteries. Large ionic radii of K and Rb are expected to cause the largest volume changes in PAHs (see Fig. 5 f). The capacity of PAHs with Li, Na, K and Rb can be ~ 1.2 – 1.3 times greater than the theoretical capacity of Li-intercalated graphite, whereas the intercalation of Mg or Ca leads to an even higher capacity, ~ 2.3 – 2.6 times higher (see Fig. 5 g). Therefore, PAH crystals may hypothetically have a greater capacity as anode material than graphite. This was confirmed by the comprehensive theoretical study by conducted by Chepkasov et al. [68] They demonstrated that PAH crystals particularly tetracene and pentacene exhibit promising characteristics as anode materials for Ca-, Li-, and Na-ion batteries.

2.4. Curved PAH architectures for anode applications

Park et al. [21] investigated the potential of contorted hexabenzocoronene (c-HBC), a novel carbon-based material, as an anode for high-performance LIBs. They found that the doubly concave molecular wire structure of c-HBC provided high conductivity and expanded interlayer spacing, facilitating efficient Li-ion diffusion through its nanopore channels while maintaining exceptional storage stability. Additionally, their conjugated aromatic framework ensured electrolyte insolubility, further enhancing its suitability for LIB applications. The c-HBC molecule was synthesized via a three-step process and purified by column chromatography [74,75]. The resulting yellow product adopted a doubly concaved structure with a distorted pentacene core. This unique crystalline arrangement was due to steric hindrance during solidification (Fig. 6a) [76]. The adoption of this semiconductor as a Li host material in LIB was proposed as a strategy based on the fact that the contorted molecular structure can potentially benefit Li-ion insertion and lead to improved conductivity. As shown in Fig. 6a, Li-ions clearly prefer to insert metals at edges of the crystal, where small aromatic molecules are present, reducing the energy barrier associated with the process. Additionally, the contortion of the crystal structure promotes efficient ion diffusion through interstitial spaces.

To identify Li-ion insertion sites in c-HBC, Park et al. conducted grand canonical Monte Carlo simulations coupled with density functional theory calculations [21]. The results revealed that the most energetically favorable insertion sites were located on the surfaces of empty pores in the $R\bar{3}$ c-HBC crystal phase. Notably, the system exhibited the highest stability (with a formation energy of -5.8 eV) when Li-ions occupied pores with negative electrostatic potential between bent edge aromatic rings (Fig. 6 b). Park et al. also calculated the formation energy of Li-inserted c-HBC and its voltage profile as a function of Li-ion content. Li ions were introduced into the unit cell in multiples of three (3, 6, 9, and 18 Li ions), corresponding to the most stable insertion sites

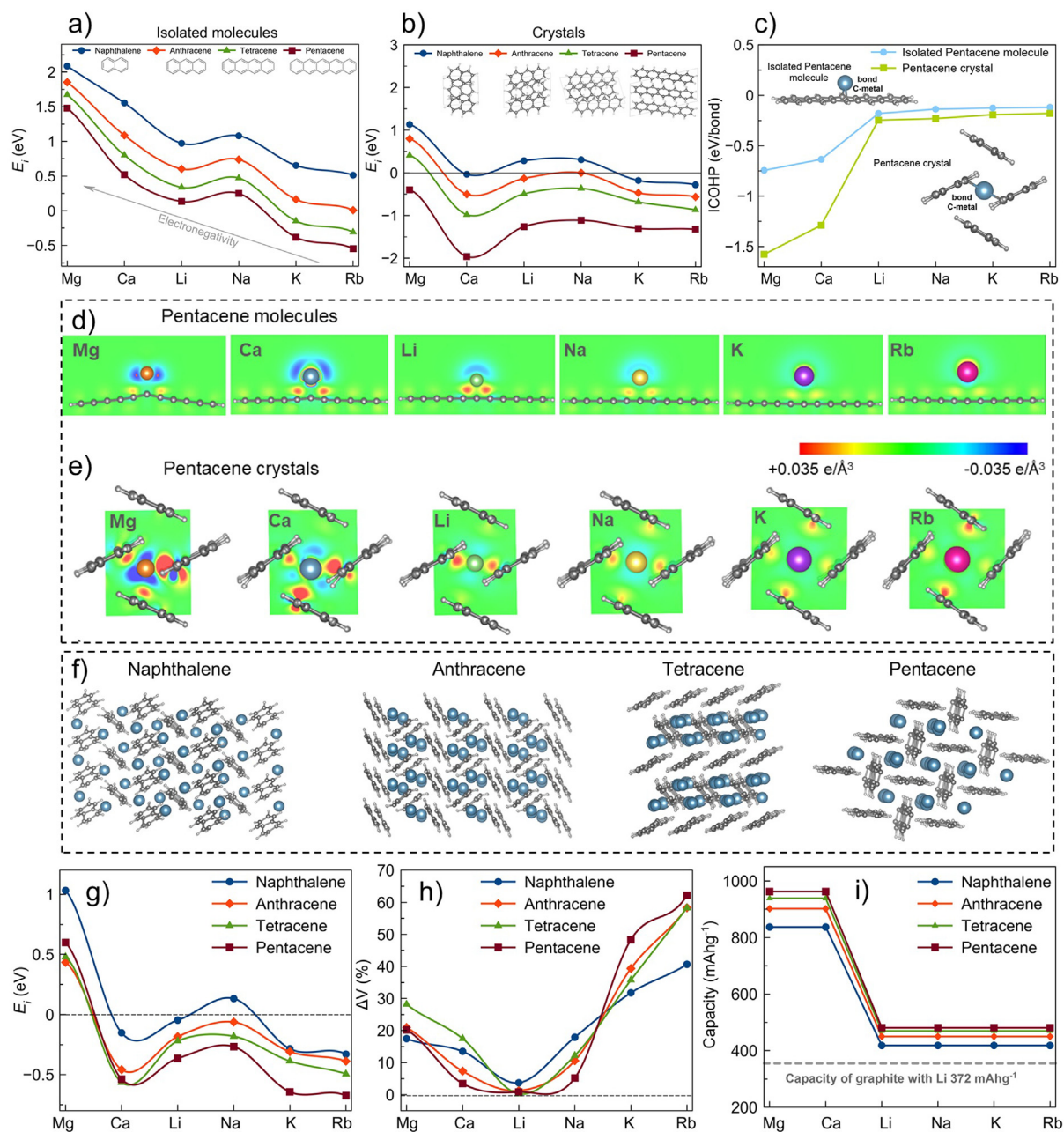


Fig. 5. (a) Energy required to remove a metal atom from the bulk crystal and adsorb it onto an isolated PAH molecule. (b) Calculated intercalation energy of a metal atom into PAH crystals. (c) The integrated COHP (ICOHP) calculated for the metal-carbon bond in the case of the pentacene molecule and crystal. (d) isolated pentacene molecule, (e) pentacene crystal, red areas correspond electron density excess, blue areas to electron density deficiency. (f) Crystal structures of naphthalene (NP), anthracene (AN), tetracene (TN) and pentacene (PN) crystals with intercalated metal atoms. The number of metal atoms per unit cell is 4, 6, 8, 10 for NP, AN, TN, PN respectively. (g) Intercalation energy as a function of the type of metal atoms. (h) Change in crystal volume as a function of the type of intercalating atom. (i) Calculated capacity of the intercalated PAH crystals. Ref. [68].

and crystallographic symmetry (Fig. 6 f,g). As shown in Fig. 6 e, the formation energies for different Li-ion contents follow an energy convex hull (red line), indicating that the $R\bar{3}$ c-HBC structure becomes unstable beyond 18 Li ions. This suggests a maximum storage capacity of 6 Li ions per molecule, consistent with the experimental capacity of 267 mAhg⁻¹ and more than the capacity of pyrene, coronene and flat HBC (Fig. 6 c). Consequently, contorted polycyclic aromatic hydrocarbon molecules represent a promising new class of organic anode materials for lithium-ion batteries. As illustrated by Park et al. [21] in their work, which establishes a foundation for innovative approaches to the development of next-generation batteries.

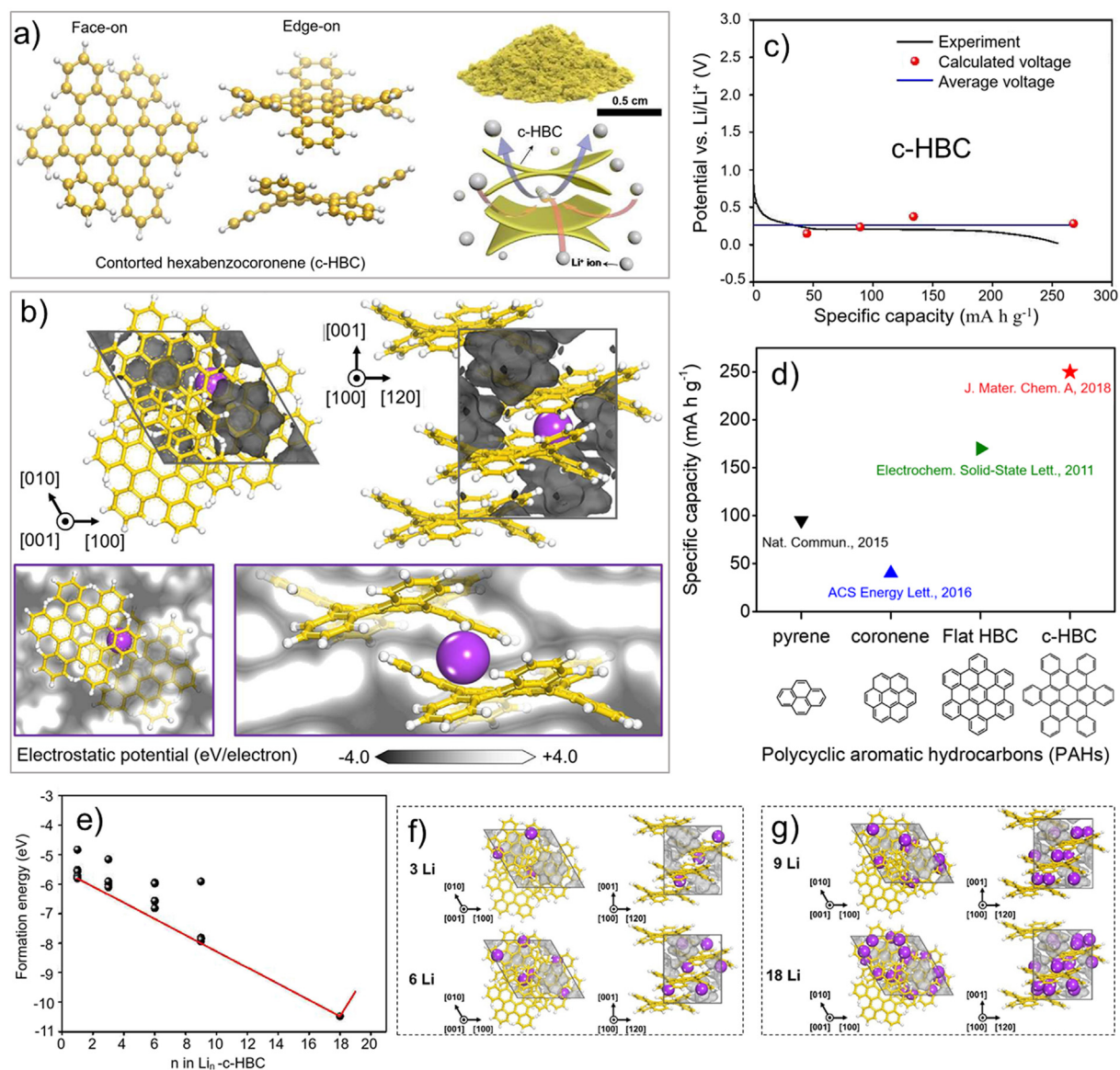


Fig. 6. (a) Face-on and edge-on chemical structure of c-HBC, the crystal structure of c-HBC exhibits two concave surfaces, digital photograph of synthesized c-HBC powder and schematic of proposed Li-ion insertion mechanism in c-HBC. (b) The optimized structure of Li-inserted c-HBC and the electrostatic potential maps of c-HBC without Li-ion. The electrostatic potential is mapped onto the Connolly surface (top) and (001) and (010) planes across the center of Li-ion (bottom left and right). Carbon, hydrogen, and lithium are colored yellow, white, and purple, respectively. (c) The experimental (black line) and calculated (red and blue lines) voltage profiles. (d) Plot of specific capacity vs. PAHs. (e) Formation energies of Li-inserted c-HBC as a function of the number of Li-ions. The calculated convex hull is shown as a red line. (f, g) Projection views of the optimized stable structures of 3, 6, 9, and 18 Li-inserted $R\bar{3}$ crystal phase. Carbon, hydrogen, and lithium are colored yellow, white, and purple, respectively. Ref. [21].

Also, such PAH as corannulene ($\text{C}_{20}\text{H}_{10}$) have attracted much research attention as potentially promising anode material for LIBs. Corannulenes are exceptional nano-sized containers capable of reversibly storing up to four electrons, thanks to their doubly degenerate lowest unoccupied molecular orbitals (see (Fig. 7 a)). The electron capacity of corannulene (five carbon atoms per one additional electron, $5\text{C}/e^-$) is higher compared to a planar analog coronene ($12\text{C}/e^-$) [82], pristine graphite ($6\text{C}/e^-$) [83], and even twice that of fullerene C_{60} ($10\text{C}/e^-$) [84]. The electron capacity of corannulene can be reduced stepwise with alkali metals ranging from Li to Cs to form mono-, di- or tri-anions [78,85–90]. However, the final step to the tetra-anion is only accessible through the reaction of corannulene with Li or the combination of Li with other alkali metals [91,92]. Therefore, the interaction of lithium ions with the tetra-reduced corannulene is crucial for the electrostatic and coordination stabilization of the highly charged corannulene tetra-anion, in addition to its distinctive electronic structure, which resembles an annulene within an annulene

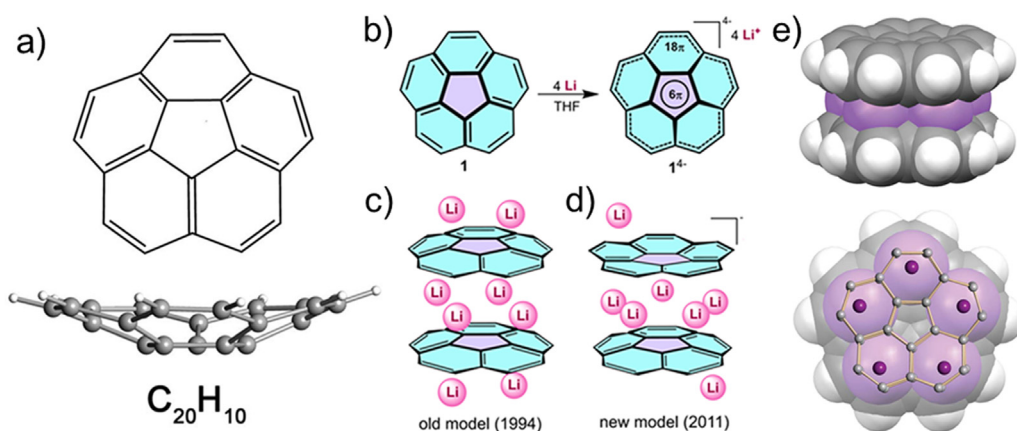


Fig. 7. (a) Corannulene and its bowl-shaped geometry, Ref. [77]. (b) Preparation of corannulene tetraanion and (c,d) two model established sandwich structure formed by corannulene tetraanion with lithium ions, Ref. [78]. (e) Space-filling depictions show side and top views of the sandwich superimposed with ball-and-stick representation, Ref. [77].

(Fig. 7b) [91]. Scott et al. [93] back in 1994 investigated the *insitu* reduction of corannulene with lithium metal using variable-temperature ^7Li NMR spectroscopy. Based on the experimental findings obtained through nuclear magnetic resonance spectroscopy and theoretical calculations, it was proposed that two corannulene tetraanions can accommodate four lithium cations between their carbon layers, with the remaining four metal ions bound to the exterior of the structure (Fig. 7 c). However, a more recent study by Zabula et al. [77] found the presence of five lithium ions trapped between two tetra-reduced bowls. In the [corannulene tetraanion/ Li_5 /corannulene tetraanion] $^{3-}$ sandwich, all ten six-membered rings of the two corannulenes participate in the binding of five lithium ions, which are positioned symmetrically between the convex surfaces of the corannulene tetraanion (Fig. 7 d). It was shown [94,95] that the anode material made from corannulene has a high initial lithium capacity of 802 mAh g^{-1} (and reversible capacity 464 mAhg^{-1}), which is nearly twice the theoretical capacity of the standard fully lithiated graphite material LiC_6 , 372 mAhg^{-1} .

2.5. Systematic analysis of heteroatom-doped PAHs

Another common and effective way to fine-tune the properties of materials is doping [96,97]. Researchers have created and synthesized a variety of PAHs with incorporated heteroatoms to alter their π -electron characteristics [17,98,99]. The influence of various dopants has been comprehensively studied using DFT calculations, establishing clear correlations between the type of dopant atom and the electrochemical performance.

Influence of Nitrogen (N). Nitrogen doping, possessing electron-withdrawing properties, generally increases the electron deficiency of the aromatic framework, leading to an increase in cell voltage (V_{cell}). For example, Nair et al. [79] investigated the adsorption of Li^+ and Li on 24 adsorbents obtained by replacing C atoms in coronene ($\text{C}_{24}\text{H}_{12}$) and circumcoronene ($\text{C}_{54}\text{H}_{18}$) with Si, N, BN, and AlN. Molecular electrostatic potential (MESP) analysis showed that such substitutions lead to a more electron-rich environment within the molecules (Fig. 8a,b). It was determined that $\text{B}_{27}\text{H}_{18}\text{N}_{27}$ is the most promising candidate for anode materials [79].

The influence of nitrogen is also pronounced for larger nanographenes such as hexa-peri-hexabenzocoronene (HBC) molecules ($\text{C}_{42}\text{H}_{18}$). Nair et al. [80] studied the adsorption of Li^+ , Li, Mg^{2+} , and Mg on twelve HBC-based adsorbents, including those doped with N, BN, and Si (Fig. 8c,d). MESP analysis confirmed that replacing C atoms with nitrogen, boron, or silicon creates a system with higher electron density compared to the pristine HBC molecule. Among the studied anode materials for lithium-ion batteries, the maximum cell voltage of 1.90 V was achieved for $\text{B}_{21}\text{H}_{18}\text{N}_{21}$, while for magnesium-ion batteries, a record value of 5.29 V was shown by $\text{m-C}_{40}\text{H}_{18}\text{N}_2$ [80]. Similar results were obtained for sodium-ion batteries: doping HBC with N atoms makes it more suitable for application, whereas B-doping significantly reduces the voltage [100]. In calcium-ion batteries, N-doping is also the most effective N-HBC demonstrates a V_{cell} of 1.01 V, significantly higher than pristine HBC (0.56 V) [81].

Influence of Boron (B) and BN-co-doping. Unlike nitrogen, boron doping often has a negative impact on cell voltage. As shown by Peyghan et al. for sodium-ion batteries, B-doping of HBC (B-HBC) leads to an unphysical negative voltage, rendering such material unsuitable for use as an anode [100]. Co-doping with boron and nitrogen (BN) yields intermediate results. For instance, for sodium-ion batteries, BN-HBC provides a V_{cell} of 1.29 V, lower than pristine HBC (1.41 V) but higher than B-HBC [100]. For calcium-ion batteries, BN-HBC (0.52 V) shows results close to pristine HBC (0.56 V) [81]. Interestingly, borazine doping (BNG) combined with halide ions can yield outstanding results. Adding a halide counterion to BNG leads to a significant improvement in LIB performance, and the calculated V_{cell} of 3.98 V for BNG/F^- is among the highest reported for nanostructured anode materials [101].

Influence of Silicon (Si) and AlN. Silicon doping also creates electron-rich systems and promotes strong ion adsorption. Si-doped coronene ($\text{C}_{12}\text{H}_{12}\text{Si}_{12}$) and circumcoronene ($\text{C}_{27}\text{H}_{18}\text{Si}_{27}$) were proposed as promising anode materials for LIBs with high V_{cell} (1.80

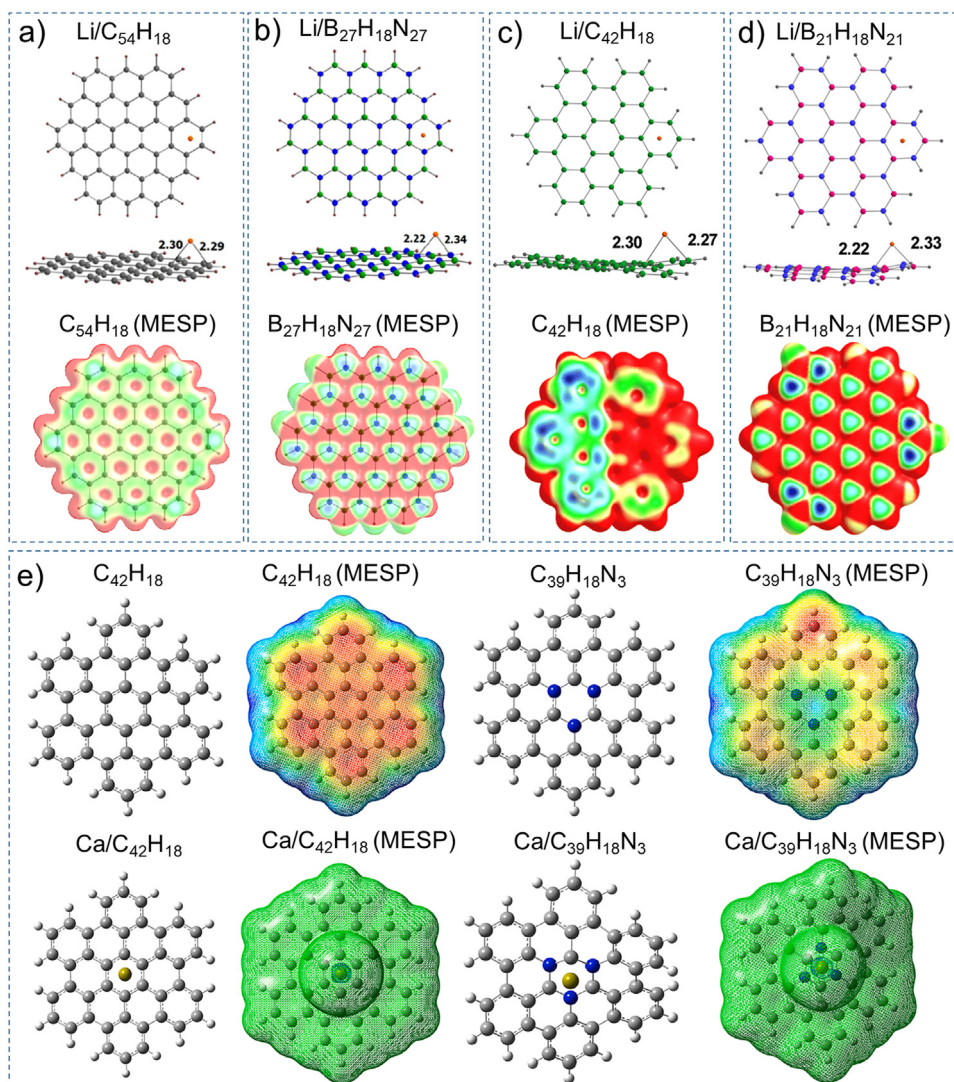


Fig. 8. (a) Optimized structure of $\text{Li/C}_{54}\text{H}_{18}$ and computed molecular electrostatic potentials (MESP) of pristine $\text{C}_{54}\text{H}_{18}$, b) Optimized structure of $\text{Li/B}_{27}\text{H}_{18}\text{N}_{27}$ and MESP of pristine $\text{B}_{27}\text{H}_{18}\text{N}_{27}$ (Ref. [79]). c) Optimized structure of $\text{Li/C}_{42}\text{H}_{18}$ and MESP of pristine $\text{C}_{42}\text{H}_{18}$, d) Optimized structure of $\text{Li/B}_{21}\text{H}_{18}\text{N}_{21}$ and MESP of pristine $\text{B}_{21}\text{H}_{18}\text{N}_{21}$ (Ref. [80]). e) Optimized structure and MESP of $\text{C}_{42}\text{H}_{18}$, $\text{Ca/C}_{42}\text{H}_{18}$ and N-doped $\text{C}_{39}\text{H}_{18}\text{N}_3$, $\text{Ca/C}_{39}\text{H}_{18}\text{N}_3$ (Ref. [81]).

V and 1.54 V, respectively) [79]. For magnesium-ion batteries, Si-doped HBC ($\text{C}_{21}\text{H}_{18}\text{Si}_{21}$) showed a voltage of 4.89 V [80]. Doping with aluminum nitride (AlN-HBC) was investigated for calcium-ion batteries and showed a voltage of 0.63 V, slightly higher than pristine HBC but inferior to N-HBC [81] (Fig. 8e).

Influence of Functional Groups. Beyond doping, functionalization plays a crucial role. Wu et al. [102] investigated the influence of -F, $-\text{NH}_2$, and $-\text{NO}_2$ substituents on the properties of HBC. It was found that adding the electron-donating $-\text{NH}_2$ group increases the cell voltage by approximately 0.25 V (to 1.95 V), while the electron-withdrawing $-\text{NO}_2$ group decreases it from 1.70 to 1.16 V. This demonstrates that not only the inductive effect but also the nature of the substituent's interaction with lithium ions plays a key role.

Thus, systematic analysis shows that nitrogen doping is the most versatile strategy for increasing the voltage of PAH anodes in various metal-ion batteries. Silicon doping opens a pathway to create anodes with record voltage for multivalent ions such as Mg^{2+} . At the same time, the use of boron and BN-co-doping requires a more nuanced approach, and functionalization with electron-donating groups like $-\text{NH}_2$ can be a surprisingly effective tool. These results underscore the importance of a systematic dopant selection to achieve target electrochemical characteristics.

Table 1

Summary of key electrochemical performance metrics for selected PAH-based anode materials in various metal-ion battery systems, including a comparison with a conventional graphite anode.

Material	Battery type	Role	Theor.capacity (mAh g ⁻¹)	Exp. capacity (mAh g ⁻¹)	Cycles/Retention	Refs.
Graphite	Li-ion	Anode	372	~350–360	>1000/>90%	[40]
Naphthalene	Li-ion	Anode	–	435 (1st discharge)/213 (1st reversible)	1500/100.1%	[20]
Biphenyl	Li-ion	Anode	–	478 (1st discharge)/220 (1st reversible)	1500/95.3%	[20]
9,9-Dimethylfluorene	Li-ion	Anode	–	522 (1st discharge)/243 (1st reversible)	1500/92.3%	[20]
Phenanthrene	Li-ion	Anode	–	458 (1st discharge)/201 (1st reversible)	1500/93.4%	[20]
p-Terphenyl	Li-ion	Anode	–	186 (1st discharge)/61 (1st reversible)	1500/93.8%	[20]
Pyrene	Li-ion	Anode	–	453 (1st discharge)/58 (1st reversible)	1500/68.4%	[20]
Naphthalene	Li/Na/K/Rb-ion	Anode	418	–	–	[68]
Anthracene	Li/Na/K/Rb-ion	Anode	451	–	–	[68]
Tetracene	Li/Na/K/Rb-ion	Anode	470	–	–	[68]
Pentacene	Li/Na/K/Rb-ion	Anode	481	–	–	[68]
Naphthalene	Ca/Mg-ion	Anode	837	–	–	[68]
Anthracene	Ca/Mg-ion	Anode	903	–	–	[68]
Tetracene	Ca/Mg-ion	Anode	940	–	–	[68]
Pentacene	Ca/Mg-ion	Anode	963	–	–	[68]
Corannulene	Li-ion	Anode	–	802 (1st discharge)/464 (1st reversible)	–	[94]
c-HBC (contorted)	Li-ion	Anode	267	270 (1st discharge)/200 (1st reversible)	–	[21]
Coronene	Li-ion	Anode	536.2	–	–	[79]

2.6. Summary and outlook for PAH-based anodes

In summary, the development of PAH-based anode materials has progressed from fundamental studies of small molecules to the rational design of crystalline structures and heteroatom-doped systems. Key achievements include the experimental validation that small-molecule PAHs such as naphthalene, biphenyl, and phenanthrene can achieve reversible capacities of 201–243 mAh g⁻¹ with exceptional long-term cyclability, retaining over 92% of their capacity after 1500 cycles [20]. The demonstration that crystalline PAHs such as pentacene can achieve theoretical capacities of 481 mAh g⁻¹, significantly exceeding graphite (372 mAh g⁻¹) with minimal volume expansion [68], hlremarkable reversible capacity of 464 mAh g⁻¹ achieved by the bowl-shaped corannulene [94], and the systematic enhancement of cell voltage through heteroatom doping further underscore their potential (Table 1).

The design principles emerging from these studies are fundamentally shaped by the distinct electrochemical roles of anodes and cathodes. Anodes require materials with low electrochemical potential and high electron affinity to facilitate efficient cation (Li⁺, Na⁺) storage through reduction reactions, typically favoring electron-deficient systems [79,80]. In contrast, cathodes operate at high potentials and rely on reversible oxidation, necessitating PAHs with low-lying HOMOs to enable stable anion (e.g., PF₆⁻, AlCl₄⁻) intercalation or p-type redox activity [9,103,104]. Consequently, while anodes benefit from nitrogen doping to enhance electron affinity and cation binding [100], cathode design often focuses on extending π -conjugation or incorporating electron-withdrawing groups (e.g., carbonyls, fluorine) to raise the redox potential [105]. These complementary requirements have driven the exploration of structurally diverse PAHs, from electron-rich heterocycles for anodes to quinones and polycyclic aromatic ketones for high-voltage cathodes. The following section examines the application of PAHs as cathode materials, where these design strategies are critically assessed.

3. PAH as cathode

3.1. PAH-based cathodes for Al-ion battery

PAHs have also been considered as promising cathode materials. In particular, recent works [6,107–109] have demonstrated the performance of PAHs as cathode materials for aluminum-ion batteries. Among secondary battery technologies, aluminum-ion batteries (AIBs) stand out as a promising candidate due to the exceptional natural abundance of aluminum in the Earth's crust, their cost-effectiveness and inherent safety derived from non-flammable ionic liquid electrolytes [110–112]. The aluminum anode offers remarkable theoretical storage capabilities, including an ultrahigh volumetric capacity of 8040 mAh cm⁻³ and a gravimetric capacity of 2980 mAh g⁻¹, enabled by its three-electron redox process [113,114]. These values surpass even those of lithium metal (3862 mAh g⁻¹ gravimetric and 2062 mAh cm⁻³ volumetric), positioning AIBs as a compelling alternative for next-generation energy storage. For this reason, selecting a cathode material with high specific capacity is critical to realizing high-performance aluminum-ion batteries (AIBs). Previous studies have identified graphite as a viable cathode material for AIBs, primarily through the formation of graphite-intercalated compounds (GICs) during charging. A typical configuration of this type consists of metallic aluminum, AlCl₃·[EMIm]Cl (1-ethyl-3-methylimidazolium chloride) ionic liquid and graphite [115]. However, the intercalation mechanism involves inserting of AlCl₄⁻ into every fourth graphite layer, resulting in stage-4 GICs at full charge. This intercalation geometry

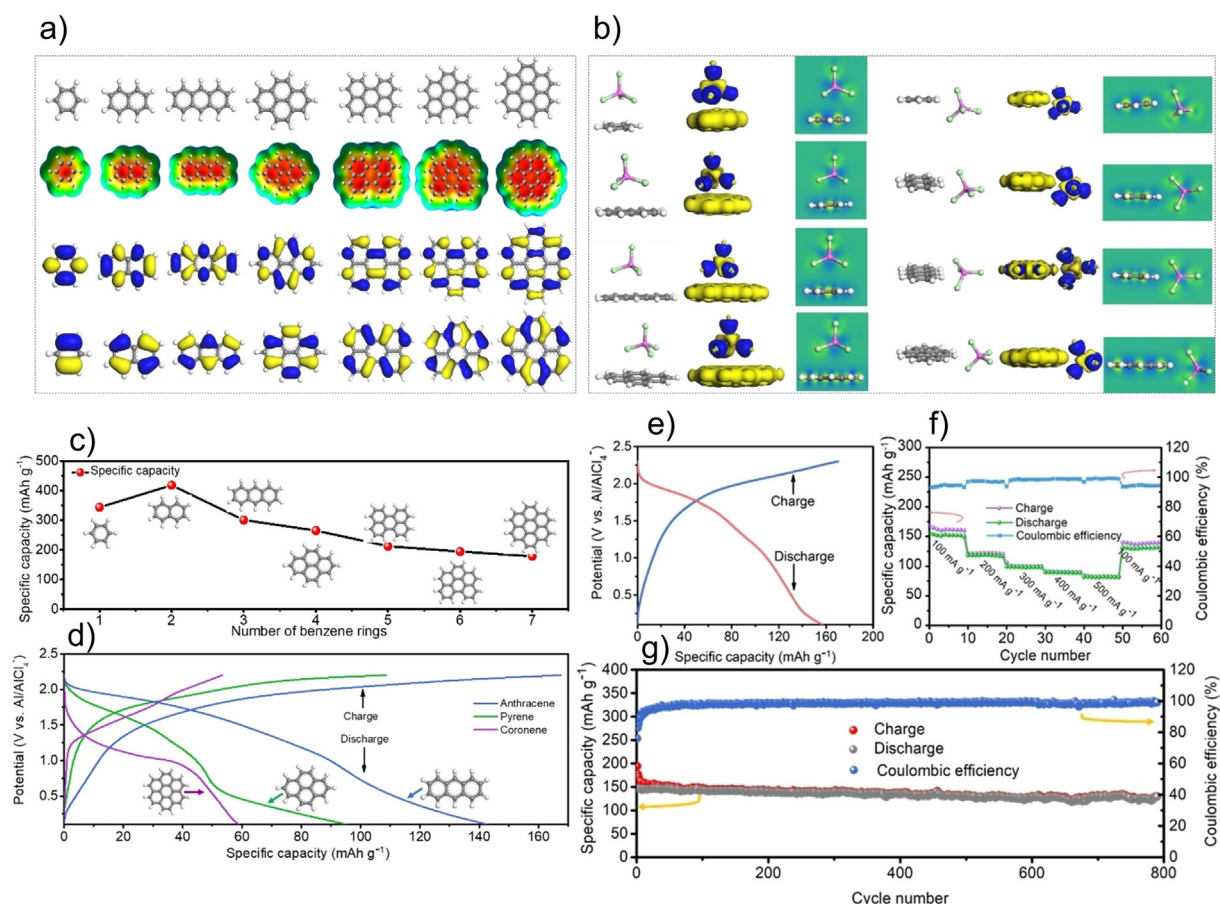


Fig. 9. (a) Structure, electrostatic potentials (the red and blue colors refer to the negative (-0.025 a.u.) and positive (0.025 a.u.) values, respectively), LUMO and HOMO of PAHs. The gray and white balls denote the C and H atoms, respectively. b) The structure, deformation electron density (DED) isosurface, and the sliced DED map for the on-plane (left) and on-side (right) adsorption of AlCl_4^- on benzene, naphthalene, anthracene and pyrene. In the DED map, the red and blue colors refer to the positive ($0.45 \text{ e}\text{\AA}^{-3}$) and negative ($-0.10 \text{ e}\text{\AA}^{-3}$) values, respectively. The gray, white, pink, and green balls denote the C, H, Al, and Cl atoms, respectively. c) Theoretical specific capacity of PAHs. d) Specific charge and discharge capacity of anthracene, pyrene and coronene as cathode materials for AIBs. e-g) The electrochemical performance of the anthracene cathode for AIBs. e) galvanostatic charge-discharge curves at 100 mA g^{-1} ; f) rate capability of anthracene at different current densities and g) capacity retention and coulombic efficiency of anthracene at 100 mA g^{-1} . Ref. [6].

limits the practical capacity of Al-graphite batteries to approximately 70 mAh g^{-1} , a critical drawback, and needs for cathode materials with higher ion storage efficiency. PAHs have been proposed as an alternative to graphite as a cathode in the AIBs [6].

Kong et al. [6] selected PAHs with small conjugated systems including benzene (C_6H_6), naphthalene (C_{10}H_8), anthracene ($\text{C}_{14}\text{H}_{10}$), pyrene ($\text{C}_{16}\text{H}_{10}$), perylene ($\text{C}_{20}\text{H}_{12}$), benzo(g,h,i)perylene ($\text{C}_{22}\text{H}_{12}$), and coronene ($\text{C}_{24}\text{H}_{12}$), as potential cathode materials for AIBs. The molecular structures of the studied PAHs and their corresponding electrostatic potential (ESP) maps are illustrated in Fig. 9a. For all PAHs, the ESP distribution reveals distinct regio-chemical features: the region above the molecular plane exhibits a low electrostatic potential (red regions in Fig. 9a), indicating a nucleophilic character, while molecular edges display a high electrostatic potential (blue regions), reflecting electrophilic behavior. However, frontier molecular orbital (FMO) analysis demonstrates a critical contrast: the lowest unoccupied molecular orbital (LUMO) and the highest occupied molecular orbital (HOMO) are both confined to the molecular plane (Fig. 9a). This orbital distribution implies that the planar region, rather than the edges, serves as the primary active site for redox reactions in aluminum-ion storage [6].

During the charging process, electrons are injected into the LUMO of PAHs [6]. The calculated LUMO energies for benzene (C_6H_6), naphthalene (C_{10}H_8), anthracene ($\text{C}_{14}\text{H}_{10}$), pyrene ($\text{C}_{16}\text{H}_{10}$), perylene ($\text{C}_{20}\text{H}_{12}$), benzo[g,h,i]perylene ($\text{C}_{22}\text{H}_{12}$), and coronene ($\text{C}_{24}\text{H}_{12}$) are 0.051 , -1.003 , -1.675 , -1.533 , -1.947 , -1.704 , and -1.451 eV, respectively. Benzene exhibits the highest LUMO energy (0.051 eV), reflecting the highest electrochemical potential required for electron injection. In contrast, naphthalene shows a markedly lower LUMO (-1.003 eV), corresponding to a reduced charging potential. As the number of aromatic rings increases to three or more (e.g., anthracene and larger PAHs), the LUMO energy decreases overall, but with a decreasing variability, suggesting

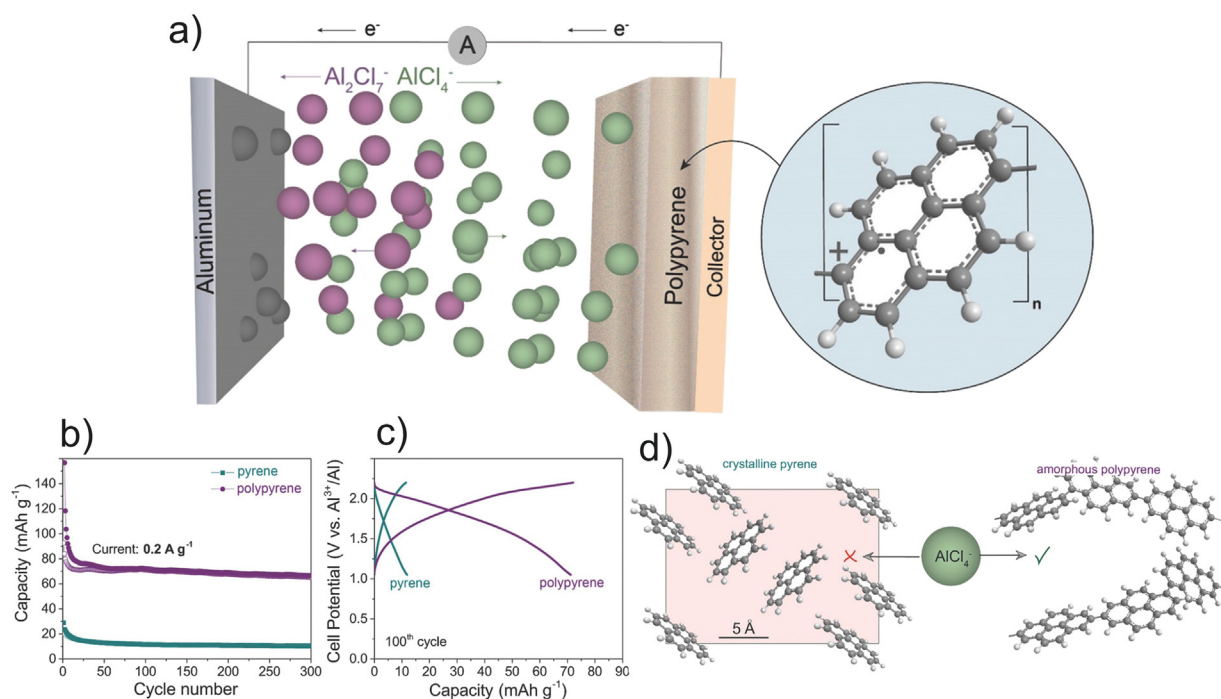


Fig. 10. (a) Schematic of the working principle of a rechargeable aluminum battery during charge with a polypyrrene cathode and chloroaluminate ionic liquid. b-c) Electrochemical performance of aluminum batteries with pyrene and polypyrrene as cathode materials: b) capacity retention upon cycling and c) corresponding charge–discharge voltage profiles for the 100th cycle. Cells were cycled at room temperature at a current density of 200 mA g^{-1} (referring to the mass of the polymer) in the potential range of 1.05–2.2 V. d) Schematic representation of the structural differences between pyrene (crystalline, e.g., dense molecular packing) and polypyrrene (amorphous, porous, and flexible), which impact their AlCl_4^- storage behavior. Ref. [106].

gradual stabilization of electronic properties. This trend implies that, while larger PAHs generally require lower charging potentials, the incremental reduction in LUMO energy becomes less pronounced as molecular size increases [6].

The adsorption of AlCl_4^- on PAHs can occur either on the molecular plane or on the side. Both configurations were investigated by Kong et al. [6], where the interaction between AlCl_4^- and PAHs is governed by donor–acceptor stabilization and electrostatic forces. In the on-plane adsorption model, strong electron donation occurs from AlCl_4^- to unoccupied π^* orbitals of PAHs, accompanied by significant back-donation from filled π orbitals in PAHs to AlCl_4^- . This synergistic π -interaction mechanism stabilizes the system more effectively than the on-sided adsorption model, where orbital interactions are less pronounced. Consequently, the on-plane configuration forms a more stable π -adsorption complex. Binding energy calculations confirmed that AlCl_4^- adsorption on the PAH plane is energetically favored over side adsorption, aligning with enhanced orbital interactions observed in planar geometry (Fig. 9b). The affinity of AlCl_4^- for benzene is significantly weaker compared to other PAHs, indicating a less favorable storage of AlCl_4^- in benzene. This finding is consistent with the findings of the LUMO analysis [6].

The theoretical specific capacity of naphthalene has been determined to be 418 mAh g^{-1} , whereas that of benzene is 343 mAh g^{-1} . As the size of PAH molecules in the crystal increases, the specific capacity gradually decreases to 300, 265, 212, 194, and 178 mAh g^{-1} for anthracene, pyrene, perylene, benzo[*g,h,i*]perylene and coronene respectively due to the increasing molecular weight of PAHs (Fig. 9c). Quantum chemistry calculations identified anthracene, pyrene, and coronene as viable cathode materials for aluminum-ion batteries, pairing an Al anode with a chloroaluminate ionic liquid electrolyte. Benzene and naphthalene were excluded due to their high volatility and propensity for sublimation under operational conditions. As illustrated in Fig. 9d, under identical testing conditions (current density: 100 mA g^{-1} ; voltage window: 0.1–2.2 V), anthracene delivered a specific discharge capacity of 143 mAh g^{-1} , outperforming pyrene (95.2 mAh g^{-1}) and coronene (56.9 mAh g^{-1}). Given its superior capacity, anthracene was comprehensively characterized [6].

Cyclic voltammetry (CV) was performed on an anthracene electrode in a voltage window of 0.1–2.3 V (Fig. 9e) [6]. Galvanostatic charge–discharge profiles revealed distinct charging (1.5–2.3 V) and discharging (2.0–1.0 V) plateaus. At a current density of 100 mA g^{-1} , the electrode delivered an ultrahigh discharge capacity of 157 mAh g^{-1} , underscoring its promising charge storage capability. Rate performance and cycling stability are critical for practical AIB applications. As shown in Fig. 9f, the anthracene cathode maintained a capacity of 82 mAh g^{-1} even at an elevated current density of 500 mA g^{-1} , demonstrating robust rate capability. Furthermore, the electrode exhibited exceptional long-term stability, retaining 130 mAh g^{-1} over 800 cycles (Fig. 9g). This remarkable cycling performance stems from the moderate interaction between AlCl_4^- ions and anthracene, which facilitates reversible ion adsorption/desorption without excessive binding that could degrade structural integrity [6].

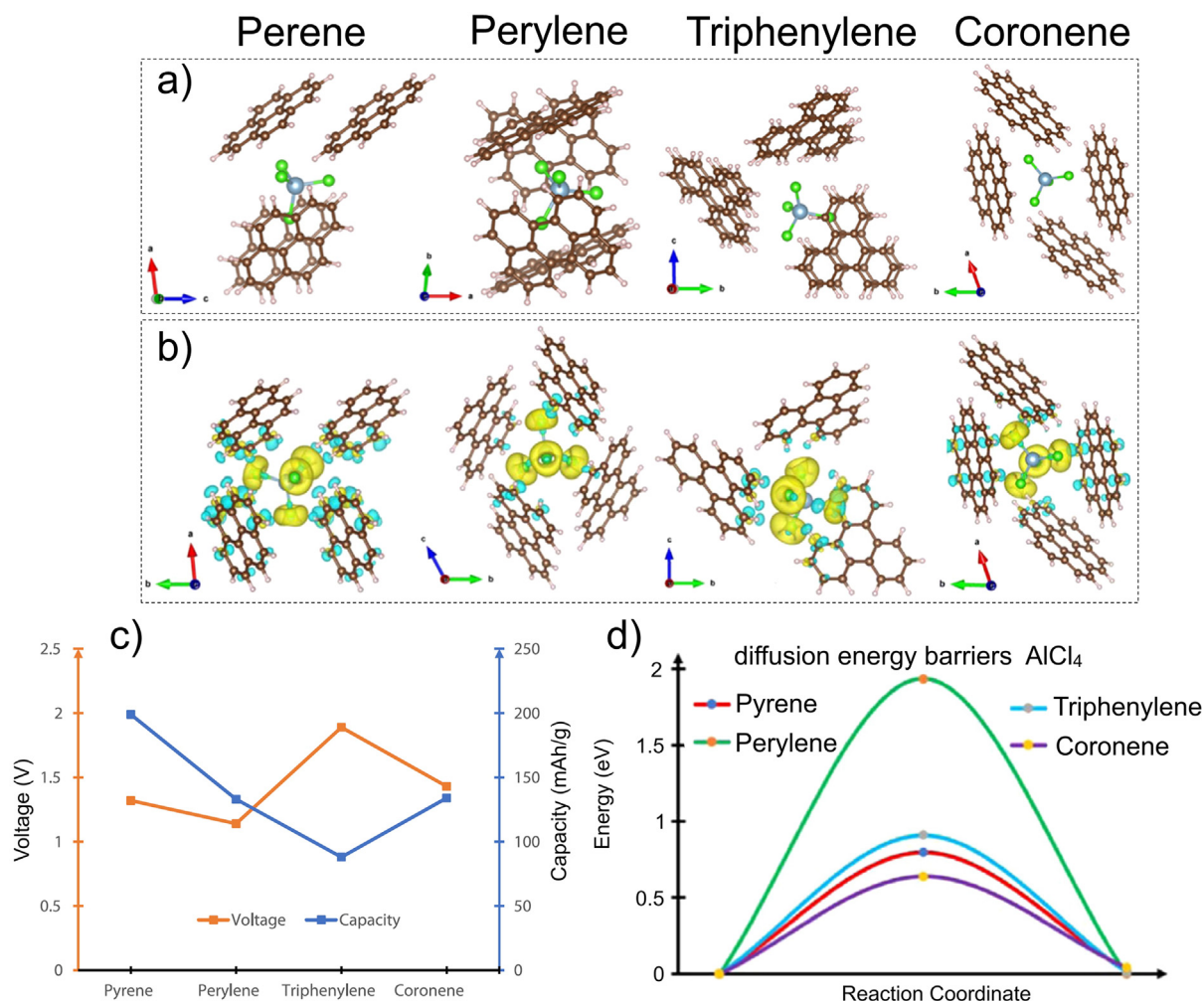


Fig. 11. (a) Optimized PAH systems of pyrene, perylene, triphenylene, and coronene with AlCl_4 intercalation at edges. (b) Charge density difference plots of PAH systems: pyrene, perylene, triphenylene, and coronene, upon AlCl_4 intercalation. Here, the isosurface value is set to 0.0018; yellow stands for $\Delta\rho > 0$, and blue stands for $\Delta\rho < 0$. (c) Variation in voltage and storage capacity for the considered PAH systems. (d) Schematic representation of the diffusion energy barriers of AlCl_4 in pyrene, perylene, triphenylene, and coronene. Ref. [46].

Walter et al. [106] have developed a novel and promising cathode material for aluminum-ion batteries using cost-effective pyrene-based polymers. During charging, the condensed aromatic structures within these polymers undergo oxidation, coupled with the absorption of aluminum tetrachloride anions (AlCl_4^-) from a chloroaluminate ionic liquid electrolyte. The discharge process involves a rapid reversal of this reaction, characterized by the reduction of the polymer framework and the subsequent release of AlCl_4^- anions. In the process of electrooxidation, pyrene can acquire up to one positive charge per four aromatic rings, forming a radical cation. This charge can be offset by the incorporation AlCl_4^- anions from an ionic liquid (Fig. 10a). As the polymer is reduced, the anions are released back into the ionic liquid. The theoretical storage capacity is 133 mAh g^{-1} assuming a single electron per pyrene unit, which is comparable to the best values for graphite. Walter et al. [106] have showed that polypyrene and its derivative, poly(nitropyrene-co-pyrene), can be used as highly reversible and high-capacity cathode materials for aluminum batteries. The experimental results showed that the capacities were 70 and 100 mAh g^{-1} , respectively, at an average voltage of 1.7 V . The poly(nitropyrene-co-pyrene) also demonstrated excellent cycling stability, with at least 1000 cycles at a current density of 200 mA g^{-1} .

As illustrated in Fig. 10b,c, the electrochemical performance of electrodes made from pyrene and polypyrene in aluminum batteries is evaluated at a high current density of 200 mA g^{-1} . In the case of polypyrene, the initial capacity decreases, but then stabilizes at approximately 70 mAh g^{-1} for 300 cycles. This is more than half of the theoretical capacity based on a one-electron redox process at the pyrene unit (133 mAh g^{-1}). Galvanostatic charge–discharge curves demonstrate that the storage capacity corresponds to a relatively high discharge voltage of $\approx 1.7 \text{ V}$ (Fig. 10c). It is crucial to highlight that the electrochemical characteristics of polypyrene are influenced by the acidity of the chloroaluminate ionic liquid, which is contingent upon the molar

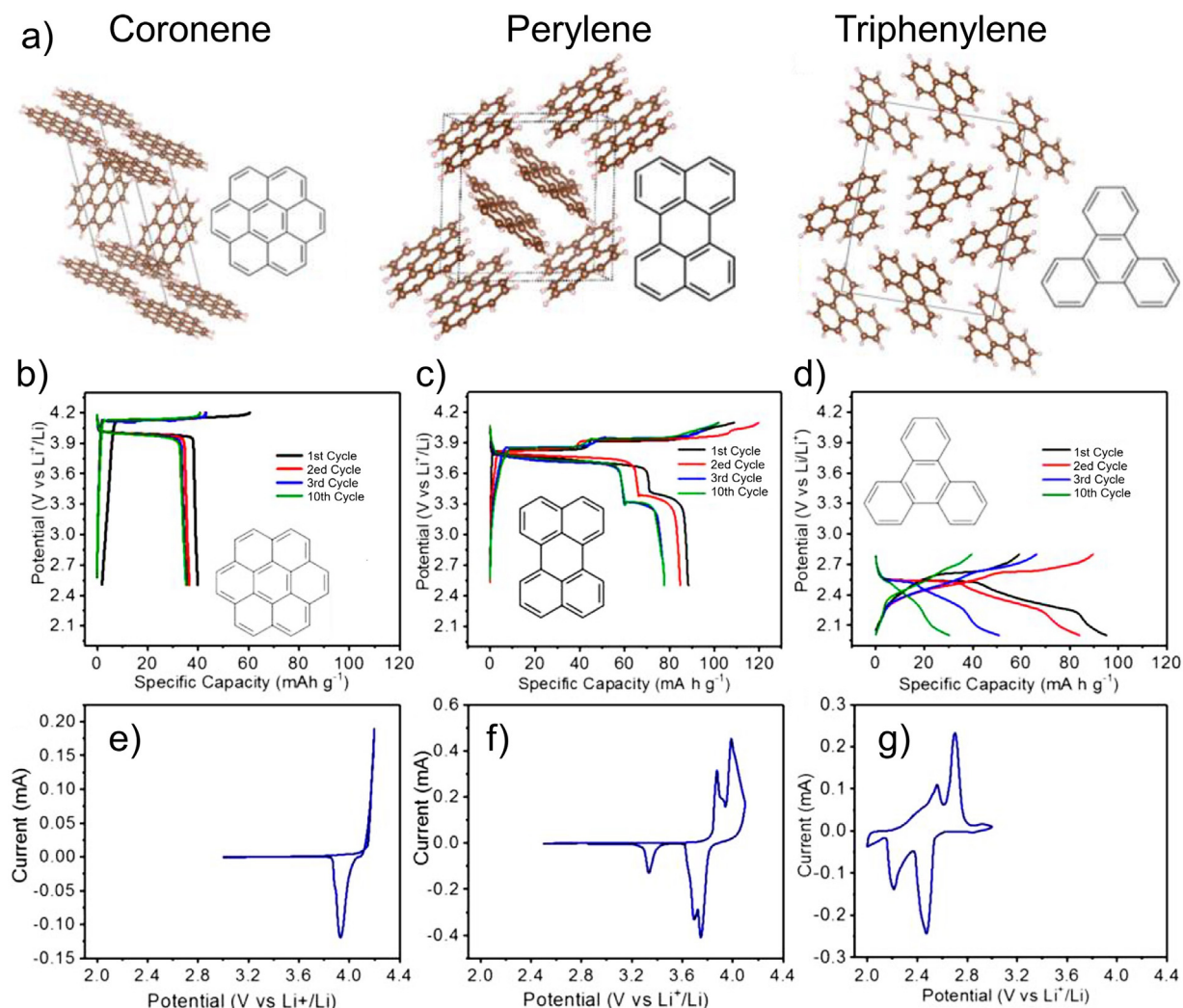


Fig. 12. (a) Crystal structures of coronene, perylene, triphenylene. GCD profiles of the (b) coronene (1st, 2nd, 3rd, and 10th cycles), (c) perylene (1st, 2nd, 5th, and 10th cycles), and (d) triphenylene (1st, 2nd, 5th, and 10th cycles) electrodes, respectively, from left to right, at a current density of 20 mA g⁻¹ and a potential range of 2.5–4.2, 2.5–4.1, and 2.0–2.8 V, respectively. Typical cyclic voltammetry profiles of (e) coronene, (f) perylene, and (g) triphenylene electrodes, respectively. Ref. [116].

ratio of AlCl₃:[EMIm]Cl. As indicated by the electrochemical data, the optimal molar ratio for AlCl₃:[EMIm]Cl is approximately 1.3. In comparison to polypyrene, the performance of monomeric pyrene is quite poor. It is worth noting that the initial discharge capacity of pyrene is relatively low, at only 20 mAh g⁻¹ and it rapidly decreases to 10 mAh g⁻¹ in subsequent cycles. The significant differences in the electrochemical behavior of pyrene and polypyrene can be attributed to a several factors. Compared to its polymeric counterpart, pyrene is more susceptible to dissolving in chloroaluminate ionic liquids and is also less conductive due to the absence of a conjugated π -system. Moreover, the efficient incorporation of AlCl₄⁻ is impeded by the compact arrangement of molecules in crystalline pyrene. The intermolecular distances of 3.5–4 Å [117] are small for diffusion of substantial AlCl₄⁻ (5.3–5.9 Å) [115,118]. On the other hand, the disordered and adaptable arrangement of polypyrene chains promotes the movement of AlCl₄⁻.

Das et al. [46] studied AlCl₄-intercalated PAH systems such as pyrene (C₁₆H₁₀), perylene (C₂₀H₁₂), triphenylene (C₁₈H₁₂), and coronene (C₂₄H₁₂) to evaluate their geometric structures, binding energies, charge transfer, voltage studies, and diffusion characteristics using density functional theory (DFT) calculations (Fig. 11). All of the considered PAHs exhibit energetically favorable binding of AlCl₄ on the edge site, potentially due to the presence of open binding regions around the edges of the molecules. All of these materials have high theoretical specific capacities of 199, 133, 88, and 134 mAh/g, respectively, for pyrene, perylene, triphenylene and coronene systems (Fig. 11c). Also Das et al. [46] calculated the diffusion energy barriers for AlCl₄ in pyrene, perylene, triphenylene, and coronene. Perylene exhibited the highest diffusion barriers and the lowest voltage, making it unsuitable

for practical cathode applications (Fig. 11d). In contrast, pyrene and coronene demonstrated optimal performance, combining competitive specific capacities with stable voltage outputs. Triphenylene, though limited by its lower capacity, achieved a notably high voltage of ~ 1.9 V, highlighting its potential for high-voltage applications. These findings underscore the promise of PAHs and analogous organic frameworks as high-performance cathode materials for aluminum-based batteries, balancing capacity, voltage, and ion mobility [46].

3.2. PAH-based cathodes for dual-ion batteries

In the realm of emerging battery chemistry, dual-ion batteries (DIBs) have recently gained significant attention [119–123]. In contrast to “rocking-chair” batteries, such as LIBs, which function by transferring a single ion charge carrier between the cathode and the anode during the charge cycle, these batteries use an alternative mechanism. In dual-ion batteries, anions and cations from the electrolyte are inserted into the cathode and anode, respectively. During the discharge process, ions migrate in opposite directions. The primary challenge facing conventional DIBs that use graphite as both the cathode and the anode is related to the high operating potential of the graphite cathode. This potential is characterized by a lower energy level compared to the HOMO energy of the electrolyte solvent, thereby inducing solvent oxidation. This insight has motivated further investigation of alternative DIB cathode materials [124,125], such as MOFs [126], organic crystalline solids [127–129], and organic polymers [130]. Recently, tremendous progress has been made in coupling such cathodes to different anodes. In particular Rodríguez-Pérez et al. [116,131] studied a group of PAH crystals such as perylene, triphenylene and coronene as DIB cathodes with the aim of uncovering the trends in their electrochemical behavior as hosts for PF_6^- anions (Fig. 12). It was shown that the insertion charge potential decreases from ~ 4.1 to ~ 3.9 V, and then to ~ 2.6 V, while the reversible (discharge) capacity increases from 39.9 to 88.4 mAh g^{-1} , and then 95.1 mAh g^{-1} for coronene to perylene and triphenylene, respectively, with decreasing molecular sizes. DFT calculations demonstrated that the binding of PF_6^- on the edge sites of coronene was energetically more favorable compared to sandwiched PF_6^- , which was between two coronene molecules [116]. In the latter case, coronene molecules are required to bend around the inserted PF_6^- in a buckling fashion, which is less favorable. It has been hypothesized that the availability of more edge sites from PAHs would result in higher capacity values [116]. Consequently, smaller PAH molecules with more edge sites per a certain number of carbons in one molecule could be more favorable for DIB.

The galvanostatic charge/discharge (GCD) curves of perylene, triphenylene, and coronene demonstrated that a reduction in the extent of conjugation in the molecules is associated with an enhancement in the reversible capacity (Fig. 12b) [116]. The charge/discharge profiles of coronene feature a single plateau, consistent with the singular pair of redox peaks observed in its cyclic voltammetry (CV) curves (Fig. 12b,e). This correlates with its modest specific capacity, which is slightly below 40 mAh g^{-1} (Fig. 12b). In contrast, perylene exhibits a higher specific capacity exceeding 80 mAh g^{-1} , accompanied by two distinct plateaus during both charging and discharging. These dual plateaus align with the two redox peak pairs seen in its CV profiles (Fig. 12c,f) [116]. The discrepancy in the number of plateaus (single for coronene versus double for perylene) suggests that PAH molecular solids may not store anions through the staged intercalation mechanism characteristic of graphite intercalation compounds (GICs). In coronene molecules, there is a single type of zigzag edge site, whereas perylene has both zigzag and armchair edge sites. Therefore, the two plateaus observed in perylene could be attributed to the presence of these two distinct edge sites. Nevertheless, it is also conceivable that the anions stored in perylene are linked to two distinct crystalline planes within the solid structure, resulting in the emergence of two plateaus. The potential profiles of triphenylene exhibit a combination of both quasi-plateaus and inclined sections, mirroring its cyclic voltammetry curves, with both sharp and broad peaks (Fig. 12d,g). The electrochemical behavior of triphenylene may be related to its “soft” structure and solubility in the electrolyte. Its pristine structure can turn amorphous upon anion insertion [116]. The insoluble PAHs, coronene and perylene, have demonstrated exceptional long-term cyclability following the initial fading process. It has been demonstrated [116] that coronene exhibits $> 100\%$ capacity retention over 4000 cycles. Perylene delivers well over 1700 cycles with a 94.5% capacity retention of the capacity obtained in the 50th cycle. It is suggested that these may be the most efficient cycling performance for anion-retaining cathodes using conventional alkyl carbonate electrolytes without any additives. The triphenylene electrodes exhibited insufficient long-term cycling stability due to their solubility [116].

3.3. PAH-based cathodes for Li/Na-ion battery

Inorganic materials have been extensively studied as cathode materials in LIB, such as transition metal oxides, phosphates, silicates, and sulfates [132,133]. LiCoO_2 and LiFePO_4 , for example, dominate the commercial lithium-ion battery market as cathode materials due to their high energy density and reliable cycling stability. However, the process of ion diffusion in these inorganic materials is typically sluggish, resulting in restricted energy densities. This is the main obstacle to full commercialization. Organic materials have recently gained significant attention as potential cathode replacements for conventional inorganic materials, due to their structural flexibility, which enhances ionic diffusion, eco-friendly properties, simpler manufacturing processes, and cost efficiency [134–139]. Recently, surface redox reactions involving lithium ions and oxygen-containing functional groups on carbon-based materials such as carbon nanotubes, graphene oxide and biomass-derived carbon, have been widely investigated for cathode applications [140–150]. DFT calculations indicated that carbonyl groups significantly enhance the redox activity of graphene oxide, while hydroxyl groups reduce the redox potential. However, controlling the surface chemistry and precise distribution of redox-active sites on these functionalized carbons remains challenging, hindering the rational design of materials to achieve precise control over desired properties.

On the other hand, small organic molecules like benzene, naphthalene, anthracene, coronene and phenalenyl and their derivatives, whose redox characteristics can be finely adjusted using established principles of organic chemistry, are seen as a promising alternative for organic materials for positive electrodes in Li/Na-ion batteries [42,103,105,137,151–162]. Notably, ketone derivatives of small organic molecules exhibit promising electrochemical performance for cathode applications due to their redox-active carbonyl groups, which undergo reversible reactions with lithium ions during discharge [163–166]. Unlike graphene oxide or functionalized carbon nanotubes, which inherently suffer from a heterogeneous oxygen functional group distribution, these molecular ketones can be tailored with precise control over their structure, enabling the optimization of desired electrochemical properties. Liang et al. [163] demonstrated that systematically designed heteroaromatic carbonyl compounds exhibit superior cell performance in lithium-ion batteries, attributed to multi-electron redox activity and enhanced cycling stability. Shimizu et al. [164] proposed that functionalizing quinones with lithiooxycarbonyl groups effectively reduces their solubility in electrolytes while maintaining voltage stability. Song et al. [167] investigated the use of polyanthraquinone as a durable organic cathode, demonstrating a near-theoretical specific capacity and impressive cycling stability. Kim et al. [151] conducted computational analyses of quinone derivatives derived from benzene, naphthalene, and anthracene. Their findings suggest that introducing carboxylic acid groups preserves cathodic behavior during discharge, enhancing both redox potential and charge capacity.

Park et al. [42] conducted a rigorous investigation of phenalenyl and anthracene derivatives (Fig. 13 a,b). The central objective of this investigation was to ascertain the extent to which the positional arrangement of carbonyl groups influences their properties as lithium-ion battery cathodes. Using DFT calculations the thermodynamic stability and electrochemical redox behavior of these ketone derivatives were analyzed. It was determined that the thermodynamic stability of ketone derivatives is characterized by a high degree of sensitivity to the spatial distribution of carbonyl groups. In addition, Park et al. [42] showed that the redox potential rises in direct proportion to the number of carbonyl groups present. The maximum observed values are 3.93 V versus Li/Li⁺ for the phenalenyl compounds and 3.82 V versus Li/Li⁺ for the anthracene derivatives (Fig. 13 c). Moreover, the study demonstrated a significant correlation between the redox potential of an organic molecule and its spin state in a thermodynamically stable configuration.

Anthracene-based compounds, including phenazine, have attracted considerable attention in research due to their redox-active characteristics. Recent studies have highlighted phenazine-based organic compounds with non-carbonyl redox-active sites as promising candidates for battery cathodes due to their conductive conjugated structures [169,170]. For example, Miao et al. investigated a series of phenazine derivatives to address the material's inherently low redox potential [169]. Their findings demonstrated that the introduction of pyridinic functional groups into the phenazine structure enhances redox activity via inductive effects. Independently, Vitaku et al. developed phenazine-based covalent organic frameworks (COFs) tailored for high-performance cathodes [170]. They revealed that conjugated phenazine ligands within the COF architecture significantly improved the efficiency of charge transport enabling enhanced electrical conductivity in the material. Kim et al. [105] conducted a study on a wide range of conjugated anthracene derivatives, including phenazine, with different numbers and arrangements of nitrogen dopants. They aimed to investigate the redox potentials and theoretical performance characteristics of these compounds. A comprehensive collection of anthracene derivatives was created by substituting carbon with nitrogen, as illustrated in Fig. 13d. The analysis revealed that, in the fully charged state, increasing the number of electron-withdrawing nitrogen atoms significantly enhanced the intrinsic redox potential of the anthracene framework. This improvement was attributed to the heightened electron-deficient character of the molecule, which boosted its reductive capacity. However, during the discharge phase, the stability of lithium interactions exhibited an inverse relationship with nitrogen content. Systems doped with one to three nitrogen atoms demonstrated reduced lithium-storage capability, leading to diminished theoretical performance. This contrasting behavior underscores the importance of balancing nitrogen incorporation: while higher doping levels (4 to 10 nitrogen atoms) improves redox properties and energy density, excessive or insufficient doping compromised structural stability. The findings by Kim et al. [105] highlight strategic nitrogen doping as an optimal design strategy to maximize the electrochemical efficiency of anthracene-based materials for energy storage applications.

There have also been significant recent advances in the study of the organic molecules as cathodes in sodium-ion batteries [47,156,166,171–176]. Kim et al. [104] analyzed seven quinone derivatives-1,4-benzoquinone (BQ), 1,4-naphthoquinone, 9,10-anthraquinone (AQ), 2-aminoanthraquinone (AAQ), 2,6-diaminoanthraquinone (DAQ), anthraquinone-2-carboxylic acid (CAQ), and anthraquinone-2,6-dicarboxylic acid (DCAQ)-to elucidate the redox properties of these organic molecules as cathodes of sodium-ion batteries (Fig. 13 f). DFT-derived correlations between redox properties, electron affinity (EA), and solvation energy reveal that maintaining a positive redox potential in quinones requires systematic optimization of both EA and solvation energy. Specifically, higher solvation energy is needed to sustain positive redox potentials as EA decreases. These correlations were validated as universal principles applicable to diverse organic molecules, including non-quinone systems [104].

Jung et al. [156] employed density functional theory (DFT) simulations to thoroughly analyze the redox characteristics and theoretical performance metrics of a range of fluoranil derivatives as potential cathode materials for sodium-ion batteries. Fluoranil, a tetrahalogenated derivative of p-benzoquinone, has an unsaturated cyclic ketone structure consisting of a six-membered carbon ring with two carbonyl groups. The electrochemical characteristics of an unaltered organic compound can be altered by altering its properties through structural and electronic modifications. Studies have demonstrated that the redox potential increases after the introduction of bulky trifluoromethyl groups into fluoranil, but this drastically reduces the specific charge capacity. Notably, fluoranil exhibits superior redox activity and theoretical performance metrics compared to its hydrogenated analogs, benzoquinone, underscoring its promise as a design strategy for high-performance sodium-ion batteries. Furthermore, the redox behavior of fluoranil derivatives is influenced not only by structural factors (e.g., steric bulk) and electronic characteristics (e.g., functional group electronegativity) but also by solvation energy, highlighting the multifaceted nature of optimizing these materials.

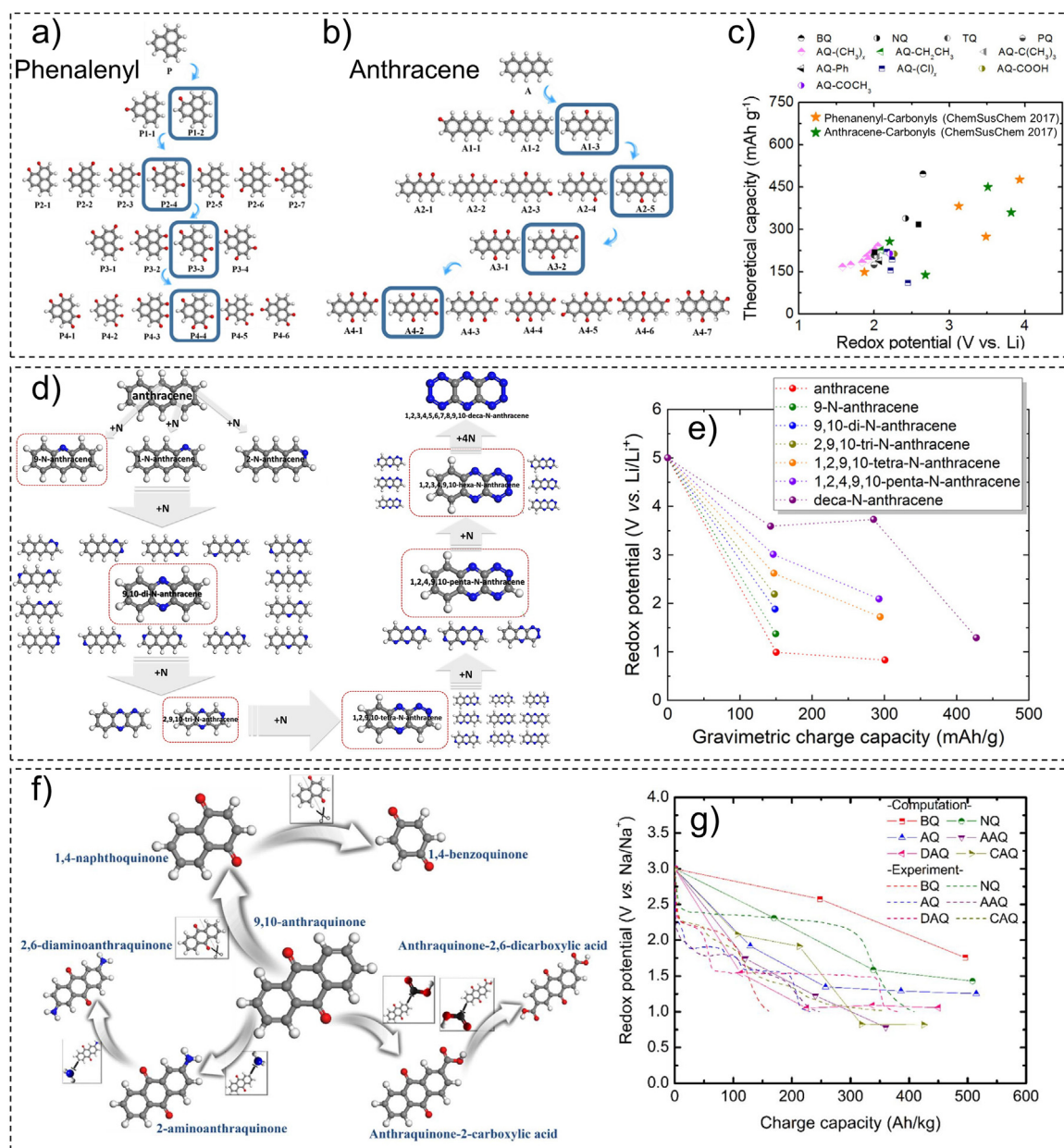


Fig. 13. (a, b) Systematic addition of up to four carbonyl functional groups onto phenalenyl (P) and anthracene (A). In “Pm-n” and “Am-n”, m and n indicates the number of carbonyl groups and the case number, respectively. The molecules wrapped by the blue boxes have the lowest energy compared to other molecules with the same number of carbonyl groups (Ref. [42]). (c) Correlations of the redox potential with the theoretical capacity based on the number of redox-active carbonyl groups for the ketone derivatives from work of Park et al. [42] and quinone derivatives such as benzoquinone (BQ), naphthalene-1,4-dione (NQ), tetracene-5,12-dione (TQ), pentacene-5,14-dione (PQ), and anthraquinone-(CH₃)_x (AQ-(CH₃)_x) from work of Bachman et al. [168]. (d) Chemical structures of anthracene and its derivatives with different numbers and configurations of introduced nitrogen atoms. An anthracene derivative with the highest redox potential is stressed in red dotted rectangular box for each given number of nitrogen atoms (Ref. [105]). (e) theoretical charge capacities for the seven compounds described in (d) (Ref. [105]). (f) Chemical structures of the seven quinone derivatives. Structures of 1,4-benzoquinone (BQ), 1,4-naphthoquinone (NQ), 9,10-anthraquinone (AQ), 2-aminoanthraquinone (AAQ), 2,6-diaminoanthraquinone (DAQ), anthraquinone-2-carboxylic acid (CAQ), and anthraquinone-2,6-dicarboxylic acid (DCAQ). The atoms in gray, white, red, and blue correspond to carbon, hydrogen, oxygen, and nitrogen, respectively (Ref. [104]). (g) Calculated redox potential vs. charge capacity for the six experimentally available quinone derivatives binding with Na atoms at metastable positions (Ref. [104]).

Table 2

Comparative performance of PAH-based cathode materials for different metal-ion battery systems.

Material	Battery type	Role	Theor.capacity (mAh g ⁻¹)	Exp. capacity (mAh g ⁻¹)	Cycles/Retention	Refs.
Anthracene	Al-ion	Cathode	300	157 (1st discharge)/143 (1st reversible)	800/83%	[6]
Pyrene	Al-ion	Cathode	265.1	95.2 (1st reversible)	–	[6]
Coronene	Al-ion	Cathode	178.5	56.9 (1st reversible)	–	[6]
Perylene	Al-ion	Cathode	212.5	–	–	[6]
Benzo[g,h,i]perylene	Al-ion	Cathode	194.2	–	–	[6]
Naphthalene	Al-ion	Cathode	418.3	–	–	[6]
Benzene	Al-ion	Cathode	343.7	–	–	[6]
Triphenylene	Al-ion	Cathode	88	–	–	[46]
Pyrene (crystalline)	Al-ion	Cathode	133	20 (initial)/10 (stabilized)	–	[106]
Polypyrene (amorphous)	Al-ion	Cathode	133	70 (stabilized)	–	[106]
Poly(nitr.-co-pyrene)	Al-ion	Cathode	133	100 (stabilized)	–	[106]
Coronene	Dual-ion	Cathode (PF ₆ ⁻)	–	39.9 (reversible)	–	[125]
Perylene	Dual-ion	Cathode (PF ₆ ⁻)	–	88.4 (reversible)	–	[125]

3.4. Summary and comparative analysis of PAH-based cathodes

The application of PAHs as cathode materials spans multiple battery chemistry, including aluminum-ion, dual-ion, and lithium/sodium-ion systems. Across these diverse platforms, several key design principles emerge that govern electrochemical performance.

For aluminum-ion batteries, PAHs with extended π -conjugated systems, such as anthracene and coronene, demonstrate efficient AlCl_4^- storage through on-plane adsorption mechanisms. Anthracene stands out with a reversible capacity of 143 mAh g⁻¹ and exceptional cycling stability exceeding 800 cycles, outperforming both pyrene and coronene under identical conditions. The transition from crystalline pyrene (which suffers from dissolution and poor conductivity) to amorphous polypyrene dramatically improves performance, achieving stabilized capacities of 70–100 mAh g⁻¹ through enhanced anion diffusion pathways.

In dual-ion batteries, the availability of edge sites proves critical for PF₆⁻ storage. Smaller PAHs like perylene and triphenylene, with higher edge-site density relative to molecular weight, deliver superior reversible capacities (88.4 and 95.1 mAh g⁻¹, respectively) compared to coronene (39.9 mAh g⁻¹). The presence of distinct edge site geometries correlates with multiple voltage plateaus, suggesting that molecular architecture directly influences anion storage behavior. Crucially, insoluble PAHs such as coronene and perylene demonstrate remarkable long-term cyclability, retaining > 94% capacity over thousands of cycles.

For lithium and sodium-ion batteries, functionalization strategies dominate performance optimization. The incorporation of carbonyl groups into phenalenyl and anthracene frameworks systematically raises redox potentials to 3.82–3.93 V vs. Li/Li⁺, with the spatial distribution of these groups critically affecting thermodynamic stability. Nitrogen doping of anthracene derivatives enhances electron deficiency and redox potential, though excessive doping compromises lithium storage capability during discharge. Similarly, quinone derivatives for sodium-ion batteries require careful balancing of electron affinity and solvation energy to maintain positive redox potentials.

Despite these advances, the practical adoption of PAH cathodes remains hindered by intrinsically low redox activity stemming from limited density of redox-active sites. Current strategies focus on integrating additional redox moieties into molecular frameworks and extending π -conjugation to improve electron transfer kinetics. To provide a clear quantitative comparison of PAH-based cathode materials across different battery systems, Table 2 summarizes the key electrochemical performance metrics discussed in this section.

4. PAH for pre-metallization

Pre-metallization is a successful method for enhancing the initial Coulombic efficiency (ICE) and energy utilization of metal-ion batteries [177–189]. The pre-metallization process acts as a compensatory strategy to address metal ion depletion caused by the formation of the solid electrolyte interface (SEI) layer [190]. By introducing supplementary ions, this method counterbalances ion losses, thereby enhancing the reversible capacity, energy density, and operating voltage of the battery system [191]. Additionally, pre-metallization induces controlled volume expansion in silicon- and tin-based anodes, which strengthens electrode stability by mitigating cracking and pulverization issues [192–196]. The rate capability is further improved due to reduced electrochemical impedance, a consequence of the thinner SEI layer formed on pre-metallized electrodes [197–201]. Moreover, in silicon-based anodes, the combined effects of lowered elastic and shear moduli and an elevated Li⁺ diffusion coefficient contribute to superior cycling performance and faster charge–discharge rates [202]. These synergistic advancements underscore the multifaceted benefits of pre-metallization in optimizing electrode durability and electrochemical efficiency.

Fig. 14 illustrates the mechanism of alkali metal-ion loss during battery operation and the role of pre-metallization in mitigating this issue. In conventional battery systems, the cathode typically contains alkali metal compounds, while the anode consists of materials such as graphite, silicon- or tin-based alloys, or non-alkali metal oxides. These anode materials work via insertion, alloy, or

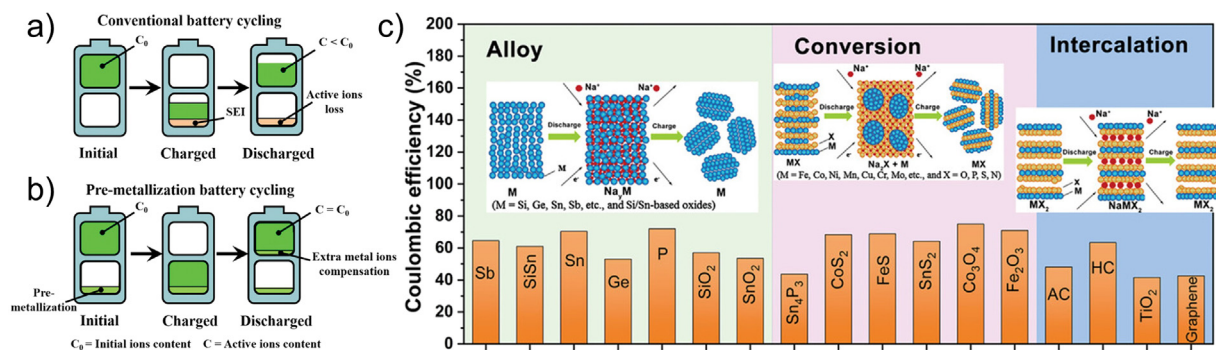


Fig. 14. (a,b) The schematic illustration of active alkali metal-ion loss and the pre-metallization strategies against alkali metal ions loss. (a) The initial ions loss of conventional battery cycling. (b) The extra metal ions compensation of pre-metallization battery cycling (Ref. [190]). (c) Schematic diagram (Ref. [190]) of three different types of anodes and corresponding initial Coulombic efficiency in sodium ion battery (Ref. [203–219]). Where the AC and HC represent amorphous carbon and hard carbon, respectively, and the inset shows a sodium-ion battery as an example.

conversion type mechanisms. Using lithium-ion batteries as an example, some Li^+ ions from the cathode become irreversibly trapped on the anode surface during the initial charge–discharge cycles, forming a SEI layer [132,220,221]. Although the SEI facilitates ion transport and insulates electrons and protects the anode, its progressive thickening, particularly in alloy and conversion type anodes prone to volume fluctuations, exacerbates active ion loss and decreases ICE due to repeated repair and accumulation of the SEI.

As shown in Fig. 14a, batteries without pre-metallization experience ion migration from the cathode to the anode during charging, accompanied by electron flow through the external circuit. During discharge, ions and electrons reverse direction. However, residual alkali metal ions remain trapped in the anode's SEI, diminishing the pool of active ions. In contrast, anode pre-metallization (Fig. 14b) involves pre-inserting alkali metal ions to form a stable SEI prior to cycling. This ensures complete recovery of cathode ions during the first cycle. Similarly, cathode pre-metallization pre-loads the cathode with excess ions, compensating for losses during initial cycles and enabling full restoration of its original state. Both strategies enhance ion retention, counteracting the cumulative losses caused by SEI dynamics and structural degradation.

For both alloy-type and conversion-type anodes, the ICE is only 55%–85%, and the initial capacity loss corresponds to ≈ 500 – 1500 mAh g^{-1} for Si anodes and ≈ 250 – 600 mAh g^{-1} for the oxide anodes (Fig. 14c) [188]. The huge volume change accompanied by the formation of the SEI layer results in structural damage and pulverization of materials, which, in turn, leads to deterioration of electrochemical performance such as a reduction in ICE. Furthermore, sluggish migration kinetics of metal ions can lead to their entrapment within bulk anode materials, resulting in active metal loss and reduction of ICE. Formation of Li_2O , Na_2O and other compounds consumes a significant number of active metal ions, remaining in the anode structure of conversion-type anodes and leading to decrease in ICE. Fortunately, these problems can be effectively resolved through pre-metallization techniques at the laboratory scale.

4.1. PAH for pre-lithiation

As far back as 1998, Owen et al. [222] introduced a chemical lithiation technique to supplement lithium in carbon anodes, aiming to offset the initial irreversible capacity loss inherent in these materials. Following this breakthrough, multiple pre-lithiation strategies emerged, including micrometer-scale lithium metal powders [223–226] and Li_xSi nanoparticles [227–231]. These were engineered as pre-lithium agents for carbon and silicon anodes. These approaches enhanced Coulombic efficiency and enabled capacity balance between electrodes, although precise control over the anode's lithiation degree remains challenging. In parallel, chemical lithiation agents such as *n*-butyllithium and lithium triethylborohydride have been explored to pre-lithiate sulfur cathodes, seeking to address significant volume expansion in traditional sulfur-based systems [232–234]. However, these compounds exhibit extreme air sensitivity, creating significant safety challenges that hinder their practical implementation. Moreover, pre-lithiated sulfur cathodes maintain the same fundamental reaction mechanisms as their conventional counterparts, resulting in similar poor cycling stability.

Shen et al. [235] has been reported a simple chemical prelithiation method using lithium naphthalenide (Li-Naph) to fully prelithiate polyacrylonitrile-sulfur composite (S-PAN) for the production of a cycle-stable Li_2S -PAN cathode. They have also shown how to partially prelithiate silicon nanopowder to create a highly efficient Li_xSi anode. Using the Li_2S -PAN cathode and the Li_xSi anode, they developed a rechargeable Si/S Li-ion battery, where Li^+ ions move back and forth between the S-PAN cathode and the Si anode during the charging and discharging processes. This approach completely sidesteps the issues arising from the disintegration of polysulfide intermediates and the dendrite formation of lithium in conventional Li/S batteries. Consequently, the Si/ Li_2S cell exhibits a nearly complete utilization of both the cathode and anode materials, resulting in a high specific energy of 710 Wh kg^{-1} . Furthermore, this prelithiation process is gentle, highly effective, and versatile, potentially offering a novel perspective for the advancement of Li-ion batteries. The usage of lithium naphthalenide (Li-Naph) as a pre-lithiation reagent, motivated by its

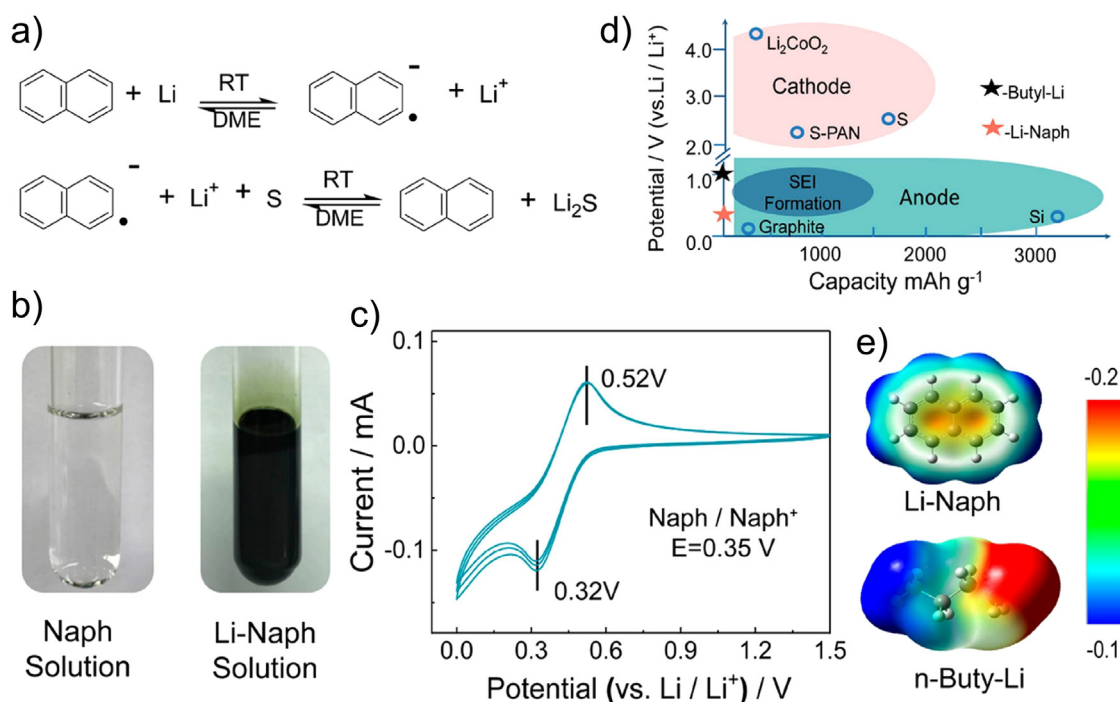


Fig. 15. (a) Synthetic reaction of Li-Naph and lithiation reaction of sulfur with Li-Naph. (b) Photographs of 1 M naphthalene and 1 M Li-Naph solution dissolved in DME. (c) Cyclic voltammograms of naphthalenide in 0.1 M LiTFSI electrolyte at 100 mV s⁻¹. (d) Potentials of selected materials related to this paper. (e) Mapped electrostatic potential (MEP) surfaces of naphthalene anion and *n*-butyl anion. Ref. [235].

advantageous properties, including straightforward synthesis, mild reactivity, and robust lithiation capability [236,237]. As shown in Fig. 15a, Li-Naph synthesis involves combining stoichiometric amounts of lithium metal and naphthalene (1:1 molar ratio) in dimethoxyethane (DME) solvent. The reaction is marked by a distinct color shift from transparent to dark green (Fig. 15b), signaling the spontaneous formation of a Li⁺-Naph⁻ charge-transfer complex. This transformation occurs via electron transfer from lithium to the naphthalene ring, generating dark green Naph⁻ radical species [237–240]. Cyclic voltammetry (CV) was employed to assess the redox activity of lithium naphthalenide (Li-Naph) in solution. As shown in Fig. 15c, the CV profile exhibits a pair of reversible redox peaks at 0.32 V and 0.52 V (vs Li/Li⁺), yielding a formal potential of 0.35 V for the Naph/Naph⁻ redox couple. This value is significantly lower than the lithiation potential of sulfur-based electrodes ($E_{S/Li_2S} \approx 2$ V, vs Li/Li⁺), confirming Li-Naph's strong reducing capacity to lithiate elemental sulfur in both S/C composites and sulfur-polymer composites (Fig. 15d). Experimental validation revealed that Li-Naph readily converts elemental sulfur to a fully lithiated Li₂S state under ambient conditions. The reaction mechanism involves electron transfer from Naph⁻ radicals to sulfur, accompanied by Li⁺ ion incorporation into the sulfur matrix to form Li₂S intermediates. Concurrently, Naph⁻ radicals are oxidized back to neutral naphthalene (Fig. 15a). Unlike conventional reagents such as *n*-butyllithium, the stabilization of Naph⁻ radicals arises from the delocalization of lone-pair electrons across naphthalene's conjugated aromatic system (Fig. 15e). This electronic delocalization confers exceptional stability to Li-Naph, enabling it to remain inert even under exposure to moist air—a critical advantage for practical battery applications [235].

A lithium-biphenyl (BP) [242] complex was previously shown to pre-lithiate phosphorus/carbon composites in ambient air by forming a protective surface film on the liquid. However, these reagents failed to chemically lithiate Si/SiO_x electrodes. Due to their higher redox potential compared to the electrodes (Fig. 16c), they instead triggered solid-electrolyte interphase (SEI) formation [241,242]. While lithiation of SiO_x was achieved by establishing electrical contact between SiO_x electrodes and Li metal via conductive Li–arene complex (LAC) solutions. This approach required either reaction times spanning several days or precise micrometer-scale spacing between Li and the electrode to overcome solution resistance. Such constraints negate the practical advantages of chemical prelithiation strategies [236,244,245].

Jang et al. [192] used LAC-based chemical pre-lithiation to enhance the electrochemical reversibility of SiO_x/Si anodes (Fig. 16a,b). This approach suppressed the formation of a solid-electrolyte interphase (SEI) and volume expansion during subsequent battery operation by addressing these issues pre-emptively prior to cell assembly. Successful chemical lithiation requires precise alignment of the reagent's electrochemical potential with the anode's operating potential. Arene molecules, with their low-energy π^* LUMO undergo reduction via the formation of a charged π system, a process that can compete with anode reduction [246]. For instance, biphenyl (BP), a model arene, is reduced at 0.33 V vs. Li/Li⁺ to form a Li-BP complex. However, this potential exceeds the ~ 0.2 V reduction threshold of Si-based anodes, preventing direct lithiation and instead promoting SEI formation (Fig. 16d). To address this limitation, Jang et al. [192] proposed the addition of electron-donating methyl groups to the benzene ring (Fig.

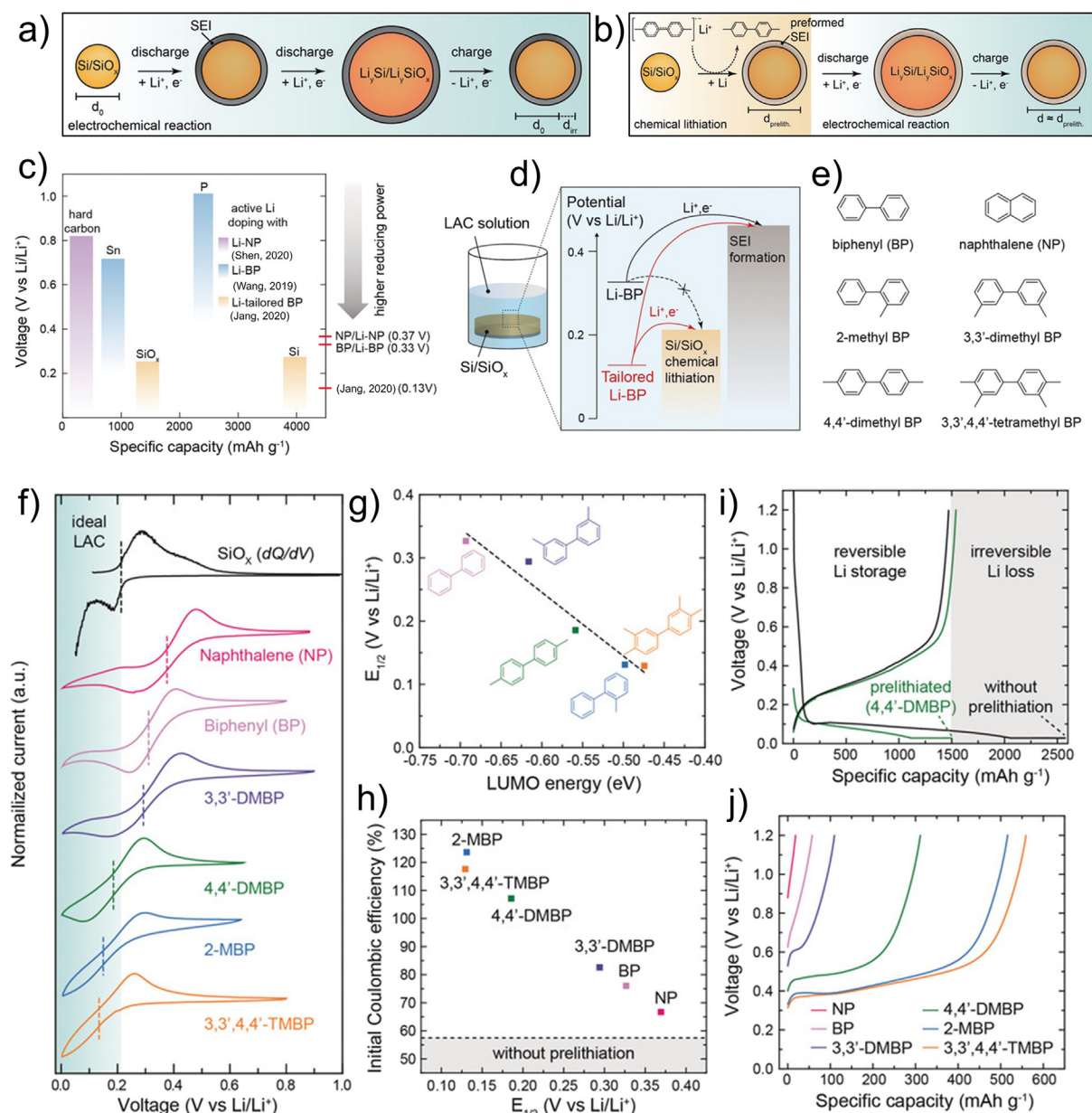


Fig. 16. (a) Lithiation/delithiation of pristine Si/SiO_x during the initial electrochemical cycle in a conventional LIB accompanied with irreversible lithium trapping and SEI formation. (b) Lithiation/delithiation of pre-lithiated Si/SiO_x using Li-arene (e.g., 4,4'-dimethylbiphenyl) complex beforehand with volume expansion and active lithium loss prior to cell assembly. (c) Capacities and potentials of various anodes pre-lithiated with different LAC solutions are taken from Shen 2020 [241], Wang 2019 [242], Jang 2020 [192]. (d) Energy diagram describing chemical lithiation and SEI formation depending on redox potential of the Li-BP complex. (e) Molecular structures of BP derivatives and NP. (f) Cyclic voltammograms of naphthalene, biphenyl, and methyl-substituted biphenyls in 0.5 m LiPF₆ in DME solution with comparison of differential capacity curves of the SiO_x anode. (g) Correlation between measured redox potential (E_{1/2}) of BP derivatives and calculated LUMO energy. (h) Correlation between redox potential and initial CE. (i) Voltage profiles of initial discharge-charge cycle of pristine (black) and pre-lithiated (green) SiO_x electrodes. (j) Electrochemical titration of active lithium stored in SiO_x electrodes after the pre-lithiation process. Ref. [192].

16e). This structural modification elevated the electron density of the arene system, thereby lowering its electron affinity and reduction potential to enable Si-compatible lithiation. BP derivatives exhibited reversible redox activity with potentials lower than pristine BP (Fig. 16f), influenced by substituent position and electronic effects (correlated with Hammett constants) [247,248]. Electron-donating groups raised LUMO energy levels, reducing reduction potentials. DFT-calculated LUMO levels aligned linearly with experimental redox trends (Fig. 16g). In pre-lithiation tests, BP derivatives with E_{1/2} < 0.2 V (e.g., 4,4'-DMBP, 2-MBP) achieved

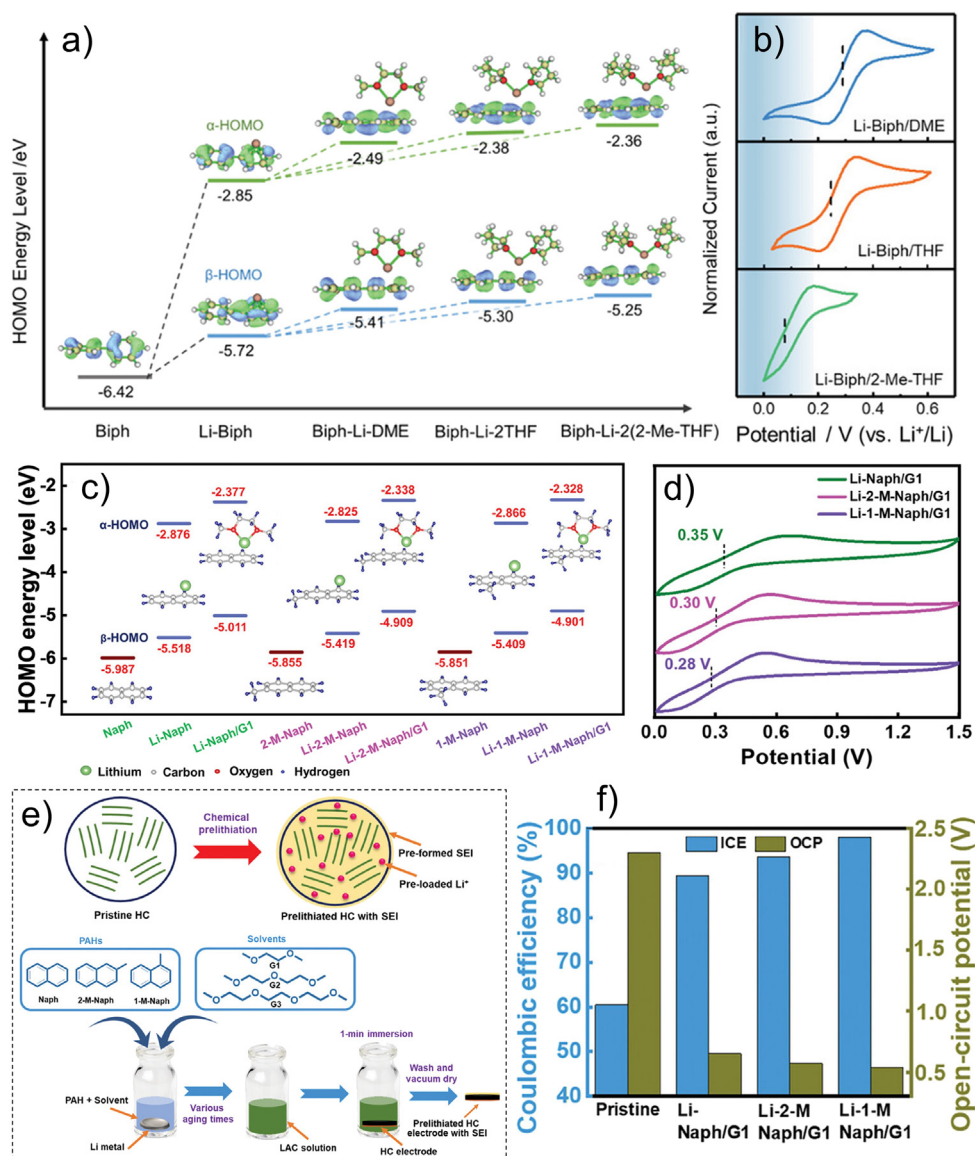


Fig. 17. (a) Geometrical configurations and HOMO energy levels of the Biph-Li-solvent complexes formed by Li-Biph in three ethers of DME, THF, and 2-Me-THF, respectively (Ref. [243]). (b) Cyclic voltammograms of Pt microelectrode in three ether solvents containing 50×10^{-3} m Li-Biph and 0.5 m LiPF_6 at a scan rate of 200 mV s^{-1} (Ref. [243]). (c) HOMO levels of Naph compounds at various coordination states. The vacuum level is aligned to 0 eV (Ref. [185]). (d) CV curves recorded in various LAC solutions at Pt electrodes (Ref. [185]). (e) Schematic illustration of chemical prelithiation process developed in study of Petra et al. [185] via optimization of PAH, solvent, and aging time. (f) ICE and OCP values of various HC electrodes (Ref. [185]).

initial Coulombic efficiencies (CE) $> 100\%$, proportional to lower $E_{1/2}$ (Fig. 16h). In this way Jang et al. [192] demonstrated that molecularly tailored Li-arene complexes, featuring optimized low redox potentials, enable precise Li integration into Si-based anodes, achieving ideal lithium content in full-cell configurations. By regulating immersion time and temperature, the method ensures controlled prelithiation levels and uniform Li distribution across electrodes, offering a scalable and cost-effective approach for industrial applications (Fig. 16).

Over the past decade, multiple strategies have been also explored to enhance the ICE of graphite (Gr) anodes in LIBs. These include optimizing electrolyte composition, incorporating solid-electrolyte interphase (SEI)-forming additives, and pre-lithiating the graphite anode [249–252]. Of these, pre-lithiation has emerged as the most effective approach, as it directly addresses the irreversible loss of active lithium caused by SEI formation and parasitic side reactions [252–255]. By compensating for this lithium depletion, pre-lithiation significantly improves Coulombic efficiencies not only during the first cycle but also in subsequent cycles, thereby boosting the overall performance and longevity of LIBs. A common pre-lithiation approach involves integrating metallic lithium, such

as lithium foil or stabilized lithium metal powder (SLMP), directly into the Gr anode during electrode fabrication [223,224,256,257]. For instance, Wang et al. [223] demonstrated that SLMP-based pre-lithiated graphite (pGr) anodes achieve near-100% ICE and a 5% increase in full-cell reversible capacity. While this method appears straightforward, it poses significant challenges: precise control over lithiation degree and uniform lithium distribution within the anode are difficult to achieve. Furthermore, SLMP's high reactivity with graphite particles can trigger excessive heat generation during processing, raising safety concerns in industrial-scale battery production. An alternative strategy, electrochemical prelithiation, employs temporary lithium/graphite half-cells discharged or short-circuited to lithiate the anode [258,259]. This method enables precise control over lithiation levels but requires labor-intensive disassembly and reassembly of cells, limiting its practicality to laboratory-scale research rather than commercial applications.

Shen et al. [243] explored chemical prelithiation as a method for Gr anodes, though this approach was historically proved ineffective for graphite electrodes. The limitations likely stem from two critical factors:

(1) redox potential mismatch: Existing lithiation reagents exhibit redox potentials higher than that of graphite (~ 0.2 V vs. Li^+/Li), rendering them thermodynamically incapable of transferring lithium to the Gr anode;

(2) solvent incompatibility: Conventional solvents used in prelithiation, such as organic ethers, are prone to co-intercalation with lithium ions. This phenomenon disrupts the layered structure of graphite during the process, compromising electrode integrity.

To overcome these two issues, Shen et al. [243] engineered a chemical pre-lithiation strategy by tailoring both the redox potential of the solution and the solvent properties. They employed a highly reductive yet air-stable molecule as the lithiation reagent and selected 2-methyl tetrahydrofuran (2-Me-THF)-a sterically hindered, strong electron-donating ether-as the solvent. This dual approach aligned the solution's redox potential with Gr lithium insertion potential while preventing solvent co-intercalation, which historically degraded the anode's structure. Through computational modeling and experimental validation, Shen et al. developed a novel solution: lithium biphenylide dissolved in 2-Me-THF (Li-Biph/2-Me-THF). This system achieved an ultralow redox potential (< 100 mV vs. Li^+/Li), exhibited exceptional compatibility with graphite, and enabled rapid, controlled pre-lithiation within minutes without disrupting the anode's lattice integrity (Fig. 17a,b). The resulting pGr anode demonstrated remarkable performance, including near-ideal ICE, enhanced rate capability, and long-term cycling stability [243].

Lee et al. explored prelithiation using Li-BP solutions with solvents of varying solvation strengths, ordered as tetrahydropyran $< 2\text{-Me-THF} < \text{THF} < \text{ethylene glycol dimethyl ether}$ [192,260]. They observed that solvents with higher solvation power, such as ethylene glycol dimethyl ether, predominantly formed solvent-separated ion pairs. This hindered Li^+ ion desolvation, causing solvent molecules to co-intercalate into the graphite structure and impairing performance. Conversely, weaker solvents like tetrahydropyran promoted desolvation, favoring contact ion pairs and aggregates, which enhanced the stability and efficiency of pre-lithiation in graphite/SiOx electrodes. The role of the LAC solvent in pre-lithiation remains unclear, and optimal solvent selection may differ across PAHs. For instance, Patra et al. [185] investigated solvent effects in Li-Naph-based pre-lithiation systems, highlighting the nuanced impact of solvent choice on electrochemical behavior (Fig. 17e). These findings underscore the importance of solvent-solvation dynamics in tailoring pre-lithiation processes for advanced battery materials. In particular, the addition of a methyl group that donates electrons to the alpha position of the naphthalene ring (Naph) was found to substantially reduce the $E_{1/2}$ value from 0.35 V to 0.28 V (vs Li^+/Li) (Fig. 17d). DFT calculations corroborated this trend, revealing that the methyl group elevates the HOMO energy level of Naph-even when accounting for coordination effects-thereby enhancing its reducing power (Fig. 17c). This electronic modulation aligns with the observed electrochemical behavior, demonstrating how targeted structural modifications can tune redox activity in battery materials. Charge-discharge analysis revealed that the ICE of pristine HC was 61%, while electrodes prelithiated with Li-Naph/G1, Li-2-M-Naph/G1, and Li-1-M-Naph/G1 solutions achieved ICE values of 89.4%, 93.5%, and 98.0%, respectively. The prelithiated HC electrodes exhibited significantly lower open-circuit potentials (OCP) compared to the pristine electrode, attributed to SEI formation and Li^+ intercalation during pre-lithiation (Fig. 17f). Notably, the Li-1-M-Naph/G1-treated electrode demonstrated the lowest OCP and highest ICE, aligning with predictions from $E_{1/2}$ measurements and DFT modeling [185].

4.2. PAH for re-lithiation

According to industry statistics, the worldwide market for lithium-ion batteries reached 259 gigawatt-hours (GWh) in 2020 and is expected to grow nearly tenfold to 2.4 terawatt-hours (TWh) by 2030, with an annual growth rate of 25% [261,262]. This rapid expansion underscores concerns over resource depletion, as large-scale LIB production demands significant raw materials. Additionally, given the typical 5–8-year lifespan of LIBs, the volume of retired batteries is expected to surge exponentially in the coming decades [263,264]. To address these challenges, direct regeneration, a non-destructive repair technique-has emerged as a promising recycling strategy. Unlike conventional methods, this approach restores degraded electrode materials to their original state, drastically reducing energy consumption while maximizing the economic value of retired batteries [265–267].

A critical factor in LIB degradation is the irreversible loss of lithium at the cathode, primarily caused by the repeated growth of the SEI during charge-discharge cycles [268–270]. To address this issue, researchers have focused on direct lithium replenishment as a regeneration strategy for degraded cathodes. For instance, spent cathode materials such as $\text{Li}_{1-x}\text{CoO}_2$ (LCO), $\text{Li}_{1-x}\text{FePO}_4$ (LFP), and $\text{Li}_{1-x}\text{Ni}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ (NCM523) [271–275] can be restored by high-temperature sintering (above 600 °C) with lithium salts like Li_2CO_3 or LiOH . However, this method raises environmental and economic concerns due to its high energy demands and significant greenhouse gas emissions. Guo et al. [276] demonstrated electrochemical re-lithiation of degraded LiFePO_4 (LFP) cathodes using pre-lithiated graphite in a regenerative cell. While effective, this method requires intricate cell assembly/disassembly steps, limiting its scalability beyond laboratory settings. Similarly, Chen's group [277–279] developed a hydrothermal-annealing hybrid process to restore degraded LCO, NCM, and LFP cathodes, employing citric acid as a reducing agent and aqueous LiOH for lithium replenishment. Park et al. [280] introduced a redox-mediated approach using 3,5-di-tert-butyl-o-benzoquinone (DTBQ), a

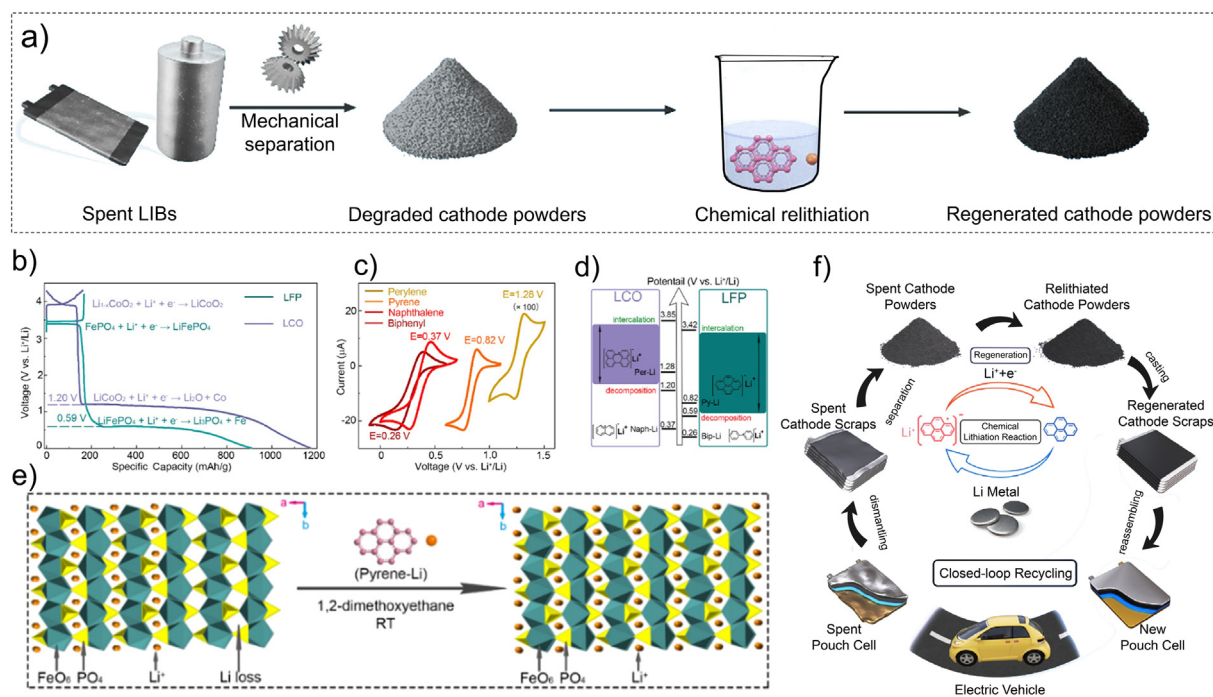


Fig. 18. (a) Schematic illustrations of chemical relithiation strategy for recycling of spent lithium-ion batteries. (b) Charge–discharge profiles of LFP and LCO cathodes in the voltage range of 0–4.2 V vs Li^+/Li (0–4.3 V vs Li^+/Li for LCO) at the current density of 20 mA/g and corresponding lithiation reaction. (c) Cyclic voltammograms of biphenyl/DME, naphthalene/DME, pyrene/DME, and perylene/DME solutions at the scan rate of 50 mV/s. (d) Scheme of selection of the ideal lithiation reagents with suitable redox potential. (e) Scheme of chemical relithiation for recycling the spent LiFePO_4 cathode. (f) Closed loop recycling process enabled by our chemical lithiation strategy during the whole life cycle of electric vehicle batteries. Ref. [282].

quinone-based molecule with lower redox potentials than transition metals, to relithiate aged NCM cathodes. Other methods, such as molten salt [275,281] and ionothermal solution treatments [276], have also been explored for direct cathode regeneration under ambient pressure. Despite these advances, most strategies fall short of green chemistry principles. Challenges include prolonged processing times, poor atom economy, and reliance on energy-intensive post-annealing steps. Consequently, there is an urgent need to develop energy-efficient, low-waste regeneration technologies that minimize chemical inputs and environmental impact.

Wu et al. [282] introduced a mild chemical re-lithiation strategy (Fig. 18a) to directly regenerate degraded LIB cathodes, aligning with circular economy and green chemistry principles. The method employs polycyclic aryllithium compounds—such as pyrene-Li and perylene-Li as reductive lithiation agents. These reagents enable a concerted Li^+ -electron transfer mechanism, efficiently restoring lithium-deficient cathode structures. Remarkably, the re-lithiation process requires only 10 min under ambient temperature and pressure, yet significantly extends the lifespan of high-value cathodes. Critically, the strategy achieves zero waste gas or solvent emissions, as the aryllithium reagents are sustainably utilized without byproducts. This combination of rapidity, ambient operating conditions, and environmental compatibility positions the method as a scalable, industrially viable solution for LIB recycling.

Used cathodes are typically depleted in lithium and contain substantial quantities of high-valence transition metal ions to maintain charge equilibrium. To address this, Wu et al. [282] proposed a chemical regeneration method aimed at replenishing Li^+ ions via redox reactions between cathode materials and reductive lithiation agents. The use of aromatic lithium complexes, such as biphenyl lithium (Biph-Li) and naphthalene lithium (Naph-Li), have low reduction potentials ($< 0.5 \text{ V}$ vs Li^+/Li) and can serve as powerful lithiation agents to counteract irreversible lithium loss in anode materials (e.g., hard carbon, Si, Sb, and SiO) [235,241,284–286], is limited when it comes to cathode materials. The strong reducing capacity of these reagents risks overlithiation in cathodes, potentially triggering irreversible structural degradation, such as crystal collapse or interfacial damage, as demonstrated in equations 1 and 2 [287–289].



Using the LiFePO_4 cathode as an example, its reversible Li^+ intercalation reaction ($\text{Li}_{1-x}\text{FePO}_4/\text{LiFePO}_4$) occurs at 3.42 V vs Li^+/Li . However, discharging the cell to a lower potential (0.59 V vs Li^+/Li) [290] triggers irreversible decomposition of LiFePO_4 , forming metallic Fe and Li_3PO_4 composites (Fig. 18b). Consequently, an optimal lithiation reagent for spent LiFePO_4 must operate

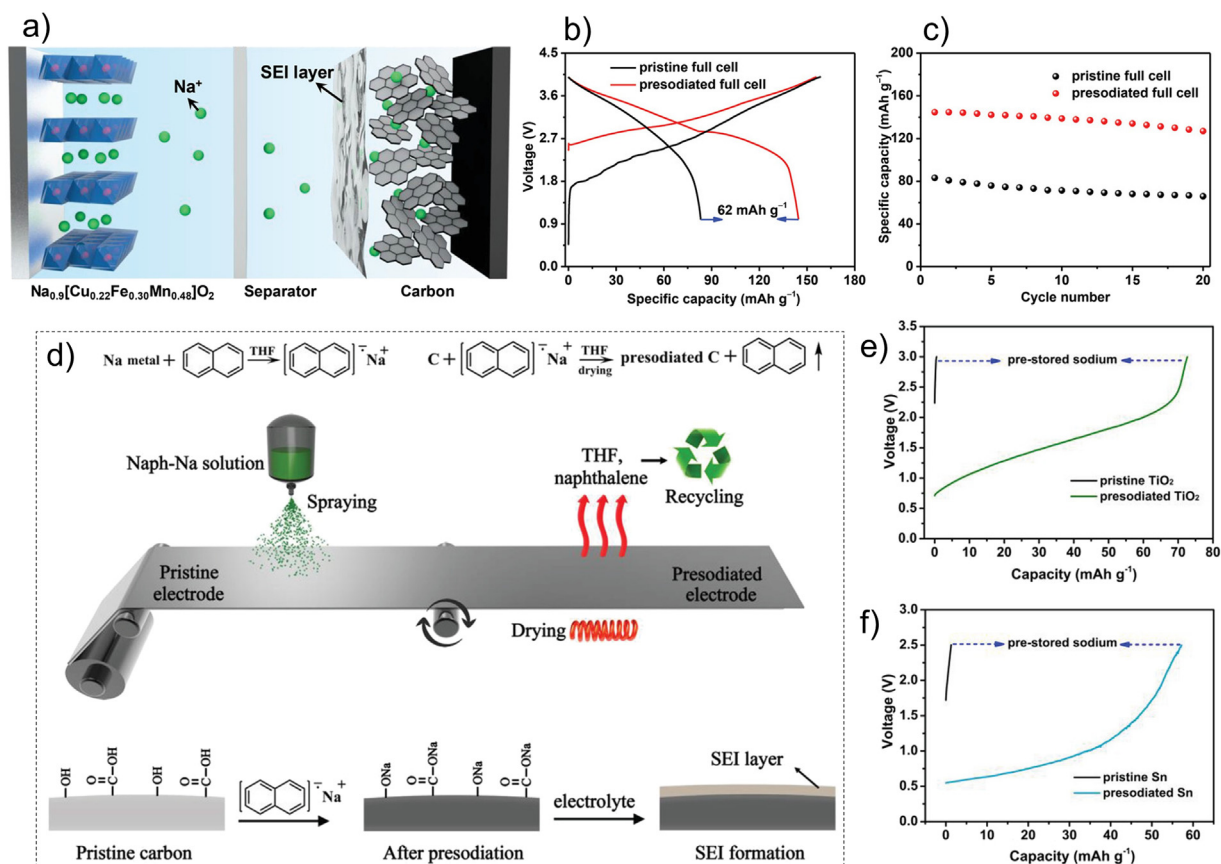


Fig. 19. (a) Schematic illustration of a sodium-ion full cell using a $\text{Na}_{0.9}[\text{Cu}_{0.22}\text{Fe}_{0.30}\text{Mn}_{0.48}]\text{O}_2$ cathode and a carbon anode with a preformed SEI. (b) The first-cycle charge/discharge curves and (c) cycling performance of sodium-ion full cells with a pristine and a presodiated carbon anode, respectively. (d) Schematic of the presodiation process at the electrode level and the presodiation mechanism of hard carbon electrodes using a Naph-Na solution as the presodiation reagent. (e,f) The de-sodiation process of the pristine and presodiated (e) TiO_2 and (f) Sn electrodes before cycling. The current density for the de-sodiation process is 20 mA g^{-1} . Ref. [283].

within the redox potential window of 0.59–3.42 V vs Li^+/Li . In contrast, the LiCoO_2 cathode decomposes at a higher potential of 1.2 V vs Li^+/Li (Fig. 18b), necessitating a milder lithiation reagent with a minimum potential of 1.2 V. Thus, identifying appropriate lithiation reagents is critical to achieving effective cathode re-lithiation while avoiding structural degradation or lattice distortion.

Through screening PAHs, Wu et al. [282] identified the pyrene lithium complex (Py-Li) as a promising candidate, exhibiting a redox potential of 0.82 V vs Li^+/Li (Fig. 18c). This potential lies safely above the decomposition threshold of LiFePO_4 (0.59 V vs Li^+/Li , Fig. 18c), enabling efficient re-lithiation of Li^+ -deficient LFP without destabilizing its lattice structure. Similarly, perylene lithium (Per-Li), with its five fused benzene rings, demonstrates a redox potential of 1.28 V vs Li^+/Li marginally higher than the decomposition potential of LiCoO_2 (1.20 V vs Li^+/Li , Fig. 18d). This makes Per-Li an ideal reagent for replenishing Li^+ in Li -deficient LCO cathodes. Based on these findings, Py-Li and Per-Li were selected as effective lithiation agents, capable of transferring both electrons and Li^+ ions into the depleted cathode lattices [282].

In the chemical re-lithiation process (Eq. (3)), $\text{Py}^{\cdot-}$ anion radicals transfer electrons to $\text{Li}_{1-x}\text{FePO}_4$ (Fig. 18e), while Li^+ ions migrate into the lattice [282]. Upon reaction completion, pyrene (Py) is regenerated as an anion radical via reaction with lithium (Eq. (5)). This allows pyrene to be recycled as a byproduct. This mechanism extends to $\text{Li}_{1-x}\text{CoO}_2$ re-lithiation (Eq. (4)) and perylene lithium (Per-Li) regeneration (Eq. (6)). Notably, the net reactions (Eqs. (7) and (8)) consume only metallic Li, with Py and Per acting as mediators to deliver Li^+ and electrons into Li -deficient LFP/LCO lattices. This closed-loop system achieves 100% atom economy, as no reagents are wasted [282].

Chemical regeneration of spent cathodes using polycyclic aryllithium reagents



Preparation and regeneration of polycyclic aryllithium reagents



Net reactions of chemical re-lithiation procedures



The process of lithiation is energetically advantageous, owing to the substantial voltage disparity between the cathode materials and the aryllithium reagent. These reactions (eqs 3 and 4) proceed efficiently under ambient conditions without inert gas protection, owing to the inherent air stability of aryllithium reagents. Leveraging the rapid kinetics of radical-mediated reactions, the process achieves full lithiation within 10 min at ambient temperature and pressure, even for completely de-lithiated cathodes. In contrast, alternative lithiation agents, such as citric acid/LiOH [279] and LiI [291], require over 6 h to restore degraded LiFePO₄ cathodes, a limitation attributed to their weaker reducing capacity [282].

Thus, Wu et al. [282] successfully developed a wet-chemical re-lithiation technique for the direct and efficient regeneration of spent LIB cathodes under ambient temperature and pressure. Their work confirmed that Li-deficient cathode materials harvested from commercial pouch cells can be restored to their original state for reuse. Beyond restoring electrochemical performance, the process achieves 100% atom economy, fully recyclable byproducts, and zero chemical waste emissions, highlighting its potential as a sustainable industrial solution (Fig. 18a).

4.3. PAH for pre-sodiation

To address the irreversible sodium loss in sodium-ion batteries (SIBs), multiple presodiation strategies have been explored. Sodium-rich cathode materials have been employed to compensate for sodium depletion during cycling, enhancing the energy density of full-cell configurations [292]. Electrochemical pre-sodiation methods, such as pre-treating hard carbon anodes in half-cells with sodium metal counter electrodes, have demonstrated improved reversible capacities in subsequent full-cell assemblies (e.g., NaNi_{0.5}Mn_{0.5}O₂ ||hard carbon systems) [258]. However, such electrochemical processes are intricate and time-intensive, limiting their scalability. Alternative approaches involve integrating sacrificial sodium agents—such as NaN₃ [293,294], Na₂C₄O₄ [295], or sodium metal powders [296], into electrodes. While these additives supply supplemental sodium, their decomposition releases hazardous gases (N₂ or CO₂) or poses safety risks due to sodium's inherent reactivity and instability. These limitations underscore the critical need for developing simplified, industrially viable pre-sodiation techniques to advance practical SIB applications.

Liu et al. [283] developed a straightforward and innovative electrode-level pre-sodiation strategy that simultaneously constructs a stable SEI layer on the hard carbon anode and pre-incorporates active sodium ions into the anode structure. This method avoids generating undesirable byproducts or leaving inert residues in the electrode. The technique involves spraying a tailored pre-sodiation solution—sodium naphthalenide in tetrahydrofuran (THF)—onto the hard carbon electrode, followed by a drying step prior to battery assembly. By modulating the amount of pre-sodiation solution applied, the sodium storage level in the anode can be precisely tuned. Crucially, this approach preserves the intrinsic structure of both the electrode and the active material, ensuring compatibility with industrial manufacturing processes.

The method involves dissolving commercial naphthalene crystals in THF, followed by the addition of sodium metal at a 1:1 molar ratio (Na/naphthalene) (Fig. 19 d). This reaction yields a 0.1 M sodium naphthalenide (Naph-Na) solution. To pre-sodiate the hard carbon electrode, the Naph-Na solution is sprayed onto its surface, initiating a pre-sodiation reaction, and then dried to eliminate residual THF and naphthalene byproducts. Notably, both THF and naphthalene remain unconsumed during this process, enabling their recovery and reuse—an advantage for cost reduction and environmental sustainability [283].

The liquid-phase pre-sodiation reagent ensures a simple and controllable procedure. As shown in Fig. 19d, Naph-Na's low redox potential and strong reducing properties drive its reaction with oxygen-containing functional groups on the hard carbon anode, forming sodium-rich functional groups. Immersing the pre-sodiated electrode in liquid electrolyte further facilitates the formation of a stable SEI layer [297,298]. In full-cell configurations, sodium ions from the cathode are partially consumed during initial charging to establish this SEI layer on the anode surface, while the remaining ions intercalate into the carbon structure, directly contributing to the battery's reversible capacity [283].

Pre-forming a SEI layer on the anode surface prior to cycling reduces the consumption of electrochemically active sodium ions from cathodes, thereby enhancing the energy density of SIBs. However, achieving the optimal pre-sodiation degree in hard carbon anodes is critical for balancing capacity between electrodes in full-cell systems. Insufficient pre-sodiation fails to fully compensate for irreversible sodium loss, while excessive pre-sodiation risks sodium dendrite formation during cycling, raising safety concerns. An ideal pre-sodiation strategy must precisely offset initial sodium loss without introducing surplus ions. Residual byproducts from pre-sodiation can degrade energy density and cycling performance, underscoring the need for residue-free methods.

In Liu et al.'s work [283], Raman spectroscopy confirmed the absence of naphthalene residues on presodiated carbon electrodes, demonstrating complete removal of byproducts during drying. This ensures that only sodium is incorporated into the anode,

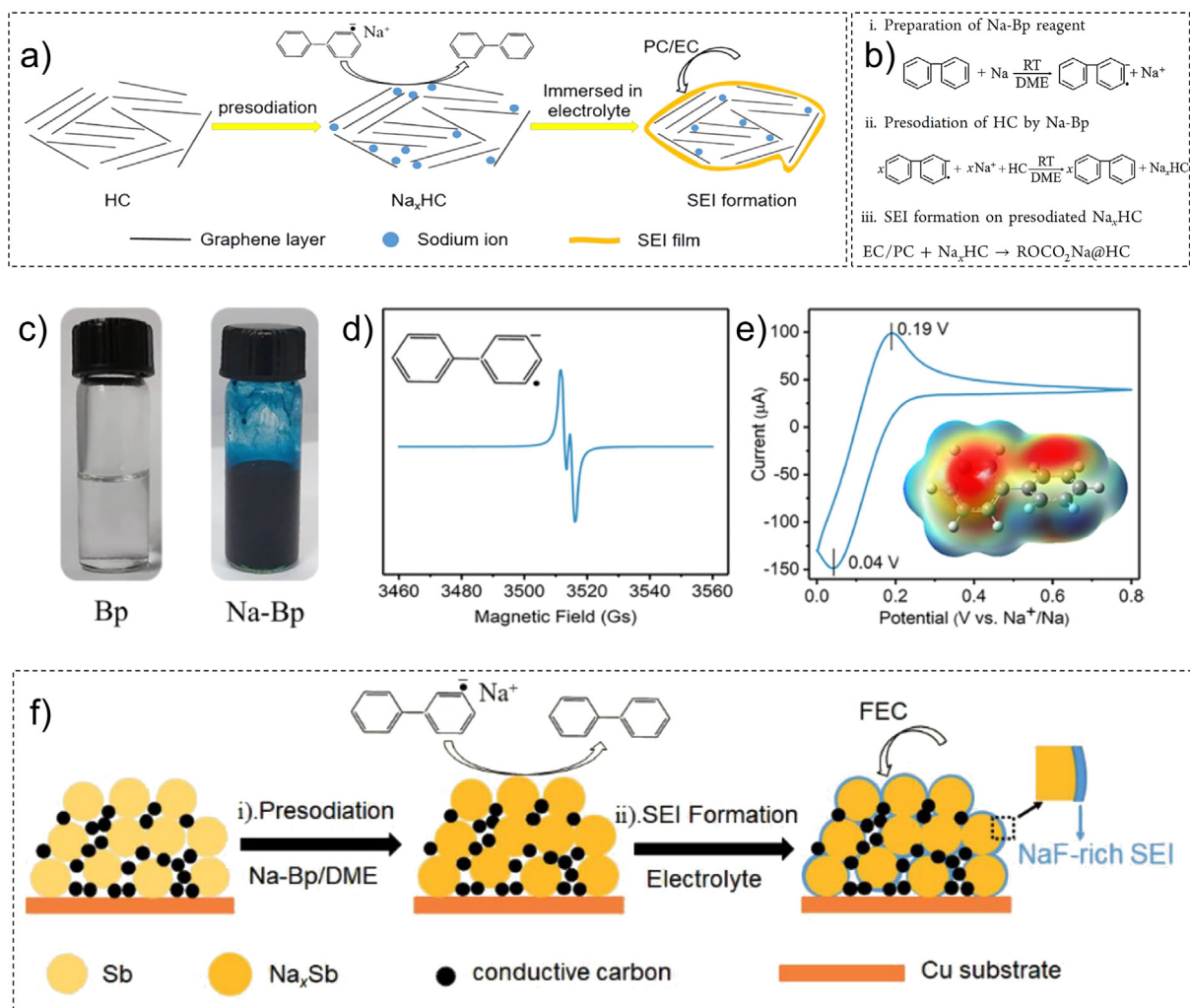


Fig. 20. a-e) Chemical presodiation mechanism of HC anodes by sodium biphenyl (Na-Bp) reagent: (a) Schematic illustration of chemical presodiation and the subsequent SEI formation process on HC anodes; (b) the overall reaction mechanism; (c) optical images of 0.5 M biphenyl and 0.5 M Na-Bp solution dissolved in DME; (d) EPR spectra of Na-Bp solution. (e) Cyclic voltammogram of Na-Bp reagent at a scan rate of 2 mV s⁻¹. The inset shows the simulated electron distribution on the biphenyl radical, as obtained by density functional theory calculations (Ref. [284]). f) A schematic mechanism illustrating the presodiation reaction of the Sb electrode in Na-Bp/DME solution and the subsequent SEI formation process on the surface of the presodiated Na_xSb electrode in a Na-ion battery electrolyte (1 M NaPF₆ EC/DEC with 5% FEC) (Ref. [285]).

maximizing energy density. Additionally, the presodiation process minimally alters the Brunauer–Emmett–Teller (BET) surface area of hard carbon, preserving its structural integrity. The simplicity, cost-effectiveness, and scalability of this approach, leveraging sodium naphthalene in a recyclable solvent system, highlight its potential for industrial adoption.

To demonstrate the role of pre-sodiation in enhancing energy density, SIBs were assembled using a $\text{Na}_{0.9}[\text{Cu}_{0.22}\text{Fe}_{0.30}\text{Mn}_{0.48}]\text{O}_2$ cathode paired with pre-sodiated and non-presodiated hard carbon anodes (Fig. 19a). Initial galvanostatic cycling at 0.1C (1C = 150 mAhg⁻¹) revealed comparable charge capacities ~156 mAhg⁻¹ for both configurations (Fig. 19 b). However, the presodiated anode exhibited a significantly higher initial discharge capacity of 145 mAhg⁻¹, surpassing the pristine anode's 83 mAhg⁻¹ by 62 mAhg⁻¹. This improvement directly correlates with sodium ions supplied during pre-sodiation, offsetting irreversible losses.

Following 20 cycles, the pre-sodiated full cell exhibited a significantly elevated specific capacity of 127 and 62 mAh g⁻¹, surpassing that of the full cell with a pristine carbon anode (Fig. 19 c). This result suggests that the pre-sodiation approach can improve the overall energy density of SIBs, while also maintaining good cycling stability. It has been demonstrated that active sodium is the only constituent that donated into the carbon anode with no other inert residues present. This phenomenon results in optimization of the overall energy density based on both anode and cathode. In order to demonstrate the universality of this pre-sodiation approach, Liu et al. [283] pre-sodiated TiO₂ and Sn electrodes using Naph-Na/THF solution as a pre-sodiation reagent following the same procedure. As with the pre-sodiated carbon electrode, both the pre-sodiated TiO₂ and Sn electrodes exhibited

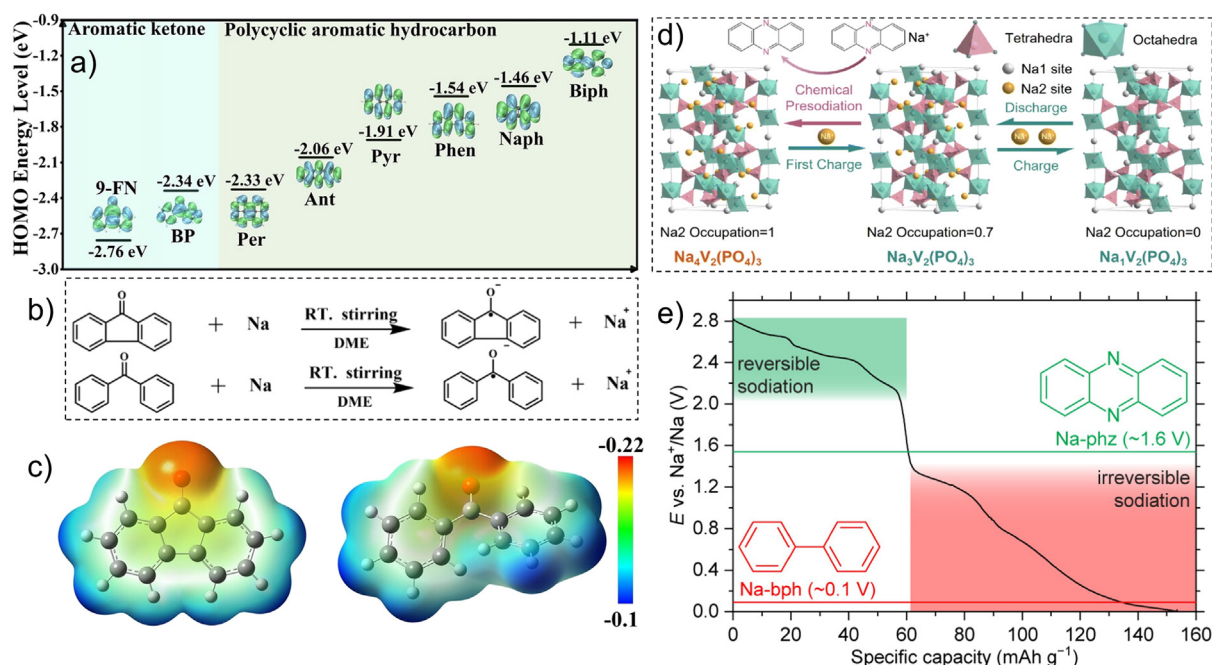


Fig. 21. (a) HOMO energy levels of anionic radicals of PAHs and Aromatic ketone. (b) the overall reaction mechanism. (c) electron cloud density diagrams of 9-FN⁻ (left) and BP⁻ (right) (Ref. [301]). (d) Schematic illustration of chemical presodiation and the subsequent cycling process of the Na₄VP cathode (Ref. [302]). (e) Discharge curve of Na_{0.44}MnO₂ vs sodium metal anode. Regions of reversible and irreversible sodiation and the redox potentials of biphenyl and phenazine are indicated (Ref. [303]).

a lower OCV than their pristine counterparts (see Fig. 19 e,f). The pre-sodiated TiO₂ and Sn electrodes exhibited a high specific charge capacities of 72 and 57 mAh g⁻¹, respectively, before cycling, indicating their effective pre-sodiation.

Sodium biphenyl (Na-Bp) was employed as a pre-sodiation reagent for HC anodes due to its operational simplicity, moderate reactivity, and robust sodiation capacity [284]. As illustrated in Fig. 20c and Formula *i* in Fig. 20 b, Na-Bp is synthesized by combining equimolar amounts of biphenyl and sodium metal in dimethoxyethane (DME) solvent. The strong electron affinity of biphenyl facilitates spontaneous electron transfer from sodium to its conjugated aromatic rings at room temperature, generating a Bp⁻ radical anion (Fig. 20d) [299]. Electron paramagnetic resonance (EPR) analysis confirms this mechanism, revealing sharp Dysonian peaks characteristic of biphenyl radicals, which underscore the electron donation from sodium to biphenyl.

Cyclic voltammetry (CV) was utilized to assess the electrochemical behavior of Na-Bp (Fig. 20e). Distinct redox peaks at 0.04 V (reduction) and 0.19 V (oxidation) vs Na⁺/Na correspond to a midpoint potential of 0.12 V for the Bp/Bp⁻ redox couple. Notably, this value represents the lowest reduction potential among known polycyclic aromatic hydrocarbon radicals (Na-PAHs) [5] closely aligning with metallic sodium's potential and highlighting Na-Bp's strong reducing capability. The conjugated aromatic system of biphenyl stabilizes radical electrons (Fig. 20e inset), imparting air and moisture tolerance, which enhances safety and scalability for industrial use [300].

Pre-sodiation of HC anodes is driven by the significant potential gap between the HC electrode (open-circuit voltage = 2.5 V vs Na⁺/Na) and Na-Bp (0.12 V vs Na⁺/Na). During this process, electrons from Bp⁻ radicals transfer to the HC electrode, enabling sodium intercalation to form Na_xHC, while biphenyl regenerates [235,283]. The intercalated Na⁺ ions occupy irreversible storage sites in Na_xHC, and the low electrode potential triggers electrolyte reduction, forming a SEI upon contact with the electrolyte (Fig. 20a). This pre-sodiation step effectively offsets irreversible sodium losses during initial cycling (Formula *iii* in Fig. 20b), improving overall anode performance [284].

Liu et al. [284] have demonstrated that the robust sodiation capability of Na-Bp enables rapid pre-sodiation of HC anodes, yielding a pre-sodiated HC (Na_xHC) with an exceptional ICE of 100%. When integrated into a sodium-ion battery full cell paired with an Na₃V₂(PO₄)₃ cathode, the Na_xHC||Na₃V₂(PO₄)₃ configuration achieves a remarkable ICE of ~95.0% and an energy density of 218 Wh kg⁻¹. These metrics significantly outperform the control cell using untreated HC (50% ICE, 120 Wh kg⁻¹), underscoring the transformative impact of chemical pre-sodiation. Critically, this reaction is both highly efficient and controllable, with broad applicability to diverse Na-storage materials. This approach establishes a versatile strategy for advancing high-performance sodium-ion battery systems towards practical implementation.

In next work Liu et al. [285] introduced a novel pre-sodiation strategy using biphenyl sodium (Na-Bp) to address irreversible sodium loss in antimony (Sb) anodes for SIBs (Fig. 20f). Through controlled chemical presodiation, the resulting Sb-pNa anode promotes the formation of a sodium fluoride (NaF)-rich SEI layer, as revealed by surface analysis. This SEI layer suppresses

detrimental side reactions with the electrolyte and alleviates mechanical stress within the Sb lattice during cycling. Consequently, the Sb-pNa electrode achieves an exceptional ICE of $\sim 100\%$ (vs. $\sim 75\%$ for pristine Sb) and delivers a remarkable capacity retention of 500 mA h g^{-1} over 300 cycles at a high current density of 2000 mA g^{-1} , far exceeding the 80 mA h g^{-1} of untreated Sb. In full-cell configurations paired with a $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ cathode, the Sb-pNa// $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ system demonstrates an ICE exceeding 95.0% and a high energy density of 232 Wh kg^{-1} , indicating efficient utilization of active materials. Liu et al. provides a transformative approach to overcoming inherent limitations of alloy-based anodes, offering a pathway to develop cost-effective, high-energy-density SIBs for sustainable energy storage [285].

Additionally, the process of chemical pre-sodiation can be enhanced by altering the structure of PAHs. Specifically Zhang et al. [301] proposed the use of aromatic ketones as gentle pre-sodiating agents by incorporating a carbonyl group ($\text{C}=\text{O}$) into PAHs to achieve a balance between the conjugated and inductive effects. Among others 9-Fluorenoneb (9-FN) and Benzophenone (BP) manifest favorable equilibrium potential of 1.55 V and 1.07 V (vs. Na^+/Na), respectively (Fig. 21a-c). Note that 9-FN demonstrates versatile presodiating capability towards multiple Na uptake hosts (tunneled $\text{Na}_{0.44}\text{MnO}_2$, layered $\text{Na}_{0.67}\text{Ni}_{0.33}\text{Mn}_{0.67}\text{O}_2$, polyanionic $\text{Na}_4\text{Fe}_{2.91}(\text{PO}_4)_2\text{P}_2\text{O}_7$, $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}-3\text{V}_2(\text{PO}_4)_2\text{F}_3$), enabling greatly improved initial charging capacity of the cathode to balance the irreversible capacity of the anode. The findings suggest that aromatic ketones are promising candidates for use as pre-sodiating cathodic reagents in high-performance sodium-ion batteries. This discovery will undoubtedly stimulate further research and development in this field.

Xu et al. [302] pioneered a novel chemical sodiation strategy to engineer sodium-rich cathode materials, leveraging phenazine-sodium (PNZ-Na) to transform $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ into a sodium-enriched $\text{Na}_4\text{V}_2(\text{PO}_4)_3$ cathode (Fig. 21d). This method drives a rapid reduction of 50% of vanadium ions from V^{3+} to V^{2+} , coupled with Na^+ insertion into the NASICON framework, achieving phase-pure $\text{Na}_4\text{V}_2(\text{PO}_4)_3$ in just 90 s. When integrated into a full cell with a hard carbon (HC) anode, the $\text{Na}_4\text{VP}||\text{HC}$ configuration delivers an exceptional energy density of 251 Wh kg^{-1} —a 58% increase over the $\text{Na}_3\text{VP}||\text{HC}$ control cell (159 Wh kg^{-1}). This breakthrough highlights the efficacy of chemical sodiation in enhancing cathode performance for high-energy-density sodium-ion batteries. Kapaev et al. [303] have presented a mild, fast, and scalable approach for chemical pre-sodiation of Na-ion battery cathodes employing a tunnel-type $\text{Na}_{0.44}\text{MnO}_2$ (NMO) as a model material to demonstrate its sodiation with sodium-phenazine solutions (Fig. 21e). Following the pre-sodiation process, the specific capacity of NMO in full cells with hard carbon anodes has increased by 80% , while the specific energy has increased by 66% .

Analysis of pre-metallization processes show that compared to conventional methods, PAH-enabled pre-lithiation and pre-sodiation demonstrate exceptional promise for advancing Li-ion and Na-ion batteries. These scalable processes are cost-effective, safe, and air-stable, enabling precise control over irreversible ion loss mitigation during initial cycles while fostering the formation of a protective SEI. Tailored PAH derivatives and solvent systems have been systematically explored, providing design principles for optimizing sodiation/lithiation efficiency across applications. Key parameters such as PAH type, solvent selection, solution concentration, and aging duration critically influence redox potential and reaction kinetics. Recent advances report substantial improvements in ICE for both anodes and cathodes, underscoring the versatility of this approach. Future research should prioritize elucidating the molar ratio between alkali metals and PAHs, which governs reaction efficiency and depth. Expanding applications such as presodiation of diverse Na-ion battery electrodes, Li/Na-ion capacitors, and conversion-type cathodes (e.g., sulfur alternatives), remains underexplored, offering fertile ground for innovation. Additionally, adapting these methods to high-energy-density, alkali metal-free systems could unlock next-generation battery architectures.

4.4. Summary and comparative analysis of PAH-based pre-metallization reagents

This section has demonstrated that PAH-based reagents represent a versatile platform for pre-lithiation and pre-sodiation, offering significant advantages over conventional methods in terms of scalability, safety, and process control. Comparison of their key characteristics reveals distinct design principles that govern their performance.

Several key design principles emerge from this comparative analysis. First, the redox potential of the PAH reagent must be carefully matched to the target electrode. Pre-lithiation of anodes requires potentials below approximately 0.2 V versus Li^+/Li , which can be achieved through electron-donating substituents or optimized solvents. Cathode re-lithiation demands milder potentials above the decomposition threshold of the active material, specifically 0.59 V for LiFePO_4 and 1.2 V for LiCoO_2 . Second, solvent selection critically influences performance. Weakly solvating ethers such as 2-Me-THF prevent co-intercalation into graphite, while DME enables rapid sodiation but may cause structural damage if not carefully controlled. Third, molecular engineering of PAH cores provides a powerful tool for tuning redox potentials and reaction kinetics. Examples include methyl substitution on naphthalene or biphenyl to achieve the desired electrochemical behavior.

Regarding efficiency and stability, the data reveal that PAH-based reagents achieve exceptional performance when properly optimized. Li-1-M-Naph delivers 98.0% percent initial CE for hard carbon anodes, while Na-biphenyl achieves 100% initial CE with remarkable air stability. The re-lithiation reagents, namely pyrene-Li and perylene-Li, demonstrate 100% atom economy and complete recyclability, addressing sustainability concerns. However, applicability remains reagent-specific, as no single PAH reagent universally suits all electrode types. Careful selection based on the target material's operating potential and chemical compatibility is therefore essential.

Future research should focus on three underexplored areas. The first is systematic investigation of the molar ratio between alkali metals and PAHs, which governs reaction depth and efficiency. The second is expansion of this methodology to emerging systems such as K-ion batteries, Li/Na-ion capacitors, and conversion-type cathodes including sulfur and selenium alternatives. The third is development of predictive models that leverage machine learning to accelerate the design of PAH derivatives with tailored redox potentials for specific electrode materials. Additionally, adapting these pre-metallization strategies to alkali metal-free systems such as Mg^{2+} , Ca^{2+} , and Al^{3+} batteries could unlock next-generation energy storage architectures beyond current lithium-dominated paradigms.

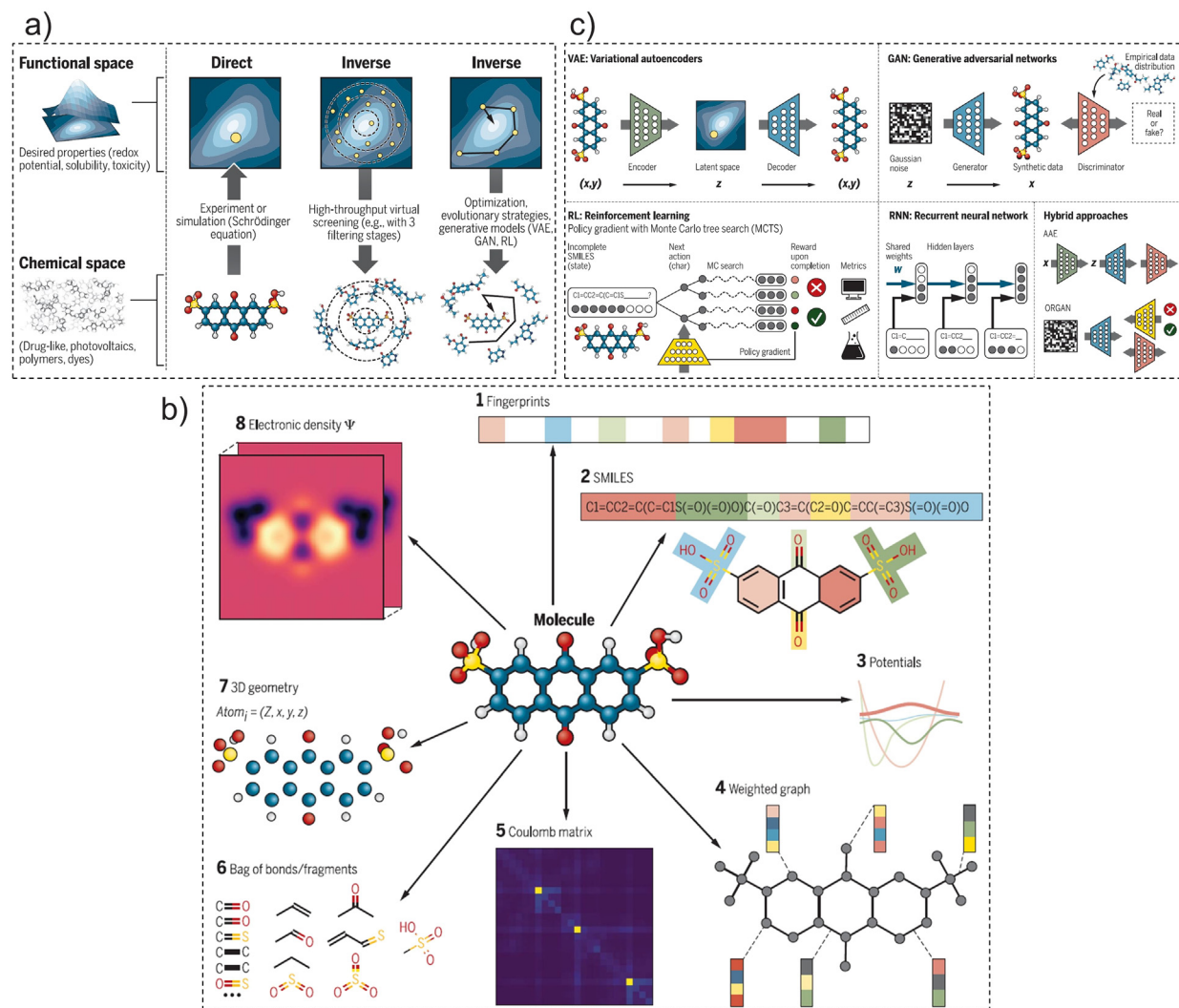


Fig. 22. (a) Schematic of the different approaches towards molecular design. Inverse design starts from desired properties and ends in chemical space, unlike the direct approach that leads from chemical space to the properties. (b) Different types of molecular representations applied to one molecule, AQDS, which is used in the construction of organic redox flow batteries. Clockwise from top: (1) A fingerprint vector that quantifies presence or absence of molecular environments; (2) SMILES strings that use simplified text encodings to describe the structure of a chemical species; (3) potential energy functions that could model interactions or symmetries; (4) a graph with atom and bond weights; (5) Coulomb matrix; (6) bag of bonds and bag of fragments; (7) 3D geometry with associated atomic charges; and (8) the electronic density. Ref. [304].

5. ML/AI in the study of PAH for batteries

5.1. General ML/AI approaches for organic molecules

Theoretical calculations have significantly advanced the discovery of energy storage material over the past decade. The synergy between computation and experiments has proven highly successful, yielding high-performance materials for metal-ion batteries [39,47,153,305–310,310–317]. Computational simulation surpasses inefficient trial-and-error experimentation by efficiently and cleanly revealing fundamental structure-performance relationships, thereby providing reliable, targeted design strategies for experiments [51,316,318–324].

Quantum chemical techniques unveil the characteristics of a molecular structure once the fundamental parameters for the atomic nuclei and their spatial positions have been determined [325,326]. Inverse design subverts this paradigm by starting the design process with desired functionality and then searching for an ideal molecular structure. In this paradigm, functionality is the input and the structure is the output. It is not necessary for functionality to correspond to a single definitive structure; it may be distributed across a range of possible structures. Inverse design (Fig. 22a) uses a range of methods including optimization, sampling, and search techniques to navigate the space of possibilities within chemical space [327,328].

High-throughput virtual screening (HTVS) is one of the earliest methodological frameworks developed for inverse design, originating within pharmaceutical drug discovery pipelines. HTVS employs computational simulation as an exploratory tool for the large-scale evaluation of molecular libraries [329,330]. The process begins with an initial curated candidate library based on heuristic chemical knowledge, which contains the search space to a computationally tractable scale. Subsequent filtering stages apply focused criteria, including synthesizability, solubility, toxicity, stability, bioactivity, and selectivity, to prioritize viable candidates. These filtered molecules undergo expert evaluation to identify potential lead compounds for organic synthesis. Critically, successful molecular motifs and substructures identified in this process are systematically incorporated into subsequent design cycles, enabling iterative functional optimization.

Another approach leverages generative models derived from machine learning. To contextualize this methodology, firstly distinguish between discriminative and generative modeling paradigms. Discriminative models estimate the conditional probability $p(y|x)$, the likelihood of observing target properties y (e.g., bandgap energy, solvation energy) given a molecular representation x . Conversely, generative models learn the joint probability distribution $p(x, y)$, capturing the co-occurrence probability of molecular structures and their associated properties. Crucially, conditioning this joint distribution enables both direct ($p(y|x)$) and inverse ($p(x|y)$) design paradigms, with the latter probabilistically generating molecular representations x that satisfy target properties y . There are several ways to build generative models, but there are three main approaches used: variational autoencoders (VAEs) [331], reinforcement learning (RL) [332], and generative adversarial networks (GANs) [333]. Representations of molecules determine which approach to use, as they determine the information that can be used in the models.

Accurate modeling of molecular systems requires solving the Schrödinger equation (SE) for the electronic Hamiltonian, yielding properties related to energy, geometry, and potential energy surface curvature. Within the SE framework, molecules are defined by nuclear charges and the Cartesian coordinates of their constituent atoms in three-dimensional space. However, ML algorithms often benefit from representations that directly encode relevant physical constraints and properties and Cartesian coordinates are not always computationally optimal. Representations facilitating direct encoding of physical priors reduce the computational burden of learning fundamental patterns from first principles. An ideal representation spanning chemical space must inherently preserve the key symmetries of the SE: permutation, rotational, reflection, and translational invariance for identical particles [334]. Mathematical constructs such as convolution, Fourier transform, and determinant are frequently integrated into representations or models to enforce these symmetries [335,336].

Molecular representation remains an active research challenge, with numerous proposed approaches exhibiting varying efficacy across different target properties [337]. Current representations broadly fall into three categories:

- Discrete: Text-based encodings (e.g., SMILES strings).
- Continuous: Vector or tensor-based descriptors.
- Weighted Graphs: Representing atoms as nodes and bonds as edges, potentially augmented with vectorized features (e.g., bond type, atomic charge, distance).

While graphs can be represented as sparse matrices, their processing within models differs fundamentally. Representations typically employ fixed-length formats via padding or dummy atoms.

For inverse design, invertibility is the ability to unambiguously map from a representation back to a synthesizable molecular structure is a critical property. If a representation lacks this property, an alternative strategy is to define an ideal target representation and evolve or screen molecules to match it quickly. Molecular graphs and Hamiltonians are examples of inherently invertible representations.

Molecular graphs offer a natural representation, interpreting molecules as undirected graphs (atoms = nodes, bonds = edges). Hydrogen atoms are often treated implicitly, inferred from standard valence rules, to reduce complexity. SMILES strings [338] represent a common text-based graph encoding governed by specific syntax. Advanced graph representations eschew text, utilizing weighted graphs with vectorized node and edge features encoding chemical attributes like bonding type, aromaticity, charge, and distance [339–341]. A significant characteristic of graph representations is their frequent non-uniqueness; this degeneracy can facilitate data augmentation [342] but may also introduce noise into models [337].

In contrast to graphs derived from bonding principles, Hamiltonian-based representations rely solely on fundamental physics and atomic constants. The Coulomb matrix [343], for instance, encodes pairwise Coulombic interactions between atomic charges. Combinations of base representations (via concatenation, summation, or differences) can model reactions [344], ensembles, or conformers. Other representations, while often used for prediction, may achieve invertibility through lookup tables or inversion techniques. Examples include bag of bonds [345], atoms [346], fingerprints [347,348], electronic density [335], symmetry functions [348], and chemical environments [334]. Fig. 22b illustrates several key molecular representations.

Molecular representations typically serve as input to deep neural networks (DNNs). Within the network, these inputs undergo sequential transformations across multiple layers. Each layer usually applies a linear operation followed by a nonlinear activation function. The weights governing these transformations are optimized via backpropagation to minimize a task-specific loss function (e.g., for regression, classification, or generation). Through this optimization, the intermediate (hidden) representations learned by the network capture transformed features of the original data, making deep learning a form of representation learning [349]. Generative models, a specific application, aim to learn the underlying distribution of training data in order to generate novel, similar data. Their loss function quantifies the difference between the generated data distribution and the observed empirical distribution. For generating sequences, recurrent neural networks (RNNs) [350,351] are a common foundation, producing sequences step-by-step by predicting the next element. RNNs can be enhanced with long short-term memory cells (LSTMs) [351,352] to handle long-range dependencies, and further augmented with attention and memory mechanisms [353]. Fig. 22c illustrates various generative model architectures.

Beyond pure generation, inverse design requires steering the generative process towards desirable properties. Variational autoencoders (VAEs) enable explicit property optimization directly within their continuous latent space. In contrast, optimizing properties with GANs and RNNs typically involves biasing the generation process itself, often using reinforcement learning (RL) to reward or penalize specific generative behaviors. VAEs [331] enable controlled generation through a structured latent space, unlike standard autoencoders that compress data into a fixed latent vector and reconstruct it, potentially memorizing inputs. VAEs enforce a probability distribution on the latent space (typically Gaussian). This probabilistic encoding represents molecules as distributions rather than points. It forces the model to learn meaningful features, enabling interpolation and sampling of novel molecules.

The VAE's continuous, differentiable latent space forms a probabilistic manifold where molecular similarity corresponds to spatial proximity. This allows:

- Interpolation: Tracing paths between molecules and decoding intermediates.
- Gradient-based optimization: Direct property tuning via latent vector adjustment.
- Semi-supervised organization: Joint training on structure and properties clusters molecules by similarity in latent space.

Initial SMILES-based VAEs have evolved to incorporate syntax/grammar rules [354,355], improving validity. Optimization challenges (e.g., local minima) are addressed via Bayesian optimization [355] or Gaussian processes [356], enabling targeted local/global property exploration across the latent manifold.

Another generative modeling approach uses adversarial training within the Generative Adversarial Network (GAN) framework. In this architecture, a generator network competes against a discriminative model (discriminator). The generator synthesizes data samples by transforming vectors sampled from a latent noise distribution, while the discriminator attempts to classify inputs as real (from training data) or synthetic (generator output). The models undergo alternating adversarial training, in which the generator's objective is to produce samples indistinguishable from real data — thereby forcing the discriminator's classification accuracy towards chance levels (50%). GAN convergence, however, presents significant challenges, including susceptibility to mode collapse (failure to capture full data diversity) and instability arising from discriminator overpowering during training. Current research actively addresses GAN optimization [357], including specialized techniques for handling discrete data representations where gradient-based optimization is complicated by non-differentiability [358,359].

Guiding the generative process in adversarial networks (GANs) or recurrent neural networks (RNNs) requires gradient-based optimization to steer molecular generation towards target properties. While standard GAN training leverages gradients to maximize the similarity between generated and real data distributions, inverse design necessitates property-conditioned gradients. These gradients must incorporate target physicochemical properties predicted by cheminformatics models, simulations, or experimental data—and propagate backward through the generator. Achieving this requires constructing a differentiable gradient estimator capable of translating property constraints into latent space updates, thereby biasing the generator towards structures exhibiting desirable characteristics.

Reinforcement learning (RL) offers several methodologies for this challenge, with Q-learning [332] and policy gradient methods [360] being particularly prominent. In this framework, the generator acts as an agent that learns optimal actions (for example, sequential token additions) in an environment (SMILES string generation) to maximize a reward signal that quantifies target molecular properties. A critical constraint arises from the structure of SMILES representations, as property-based rewards can only be assigned when completion of valid strings are completed. To address this delayed reward attribution problem, Monte Carlo Tree Search (MCTS) is frequently used. MCTS constructs a probabilistic decision tree by simulating potential sequence completions and evaluating their associated rewards. The path weights are updated according to task performance. This search process trains a neural network that parametrizes the generation policy.

Inverse design is a critical enabler for accelerating the discovery and engineering of novel materials. Recent advances in inverse design methodologies, particularly machine learning-driven approaches, have facilitated probabilistic, data-driven mappings of chemical space. Generative models exemplify this progress by producing extensive libraries of candidate molecules. However, realizing the potential of these computational approaches requires concomitant development of automated high-throughput experimental pipelines for physical synthesis and characterization. Such validation is essential to establishing the practical utility of generative inverse design frameworks.

The integration of machine learning into the design of metal-ion batteries is progressing at an exponential pace, rapidly transforming from a niche methodology into a cornerstone of modern materials discovery [361–369]. This explosive growth is vividly illustrated by the remarkable surge in high-impact publications over just the past year, which spans the entire battery landscape, from AI-powered molecular discovery and the design of Li-rich cathodes [361–363] to the optimization of electrolytes and the prediction of their solvation dynamics [370–372], as well as the accelerated screening of materials for solid-state and lithium-sulfur batteries [365,373,374]. Furthermore, the field is witnessing the emergence of sophisticated, interpretable models and even dedicated AI-driven software platforms, underscoring a clear shift towards practical, automated, and scalable battery design solutions [375–378].

5.2. High-throughput screening

A particularly promising advancement is machine-learning-assisted high-throughput screening, an emerging method that further accelerates the design and refinement of next-generation metal ion battery materials [363,370,374,377,379–383]. Allam et al. [384–387] employed a combined DFT and machine learning (ML) framework to create a method for rapidly testing a wide range of materials for use as electrodes in devices that employ PAHs (Fig. 23). Their DFT modeling predicted fundamental quantum mechanical properties (redox potentials, electron affinity, HOMO, LUMO) for selected organic materials. These properties, along with

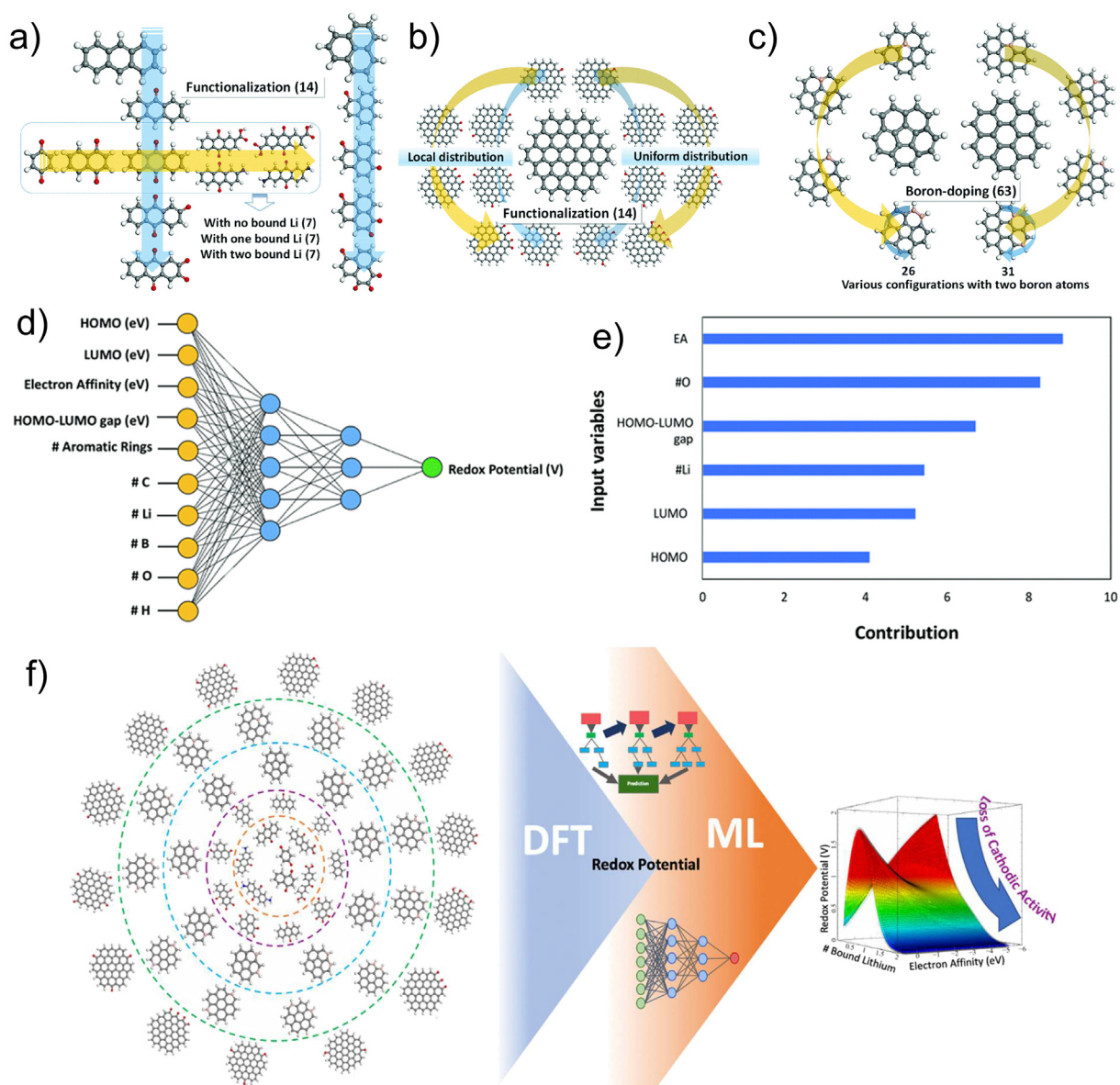


Fig. 23. a-c) Molecules used to train an artificial neural network. d) Artificial neural network with 10 input variables and two hidden layers. e) Contributions of each input variable to the redox potential (target property) in the machine learning. Ref. [384] f) High-throughput screening workflow integrating DFT and ML. Ref. [386].

structural data (atom counts for O, Li, B, C, H; number of aromatic rings), serve as inputs for ML-based redox potential prediction. Linear correlation analysis reduced this large input set to six core variables: electron affinity, HOMO, LUMO, HOMO–LUMO gap, number of oxygen atoms, and number of lithium atoms. An artificial neural network, trained using the quasi-Newton method, accurately estimates redox potentials. The analysis of contributions showed that electron affinity was the most important factor, followed by the number of oxygen atoms, the energy gap between HOMO and LUMO, the number of lithium atoms, LUMO, and HOMO.

Building on this foundation, Allam et al. [385,386]. significantly advanced the methodology by introducing more sophisticated models like Kernel Ridge Regression (KRR) and Gradient Boosting Regression (GBR), benchmarked against the initial ANNs. KRR, combined with an advanced feature engineering “pipeline” and LASSO feature selection, achieved the highest predictive accuracy, reaffirming electron affinity (EA) and the number of bound lithium atoms (#Li) as the most critical features for determining redox potential [386]. The practical utility of these computational insights was further demonstrated in a combined experimental and theoretical study on carbon quantum dot and reduced graphene oxide hybrids for alkali-ion storage [387].

To elucidate quinone instability in water, Tabor et al. [388] integrated DFT and experimental studies, probing links between structure, reduction potential, and decomposition pathways. Their virtual screening of 140,000 quinone pairs (over 1 million calculations) applied learned design principles. Stability against water was evaluated using the gem-diol constant ($\log K_{hyd} < 1.0$) and Michael addition energy ($\Delta E_{mic} < 0.03$ eV) (Fig. 24a-f). They found stable low-reduction-potential quinones to be common, but examples of stable high-reduction potential examples are scarce [388]. The highest reported stable reduction potential in acid (0.85 V vs. SHE) corresponds to a predicted stability cliff (0.90–0.95 V). Critically, quinones with high reduction potential (> 1.0 V OCV) and thermodynamic resistance to Michael addition and gem-diol formation are absent. Mitigating Michael addition kinetically may be feasible, but addressing gem-diol formation is uncertain. Modulating redox-site pK_a is another avenue for investigation.

The progression from high-throughput screening to targeted material design is exemplified by recent work from the Tabor group. Building on stability screening principles, Li and Tabor [389] conducted a two-objective virtual screening campaign for low-potential radical candidates in organic polymer batteries. Their approach integrated a Gaussian Process regression model to efficiently predict reduction potentials across 660 candidates, with top performers subsequently evaluated for electronic coupling—a key property for solid-state electron transport. This screening identified N-methylphthalimide derivatives as promising “lead” candidates for further study (Fig. 24g,h). [389].

Subsequently, Alessandri et al. [390] validated these predictions using multiscale modeling that combined atomistic molecular dynamics, electronic structure calculations, and machine learning surrogate models. Their work predicted apparent electron diffusion coefficients for phthalimide-based polymers—three orders of magnitude higher than the reference nitroxide radical polymer PTMA—with the PTMA prediction validated against experimental data [390]. This demonstrates a complete pipeline from computational screening to quantitative property prediction for complex material systems.

While DFT implementations differ, standard high-throughput computational (HTC) screening involves: (1) Selecting a large, diverse database; (2) Benchmarking methods for cost/accuracy balance; (3) Establishing a criteria-driven workflow (including properties, cost, toxicity); (4) Executing calculations; (5) Selecting top candidates; and (6) Extracting trends and synthesis guidance. Current limitations of HTC include reliance on brute-force DFT scaling, analysis of only a tiny chemical subset, and dependence on researcher intuition. Machine learning presents a more robust alternative for leveraging larger datasets and deeper analysis.

A more strategic approach to the successful identification of promising candidates is to create a high-throughput virtual screening (HTVS) system that incorporates a variety of mathematical or proxy models with varying degrees of precision and computational expense. Typically, computationally faster and less accurate models are used at the initial stages to quickly filter out candidates that are unlikely to have the desired properties. The surviving candidates then move on to subsequent stages, where they are evaluated using more accurate, but also more expensive, models. Finally, molecules that reaches the penultimate stage are verified using the most accurate model. The core challenge in HTVS is optimizing the computational resource allocation across these stages in order to maximize the return on investment (ROCI) in computational resources. Due to their ability to efficiently screen vast candidate libraries, HTVS pipelines are widely used in biology [391], chemistry [392], and materials science [393]. Historically, operational strategies for HTVS pipelines relied on expert intuition, often yielding suboptimal performance. Recently, a mathematical optimization framework has been developed to address this limitation [394,395]. This framework systematically optimizes screening policies for throughput and efficiency by modeling the joint distribution of predictive scores across stages. Based on this model, it jointly optimizes screening thresholds to maximize throughput and accuracy while minimizing computational costs. This optimized approach significantly boosts pipeline efficiency while effectively meeting screening objectives. Although this framework provides a systematic alternative to heuristic policies, it assumes the HTVS pipeline structure itself is fixed and predetermined, focusing solely on optimizing the screening policy.

Woo et al. [394] apply their proposed optimal HTVS design and operational strategies to address the challenge of efficiently and accurately screening redox-active organic materials for next-generation energy storage devices (Fig. 25). To achieve this, the authors propose an effective strategy for constructing an HTVS computational pipeline. This pipeline integrates the highest-fidelity redox potential (RP) predictor, based on DFT calculations, with ML surrogate models designed to balance computational efficiency against predictive fidelity. Specifically, the high-fidelity DFT model is decomposed into four sequential ML surrogate models. Each surrogate computes an intermediate electronic property—namely HOMO energy, LUMO energy, the HOMO-LUMO gap, and the Electron Affinity (EA)—required for the final RP prediction within the DFT framework. These sub-models constitute the foundational components of the HTVS pipeline. Subsequently, five distinct surrogate models are learned to serve as sequential screening stages. Each stage predicts RP utilizing different combinations of these intermediate properties, operating at varying levels of computational complexity and fidelity. Furthermore, the authors explore the incorporation of “sub-surrogate” models that predict the next available intermediate properties based on features accessible at a given stage. These predicted properties function as “virtual” features for downstream surrogate models, enhancing overall predictive accuracy. Finally, the HTVS pipeline optimization framework is generalized to enable the identification of materials exhibiting RP values within a specified target range, rather than solely based on a minimum or maximum RP threshold. Woo et al. [394] rigorously evaluated the performance of their optimized HTVS pipelines under various scenarios, demonstrating significant improvements over baseline methods in terms of screening efficiency, predictive accuracy, and result consistency. Consequently, the approach developed by Woo et al. [394] offers significant potential for accelerating the design of organic materials, including PAHs, for advanced energy storage applications.

Beyond screening, more sophisticated AI paradigms are emerging. Given that chemical space exceeds 10^{60} potential molecules [396], exhaustive screening is impractical. As highlighted by Cao and Aspuru-Guzik [397], computation-experiment synergy is essential for accelerating organic energy storage discovery. Shree Sowndarya et al. [398] recently demonstrated a reinforcement learning framework for molecular design that generates novel structures by learning which building blocks yield desired properties. This paradigm shift from passive screening to active design, summarized by Cao et al. [396], offers significant potential for discovering new PAH-based battery materials beyond those conceived through intuition alone.

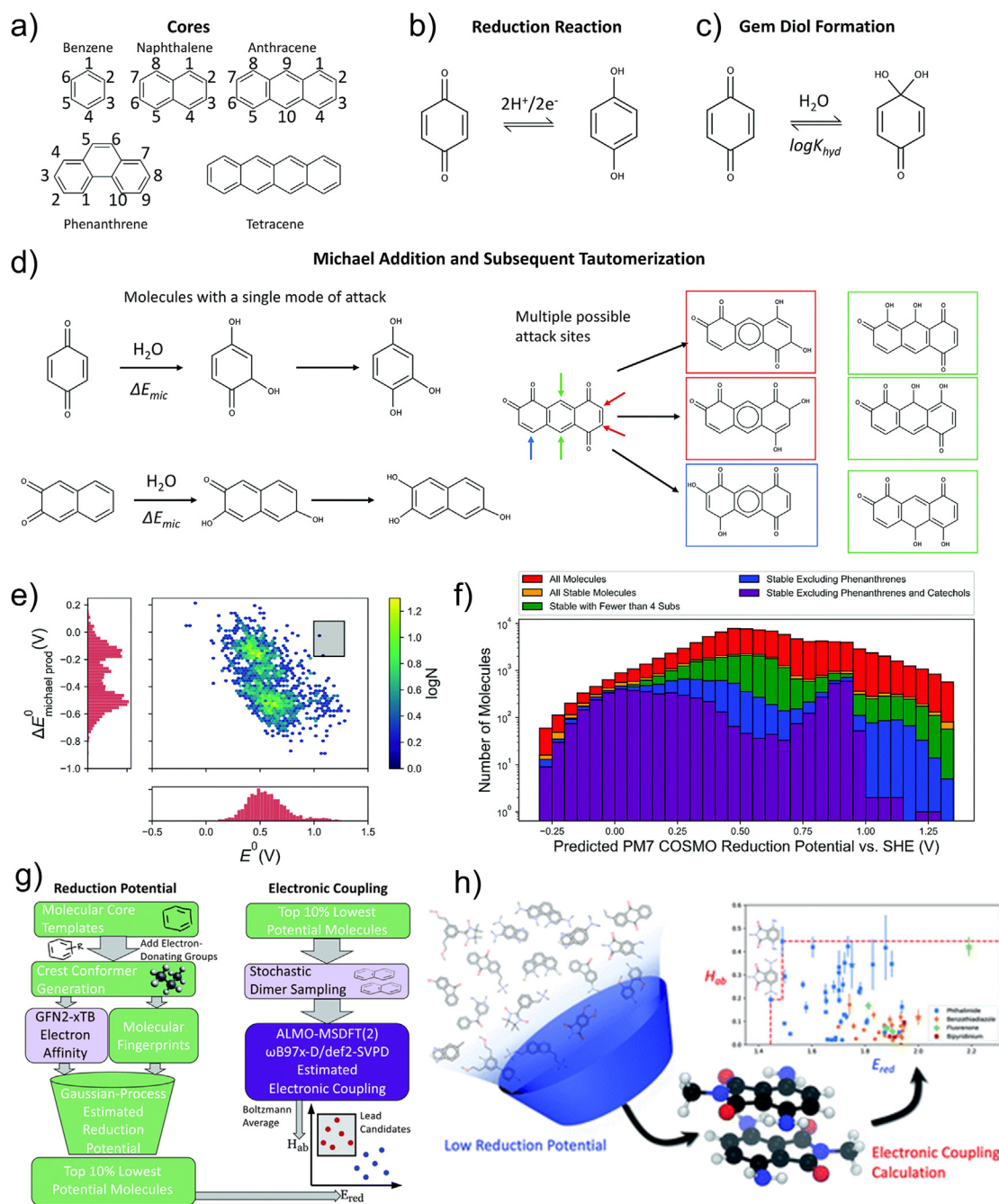


Fig. 24. (a) PAH molecule templates and possible bonding points. (b) The redox reaction of quinones in acidic aqueous flow batteries: the two proton two-electron reduction of two ketones to two hydroxyl groups. (c) mechanism for gem-diol formation, which is reversible but hampers kinetics. (e) change in reduction potential of a molecule upon Michael addition, 1763 molecules pictured, where desired region is shaded in black box and contains only two molecules (sulfonated 1–4 benzoquinones), $E_{\text{Michael prod}}^0$, N are Michael addition/substitution energy and number of molecules, respectively. (e) plot of redox potential distributions of various sets of molecules, where molecules stability criteria follows (1) PM7 COSMO Michael addition energy < 0.03 eV and (2) $\log K_{\text{hyd}} < 1.0$. Ref. [388]. (g,h) Workflow for screening of low reduction potential and high electronic coupling molecule targets. Ref. [389].

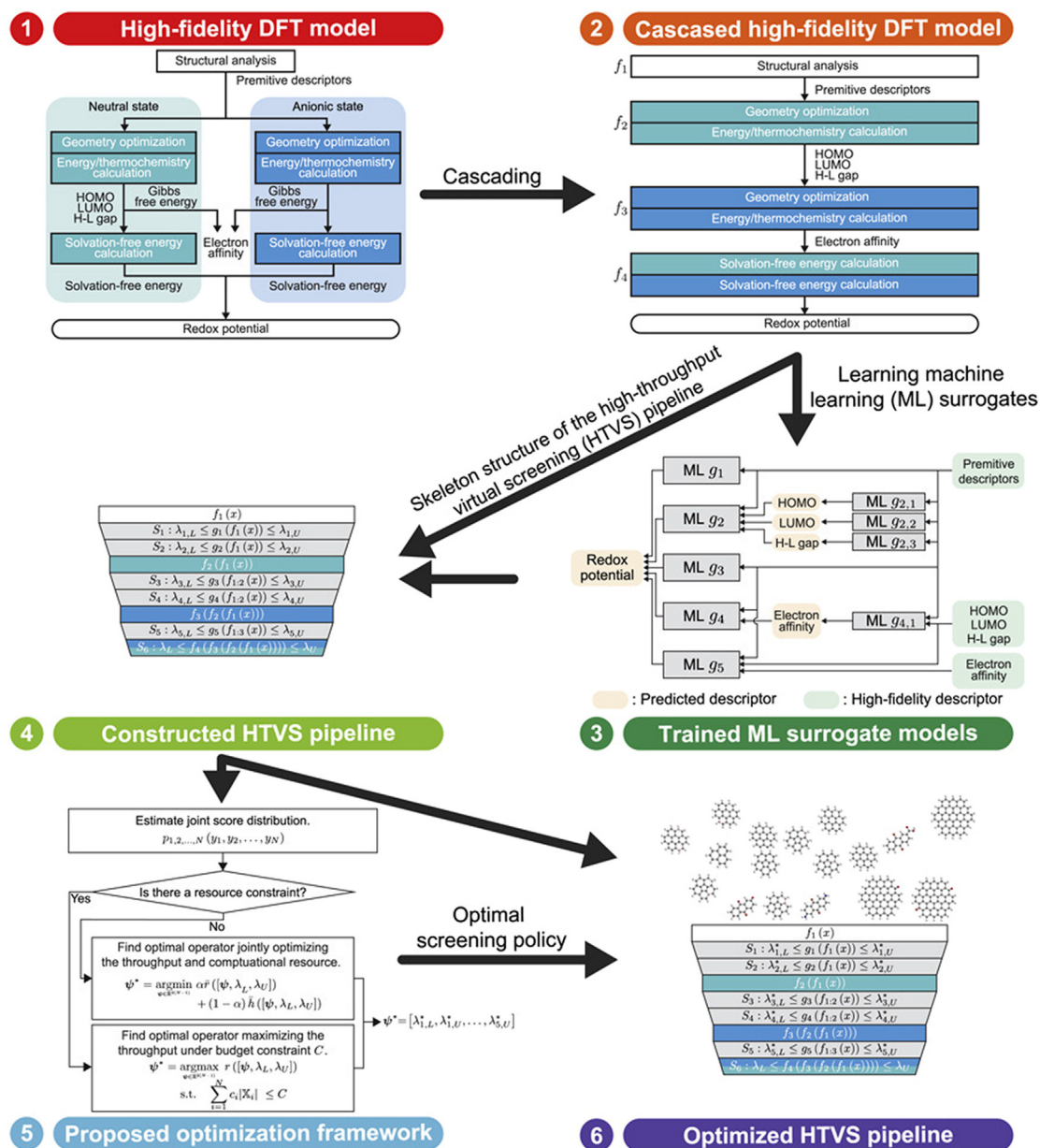


Fig. 25. Scheme of optimal high-throughput virtual screening pipeline for efficient selection of redox-active organic materials. Ref. [394].

5.3. Machine learning

Das et al. [45] proposed a robust supervised ML approach to accurately predict voltage and volume changes during anion intercalation from diverse DIB salts (Fig. 26a) into eight PAH cathode materials (Fig. 26b). They systematically constructed a comprehensive dataset by modeling the intercalation of 1–8 anions per salt into each cathode. To ensure reliability, 15% of voltage data was generated via precise DFT calculations. After evaluating multiple ML models using performance metrics, the optimal model was selected for voltage prediction and applied to the remaining 85% of systems. The model demonstrated exceptional accuracy, confirming its efficacy for predicting voltages across a wide range of intercalation scenarios.

To train their ML models, Das et al. [45] first calculated voltages for selected salt-electrode combinations using DFT. Given eight PAH cathode materials and eleven DIB salts—with intercalation of up to eight anions per system—the complete dataset comprised 704 entries (Fig. 26c). To ensure robust sampling, they strategically generated DFT-calculated voltages for 15% of systems (176 entries), covering two distinct intercalation states per salt–cathode combination. This representative subset captured key features of the full dataset, providing a solid foundation for model training. The authors assert that training on this strategically sampled

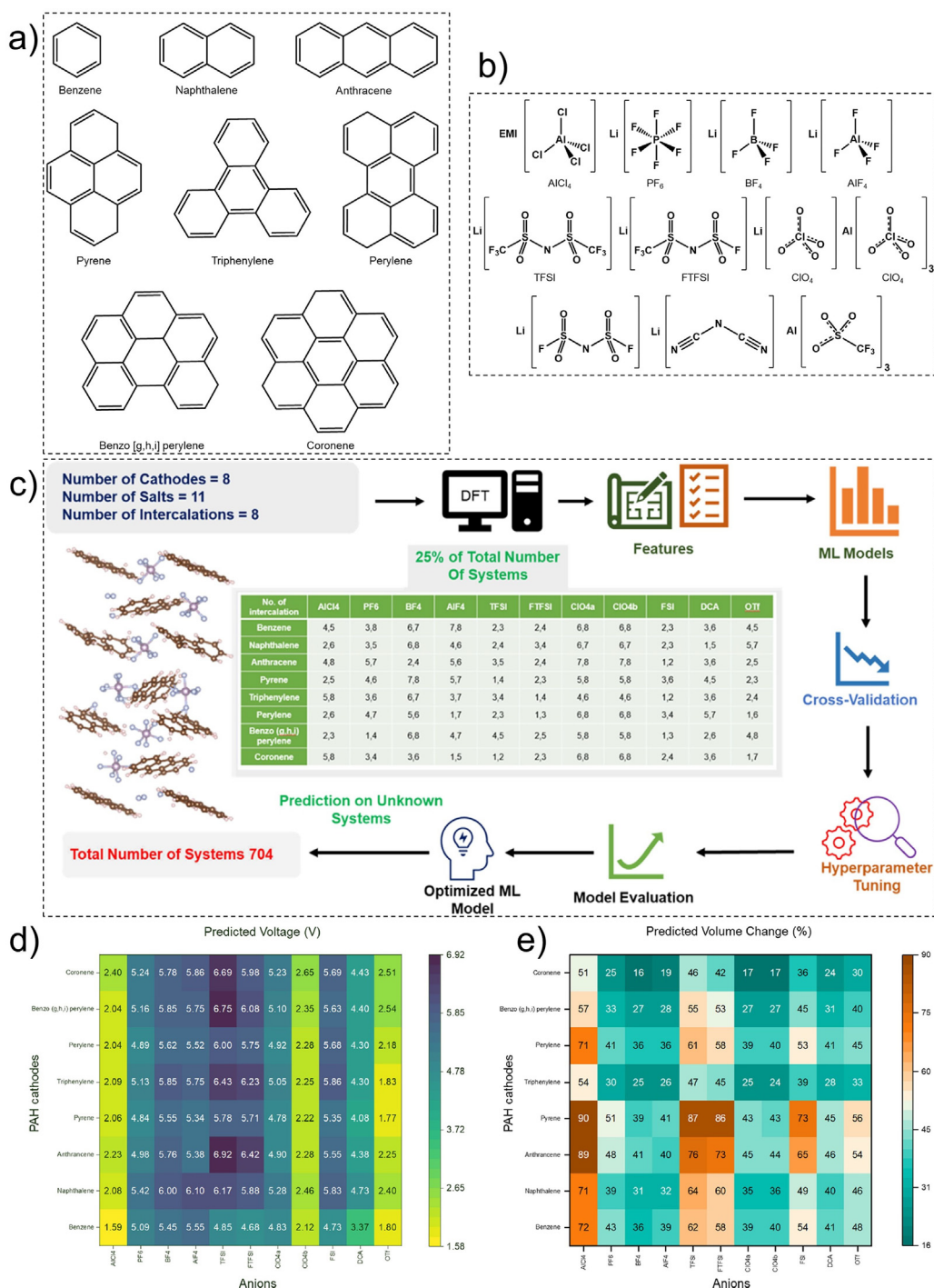


Fig. 26. (a) Considered PAH cathodes. (b) Considered DIB salts for intercalation into PAH cathodes. (c) Workflow for ML-based study of voltage and volume change of cathode materials in DIBs including data processing, feature engineering, model training, selection, and property prediction. (d) Heatmap indicating the predicted voltage for all combinations of anions and PAH cathodes. Here, ClO₄-a represents LiClO₄ and ClO₄-b represents Al(ClO₄)₃ salt. (e) Heatmap indicates the predicted volume change for all combinations of anions and PAH cathodes. Here, ClO₄-a represents LiClO₄ and ClO₄-b represents Al(ClO₄)₃ salt. Ref. [45].

DFT data enabled their ML models to effectively generalize across all 704 anion–electrode combinations. Volume changes during intercalation were similarly calculated for the same dataset.

Das et al. [45] used various regression models: decision trees regression (DTR), extra trees regression (EXTR), gradient boosting regression (GBR), light GBM regression (LGBR), random forest regression (RFR), and XGBoost regression (XGBR), and determined the GBR model as the best-fitted. Using GBR average voltages for all 88 cathode–salt combinations was calculated [45](Fig. 26d). The resulting heatmap reveals that voltage variation depends more strongly on salt choice than cathode material. This arises because each salt has distinct electrochemical behavior governed by, anion–cathode interactions, cation–anode interactions, solvation energy of the intact salt. Consistent with prior observations [399], Al-based DIBs universally deliver lower voltages than Li-based systems across all cathodes. However, Al-salts offer other advantages warranting further study [400]. Among Li-salts, TFSI and FTFSI anions yield the highest voltages (> 4 V), likely due to their large size requiring greater energy for intercalation. This energy penalty stems from overcoming π - π interactions while expanding cathode lattice spaces. Regarding cathodes, benzene consistently yields the lowest voltages for most anions, while coronene delivers the highest. Larger PAHs like coronene exhibit stronger π - π stabilization, requiring higher charging voltages to overcome these interactions during intercalation.

The results obtained within the framework of ML calculations [45] are in good agreement with experimental data. For example, anthracene as a cathode for AlCl_4 -based DIB gives a discharge voltage of 1.5–2.3 V which agrees with result from ML where average voltage of 2.23 V [6]. Crystalline pyrene and polypyrene as cathode materials deliver around 1.7 V in the case of the AlCl_4 -based battery [106], whereas Das et al. [45] in ML calculations has been obtained 2 V in case of pyrene cathode material. A multi-ions battery involving LiPF_6 and AlF_3 salt in EMC-DMC solvent with graphite cathode to show maximum voltage around 5 V [401] while results from ML was values are above 5 V [45]. An $\text{Al}(\text{ClO}_4)_3$ /propylene carbonate-fluoroethylene carbonate electrolyte with the graphene-based cathode also reported maximum voltages around 2 V [402] which agrees with ML results.

Furthermore, the employment of the most suitable XGBR model for the prediction of volume change has enabled Das et al. [45] to predict the volume change upon full intercalation for all 88 combinations of salt and PAH cathode. As demonstrated by the heatmap, specific PAH materials, including anthracene and pyrene, demonstrate heightened volume alterations in response to the majority of the anion intercalations (Fig. 26e). Conversely, larger PAHs, such as benzo[g,h,i]perylene and coronene, exhibit a comparatively minimal volume change upon intercalation by most anions. It has been demonstrated that, among the anions, AlCl_4 , TFSI, FTFSI, and FSI induce a greater volume change upon intercalation into the cathodes in comparison to the other salts. This phenomenon can be attributed to the larger size of the former, which necessitates greater spatial requirements for intercalation when compared to smaller ions such as AlF_4 , BF_4 , PF_6 , and ClO_4 , among others. The values for both LiClO_4 and $\text{Al}(\text{ClO}_4)_3$ are found to be similar, as the intercalating ion is virtually the same (see Fig. 26e).

The outcomes of the ML simulation conducted by Das et al. [45] demonstrated that amalgamations of $\text{Al}(\text{ClO}_4)_3$ and $\text{Al}(\text{OTf})_3$ salts with PAHs, including coronene and benzo[g,h,i]perylene, have the potential to satisfy low voltage requirements (approximately 2.5 V). Conversely, for higher voltage requirements (approximately 5 V), the most favorable combinations are hypothesized to be LiBF_4 and LiAlF_4 salts with coronene, benzo[g,h,i]perylene, naphthalene, and LiFSI with coronene. The low-volume change exhibited by these systems upon intercalation, in comparison to graphite, suggests their long-term cyclic stability. This is a significant consideration in the context of DIBs, as their stability is a primary concern. In conclusion, the work of Das et al. [45] establishes a robust and widely applicable machine learning (ML)-based workflow for efficient and accurate screening of PAH cathode materials and salts based on voltage predictions. It is evident that this approach facilitates rapid assessment and exploration, thereby contributing to the accelerated design and development of advanced PAH cathodes and salts.

5.4. Summary and future directions for ML/AI in PAH-based batteries

The expanding role of machine learning in battery materials discovery is not limited to the direct screening of PAH cathodes. As highlighted in recent comprehensive reviews and articles, ML is accelerating the design and performance prediction of a wide range of electrode and electrolyte materials for lithium-ion batteries [403–406]. Xu et al. [407] provide an extensive overview of how data-driven methods are being used to capture complex structure–activity–process–performance relationships, allowing for the rapid screening and compositional optimization of cathode, anode, and electrolyte materials [407]. This broader methodological approach provides a powerful toolkit that can be directly address the challenges faced by PAH-based systems, such as optimizing doping strategies and predicting the stability of novel PAH derivatives in complex electrolyte environments.

Furthermore, the principles of computer-aided design are being powerfully applied to another key battery component, the electrolyte. Pi et al. [408] reviewed the use of multi-scale molecular simulations and artificial intelligence for designing ionic liquids as green solvents in chemical processes, including their application in lithium-ion batteries. This is particularly relevant to PAH-based batteries where solubility of active materials and the nature of the electrode-electrolyte interfaces are critical challenges. The framework described by Pi et al. combining quantum chemistry, molecular dynamics, and machine learning for high-throughput screening of IL properties, offers a direct pathway to computationally design optimal electrolytes that are compatible with specific PAH electrodes. This mitigates dissolution issues and enhances solid-electrolyte interphase formation.

These studies collectively demonstrate that the integration of ML with experimental validation is rapidly becoming a standard paradigm in energy storage research. The frameworks developed for general organic electrodes and ionic liquids are not just abstract concepts but are ready-to-use methodologies. For the PAH community, this means that the computational insights from studies like those of Das et al. [45] and Allam et al. [386] can now be combined with advanced electrolyte design tools to tackle multi-faceted challenges. For instance, one could envision a future workflow where an ML model first screens thousands of PAH derivatives for a desired redox potential and stability, and a second, electrolyte-focused ML model simultaneously screens for the optimal IL solvent that maximizes the candidate's capacity retention and rate performance. This integrated, ML-guided approach promises to accelerate the development of complete, high-performance PAH-based battery systems, directly addressing the need for a more cohesive strategy from molecular design to full-cell realization.

6. Conclusions and perspectives

PAHs have firmly established themselves as a versatile and promising class of materials for next-generation metal-ion batteries. This review has systematically charted the progress from fundamental molecular studies to the rational design of high-performance anodes, cathodes, and pre-metallization agents. The unique advantages of PAHs, including their structural tunability, economic viability through the use of abundant precursors, environmental sustainability, and exceptional redox chemistry, position them as compelling alternatives to conventional inorganic materials.

As anode materials, PAHs have demonstrated capabilities that transcend traditional limitations. Experimental validation confirms that small-molecule PAHs like naphthalene and phenanthrene achieve reversible capacities of 200–240 mAh g⁻¹ with extraordinary longevity, retaining > 92% capacity over 1500 cycles. More significantly, crystalline PAHs such as pentacene offer theoretical capacities up to 963 mAh g⁻¹ for multivalent systems, far exceeding graphite's 372 mAh g⁻¹, while maintaining minimal volume expansion (< 8% for Ca-ion intercalation). The curved architecture of corannulene, with its remarkable reversible capacity of 464 mAh g⁻¹, exemplifies how molecular topology can unlock new storage mechanisms.

For cathode applications, PAHs have proven equally versatile. In aluminum-ion batteries, anthracene delivers 143 mAh g⁻¹ with exceptional stability over 800 cycles, while polypyrrene's amorphous architecture overcomes the dissolution and conductivity limitations of its crystalline counterpart. In dual-ion batteries, the strategic advantage of edge-site availability becomes paramount. Smaller PAHs like perylene and triphenylene, with higher edge-site density, achieve superior reversible capacities (88–95 mAh g⁻¹) compared to coronene. Functionalization strategies have systematically elevated redox potentials, with carbonyl-substituted phenalenyl derivatives reaching 3.93 V vs. Li/Li⁺, demonstrating that molecular engineering can rival inorganic cathodes in voltage.

A particularly powerful demonstration of PAH utility lies in pre-metallization, where compounds like lithium naphthalenide and sodium biphenyl provide a scalable and air-stable route to mitigate irreversible ion loss. Their performance can be atomically engineered, as shown by methyl-substituted derivatives that lower the redox potential to achieve a near-theoretical initial Coulombic efficiency of 98.0% in hard carbon anodes. The field reaches an even higher level of sophistication with closed-loop re-lithiation. By employing reagents such as pyrene-Li and perylene-Li, spent cathodes can be restored with perfect 100% atom economy and full recyclability, creating a zero-waste process that perfectly embodies the principles of sustainable development.

The integration of machine learning and high-throughput screening has fundamentally accelerated the discovery and optimization of PAH-based materials. Seminal computational studies have established electron affinity as the primary descriptor governing redox behavior, enabling rapid initial assessment of candidate molecules. Systematic investigations combining density functional theory with machine learning regression models have successfully predicted voltage profiles across hundreds of salt–cathode combinations, demonstrating that computational approaches can now reliably guide experimental efforts. Optimized high-throughput virtual screening pipelines, employing multi-fidelity surrogate models, efficiently navigate vast chemical space by balancing computational cost against predictive accuracy. Perhaps most significantly, emerging generative models are transitioning the field from passive screening to active inverse design, where molecules are created specifically to meet target electrochemical parameters such as redox potential, capacity, or stability. This paradigm shift promises to dramatically accelerate the development of next-generation PAH-based energy storage materials.

Beyond the remarkable progress in molecular design and electrochemical performance, a critical examination reveals that the path to the commercial viability of PAH-based batteries is contingent upon overcoming several persistent and interconnected challenges. While this review has highlighted the success of doping and functionalization in tuning redox potentials, the fundamental issue of intrinsically low volumetric energy density remains a significant hurdle. The organic nature of PAHs, with their low density and reliance on often-heavy counterions for charge balance in cathodes (e.g., AlCl₄⁻, PF₆⁻), results in energy densities that, while competitive gravimetrically, lag behind their inorganic counterparts on a volumetric basis. Future research must therefore pivot from a purely gravimetric focus to a holistic optimization of both gravimetric and volumetric metrics. This could involve the strategic design of dense, three-dimensional PAH frameworks or the development of hybrid electrode architectures that integrate high-capacity PAHs with conductive, high-density carbon scaffolds, thereby maximizing active material loading within the electrode.

Furthermore, the dual challenges of dissolution in electrolytes and long-term structural stability require a more integrated, systems-level approach. As evidenced by the contrasting behaviors of crystalline pyrene and amorphous polypyrrene, molecular engineering alone is insufficient; the mesoscale architecture of the electrode and its interface with the electrolyte are paramount. The very solubility that enables elegant pre-metallization strategies also threatens the longevity of cathode materials. Future advancements will likely depend on the development of in-situ polymerization techniques, the use of highly compatible solid or gel-polymer electrolytes to physically anchor active molecules, and the creation of protective artificial interfaces. The insights gained from machine learning studies, which have identified electron affinity as a key descriptor for redox potential, must now be extended to model and predict solubility parameters and SEI formation dynamics on complex PAH surfaces. This would allow for the co-design of active materials and electrolytes, moving beyond isolated component optimization.

Finally, the integration of machine learning into the discovery pipeline must evolve from a tool for screening existing libraries to a true engine for inverse design and predictive synthesis. The current high-throughput virtual screening, while powerful, is often constrained by the computational cost of accurately modeling the dynamic behavior of PAH systems in a full-cell environment. The next frontier lies in developing multi-fidelity models that can bridge the gap between gas-phase quantum mechanical calculations and the complex reality of solvated ions, fluctuating potentials, and evolving interfaces over thousands of cycles. By training generative models on both successful and failed experiments, the community can begin to predict not only the ideal redox potential and capacity of a novel PAH but also its synthetic accessibility, processability, and long-term stability *a priori*. This paradigm shift from passive screening to active, goal-oriented design will be essential to navigate the vast chemical space of PAHs and deliver a new generation of sustainable, high-performance energy storage systems that are ready for practical deployment.

CRediT authorship contribution statement

Ilya V. Chepkasov: Writing – original draft, Conceptualization. **Stanislav A. Evlashin:** Writing – original draft. **Alexander G. Kvashnin:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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